Class Notes

PH3201: Basic Statistical Mechanics

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Course Information

§1 Marks Distribution

Mid-Semester Exams: 20 Marks.

End-Semester Exams: 50 Marks.

Internal Assisment: 30 Marks.

- ▷ Class Test: Two tests of 15 marks each.
- ▷ Assignments: As practice problems.

§2 Syllabus

Basic overview of this course is as follows:

1. Review of Thermodynamics.
2. Probability Theory.
3. Microcanonical Ensemble.
4. Canonical Ensemble.
5. Grand Canonical Ensemble.
6. Quantum Statistics.

§3 References

1. *Statistical Physics of Particles* by Mehran Kardar[3]. For the first half of the course (*i.e.* classical statistical mechanics).
2. *Statistical Mechanics* by Kerson Huang[2]. For the second half of the course (*i.e.* quantum statistical mechanics).

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1 Lecture 01: (02/01/2025)

§1.1 Introduction

If we want to summarize the whole field of statistical mechanics in one line, we can say that "it is the study of *coarse-grained* properties of *very large* systems". This description is very easy to understand, but it is not very precise. Let's try to understand the keywords in this description.

- **Coarse-grained:** This means that we are not interested in the microscopic details of the system. We are interested in the properties of the system which are averaged over the microscopic details. For example, we are not interested in the position and momentum of each particle in a gas, but we are interested in the pressure, temperature, and volume of the gas.
- **Very large:** This means that we are interested in systems which have a very large number of particles. For example, a mole of gas contains 6.022×10^{23} particles.

This also carries another very important property of the systems *i.e.* interactions between the constituents particles. For our gas example, we assume that the gas is diluted enough that the interactions between the particles are negligible.

Building upon the above example, we can think of a gas particle embedded in a 'field of interactions' which is created by all the other particles. This description of average interactions is now know as the *mean field approximation*.

In our discussion, we will only focus on the equilibrium properties of the system. We will not discuss the dynamics of the system which comes under the domain of *non-equilibrium statistical mechanics*.

For most part of the course, we will use following systems as examples:

- We will use gas system for our study and mainly focus on the density of the gas.
- We will also study magnetic systems (made up of spins *i.e.* spin chains) and focus on the state of the system like ordered or disordered.
- Biological systems are also a good example of systems which can be studied using statistical mechanics.
- Social Interactions can also be studied using statistical mechanics.

§1.2 Review of Thermodynamics

Thermodynamics: It is a phenomenological description of properties of macroscopic systems in thermal equilibrium. It is based on a few empirical laws which are based on observations. These laws are not derived from any microscopic theory. On the other hand, statistical mechanics tries to derive these laws from microscopic theory.

Statistical Mechanics: It is a microscopic theory of the thermodynamic properties of systems. It tries to derive the laws of thermodynamics from the microscopic description of the system *i.e.* $\{\mathbf{r}, \mathbf{p}\}$. We can think stat mech as a bridge between macroscopic properties and microscopic description.

This was tried by various people using different description of '*entropy*' like

$$\underbrace{S}_{\text{Clausius}} = \overbrace{k_B \ln \Omega(E, V, N)}^{\text{Boltzmann}} = \underbrace{-k_B \sum_i p_i \ln p_i}_{\text{Shannon}} \tag{1.1}$$

§1.2.1 The Zeroth Law

Law 1.1 (Zeroth Law)
If two systems, A and B, are separately in equilibrium with a third system, C, then they are also in equilibrium with one another.

Let the equilibrium state of systems A, B, and C be described by the variables $\{A_1, A_2, \ldots\}$, $\{B_1, B_2, \ldots\}$, and $\{C_1, C_2, \ldots\}$ respectively. The assumption of equilibrium implies constraints on the variables of the systems.

$$f_{AC}(A_1, A_2, \ldots, C_1, C_2, \ldots) = 0 \quad \text{and} \quad f_{BC}(B_1, B_2, \ldots, C_1, C_2, \ldots) = 0 \tag{1.2}$$

Due to fact that these constraints describe real physical systems, they must be solvable for any arbitrary variables, so we can write

$$C_1 = F_{AC}(A_1, A_2, \ldots, C_2, \ldots) \quad \text{and} \quad C_1 = F_{BC}(B_1, B_2, \ldots, C_2, \ldots) \tag{1.3}$$

Since C is in equilibrium with both A and B independently, we can write

$$F_{AC}(A_1, A_2, \ldots, C_2, \ldots) = F_{BC}(B_1, B_2, \ldots, C_2, \ldots) \tag{1.4}$$

According to law 1.1, A and B are also in equilibrium with each other, so we can write

$$f_{AB}(A_1, A_2, \ldots, B_1, B_2, \ldots) = 0 \tag{1.5}$$

So we can find a set of variables $\{A_i, B_j\}$ which satisfy eqn. (1.5), and substituting these variables in eqn. (1.4) we can find an equality which must holds for any set of variables $\{C_i\}$. We can then change these parameters, moving along the manifold constrained by eqn. (1.5), and eqn. (1.4) will remain valid irrespective of the state of C.

Therefore, we can simplify eqn. (1.4) by cancelling out the C terms, and we get

$$\Theta_A(A_1, A_2, \ldots) = \Theta_B(B_1, B_2, \ldots) \tag{1.6}$$

And this function Θ characterizes the thermal equilibrium of the system. This function specifies the *equation of state* of the system.

§1.2.2 The First Law

Law 1.2 (First Law)
The change in the internal energy of a system is equal to the heat added to the system minus the work done by the system.

$$\mathrm{d}U = đQ - đW \tag{1.7}$$

Here, $đQ$ is the heat added to the system and $đW$ is the work done by the system.

In all general terms, the first law of thermodynamics is the law of conservation of energy for a macroscopic system.

Few important points to note about the first law are:

- The differential used with internal energy U is *exact*, *i.e.* $\int \mathrm{d}U$ is *path independent*.
- The differential used with heat and work are *inexact*, *i.e.* $\int đQ$ and $\int đW$ are *path dependent*.

Using our knowledge of generalized forces and generalized displacements, we can write work done as

$$đW = \sum_i \mathbf{J}_i \cdot \mathrm{d}\mathbf{x}_i \,. \tag{1.8}$$

We can tabulate the generalized forces and generalized displacements for various systems as follows:

System	Generalized Force	Generalized Displacement
Wire	Tension T	Length L
Film	Surface Tension $\mathcal{S}$	Area A
Fluid	Pressure P	Volume V
Magnet	Magnetic Field $\mathbf{B}$	Magnetization $\boldsymbol{\mu}$
Chemical Reaction	Chemical Potential μ	Number of Particles N

Table 1.1: Generalized Forces and Displacements for Various Systems

Usually displacements are *extensive* quantities, that is, proportional to system size. Forces are usually *intensive* quantities and are independent of system size.

§1.2.3 The Second Law

Law 1.3 (Second Law)
We can state the second law in many ways, but the most common statements are:

Kelvin's Statement: No process is possible whose sole result is the complete conversion of heat into work.

Clausius' Statement: No process is possible whose sole result is the transfer of heat from a colder to a hotter body.

It is easy to show that both statements are equivalent. Another way to state the second law is based on the concept of reversible engines (*i.e.* Carnot engines). The efficiency of an engine is given by

$$\eta = \frac{W}{Q_H} = \frac{Q_H - Q_C}{Q_H} = 1 - \frac{Q_C}{Q_H} \leq 1 \tag{1.9}$$

where Q_H is the heat absorbed from the hot reservoir, Q_C is the heat rejected to the cold reservoir, and W is the work done by the engine. The efficiency of a Carnot engine can be shown to be dependent only on the temperatures of the reservoirs and is given by

$$\eta = 1 - \frac{T_C}{T_H}. \tag{1.10}$$

Remark 1.4 (Basic form of Clausius Inequality for Carnot Engines). For a Carnot engine, the efficiency can be written in following form (see eqns. (1.9) and (1.10)). Thus, we can write

$$\eta = 1 - \frac{Q_C}{Q_H} = 1 - \frac{T_C}{T_H} \quad \Rightarrow \quad \frac{Q_C}{T_C} + \frac{Q_H}{T_H} = 0 \tag{1.11}$$

Theorem 1.5 (Carnot's Theorem)
No engine operating between two reservoirs (at temperatures T_H and T_C) is more efficient than a Carnot engine operating between them.

$$\eta_{\text{non-Carnot}} \leq \eta_{\text{Carnot}} = 1 - \frac{T_C}{T_H} \tag{1.12}$$

Proof:
Since Carnot engine is reversible, we can use it as a refrigerator. Let two reservoirs at T_H and T_C with $T_H > T_C$. Let a non-Carnot engine is used to run Carnot refrigerator.

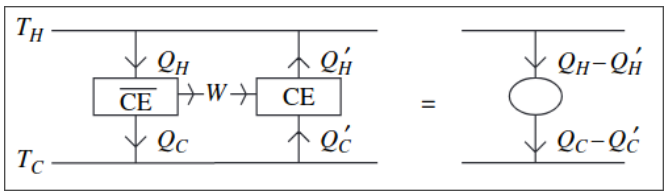


Figure 1.1: A non-Carnot engine running a Carnot refrigerator

From the figure, we can say that the net effect of the process is to transfer heat equal to $Q_H - Q'_H = Q'_C - Q_C$ from the hot reservoir to the cold reservoir. According to the second law, this heat transfer can't be negative, so $Q_H \geq Q'_H$, since both engines uses same amount of work, so

$$\frac{W}{Q_H} \leq \frac{W}{Q'_H} \quad \Rightarrow \quad \eta_{\text{non-Carnot}} \leq \eta_{\text{Carnot}} \tag{1.13}$$

Q.E.D.

Referring to Carnot engine, anytime we want to talk about Second Law is a bit inconvenient, so we need a new quantity which implies same conclusions.

§1.2.3.1 Entropy

We require following theorem to define the entropy.

Theorem 1.6 (Clausius Theorem)
For any cyclic process (reversible or not),

$$\oint \frac{đQ}{T} \leq 0 \tag{1.14}$$

where $đQ$ is the heat increment supplied to the system at temperature T.

Proof:
Let at any point in cycle a small amount of heat $đQ$ is added to the system at temperature T and let a total work of ΔW is done by the system. Since the process is cyclic, from First Law we can write

$$\Delta U = 0 \quad \Rightarrow \quad \oint đQ = \Delta W \tag{1.15}$$

Let's construct a new system ('master' system) in which this heat $đQ$ is added via a Carnot engine which is connected between a reservoir at temperature T_0 and temperature T (see fig. 1.2).

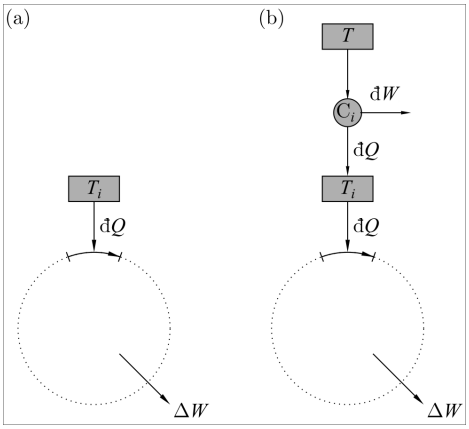


Figure 1.2: Construction of a master system for proof of Clausius Theorem

The reservoir at T_0 is common for all the Carnot engines connected at all points of the cycle. Each Carnot engine produces work $đW$, from remark 1.4 we can write

$$\frac{đQ}{T} = \frac{đQ + đW}{T_0} \quad \Rightarrow \quad đW = đQ\left(\frac{T_0}{T} - 1\right) \tag{1.16}$$

For the master system (in (b)), the total work done is $\Delta W + \oint đW$. If we look closely, we can see that the master system is converting heat to work, which is not possible according to Second Law. So, we can write

$$\Delta W + \oint đW \leq 0 \quad \Rightarrow \quad \Delta W + \oint đQ\left(\frac{T_0}{T} - 1\right) \leq 0 \quad \Rightarrow \quad T_0 \oint \frac{đQ}{T} \leq 0 \tag{1.17}$$

Since $T_0 > 0$, we can write

$$\oint \frac{đQ}{T} \leq 0 \tag{1.18}$$

Q.E.D.

Corollary 1.7 (Clausius Theorem for Reversible Processes)
For a reversible process,

$$\oint \frac{đQ}{T} = 0 \tag{1.19}$$

Pf: Running a reversible cyclic in opposite direction, we get $đQ_{\text{rev}} \to -đQ_{\text{rev}}$. Thus, for both these cyclic processes, we can write

$$\oint \frac{đQ_{\text{rev}}}{T} \leq 0 \quad \text{and} \quad \oint \frac{-đQ_{\text{rev}}}{T} \leq 0$$

From these two inequalities, we can write

$$\oint \frac{đQ_{\text{rev}}}{T} = 0 \tag{1.20}$$

□

Let a thermodynamic system goes from state A to state B via a reversible path and come back to state A via another reversible path. In total this system has undergone a reversible cyclic process. So, we can write

$$\oint \frac{đQ_{\text{rev}}}{T} = 0 \quad \Rightarrow \quad \int_A^B \frac{đQ_{\text{rev}}^{(1)}}{T} + \int_B^A \frac{đQ_{\text{rev}}^{(2)}}{T} = 0 \quad \Rightarrow \quad \int_A^B \frac{đQ_{\text{rev}}^{(1)}}{T} = \int_A^B \frac{đQ_{\text{rev}}^{(2)}}{T} \tag{1.21}$$

This implies that $\int \frac{đQ_{\text{rev}}}{T}$ is path independent, thus it is an exact differential.

Definition 1.8 (Entropy):
Entropy is defined as

$$\mathrm{d}S = \frac{đQ_{\text{rev}}}{T}. \tag{1.22}$$

Remark 1.9. Entropy is a state function, and it is path independent.

$$S(B) - S(A) = \int_A^B \frac{đQ_{\text{rev}}}{T} \tag{1.23}$$

With this we can find a bound on $\int \frac{đQ_{\text{irrev}}}{T}$,

Example 1.10 (Irreversible Change)
Consider a thermodynamic system which goes from state A to state B via an irreversible path and come back to state A via a reversible path. We can write

$$\oint \frac{đQ}{T} \leq 0 \quad \Rightarrow \quad \int_A^B \frac{đQ_{\text{irrev}}}{T} + \int_B^A \frac{đQ_{\text{rev}}}{T} \leq 0 \quad \Rightarrow \quad \int_A^B \frac{đQ_{\text{irrev}}}{T} \leq \int_A^B \frac{đQ_{\text{rev}}}{T}$$

Thus, we can write

$$S(B) - S(A) \geq \int_A^B \frac{đQ_{\text{irrev}}}{T} \tag{1.24}$$

This even holds for states A and B which are very close to each other. So, we can write

$$\mathrm{d}S = \frac{đQ_{\text{rev}}}{T} \geq \frac{đQ_{\text{irrev}}}{T} \tag{1.25}$$

§1.2.3.2 Approach to Equilibrium

Here we will to establish that the entropy of an isolated system always increases. Let's consider a bipartite system (namely subsystem A and subsystem B) which are thermally isolated from each other and weakly interacting. Let the total internal energy of the system be $U = U_A + U_B$ and total entropy be $S = S_A + S_B$. Since the system is weakly interacting, we can write $S_A = S_A(U_A)$ and $S_B = S_B(U_B)$. Thus,

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{\mathrm{d}S_A}{\mathrm{d}t} + \frac{\mathrm{d}S_B}{\mathrm{d}t} = \frac{\partial S_A}{\partial U_A}\frac{\mathrm{d}U_A}{\mathrm{d}t} + \frac{\partial S_B}{\partial U_B}\frac{\mathrm{d}U_B}{\mathrm{d}t}$$

Since the system is isolated, we can write $\frac{\mathrm{d}U_A}{\mathrm{d}t} = -\frac{\mathrm{d}U_B}{\mathrm{d}t}$. Thus, we can write

$$\frac{\mathrm{d}S}{\mathrm{d}t} = \frac{\mathrm{d}U_A}{\mathrm{d}t}\left(\frac{\partial S_A}{\partial U_A} - \frac{\partial S_B}{\partial U_B}\right) = \frac{\mathrm{d}U_A}{\mathrm{d}t}\left(\frac{1}{T_A} - \frac{1}{T_B}\right)$$

Here, we have three possibilities:

- If $T_A = T_B$, then both subsystems are in thermal equilibrium and $\frac{\mathrm{d}S}{\mathrm{d}t} = 0$.
- If $T_A > T_B$, then energy will flow from A to B. Thus, $\frac{\mathrm{d}U_A}{\mathrm{d}t} < 0$ and $\frac{\mathrm{d}S}{\mathrm{d}t} > 0$.
- If $T_A < T_B$, then energy will flow from B to A. Thus, $\frac{\mathrm{d}U_A}{\mathrm{d}t} > 0$ and $\frac{\mathrm{d}S}{\mathrm{d}t} > 0$.

Thus, we can conclude

$$\frac{\mathrm{d}S}{\mathrm{d}t} \geq 0, \tag{1.26}$$

which implies that the entropy of an isolated system always increases.

§1.2.4 The Third Law

Law 1.11 (Third Law)
The entropy of all systems at zero absolute temperature is a universal constant that can be taken to be zero.

$$\lim_{T \to 0} S(\mathbf{X}, T) = 0 \tag{1.27}$$

This law is a direct consequence of quantum mechanics. And a direct consequence of this law is that "It is impossible to cool any system to absolute zero temperature in a finite number of steps."

For example, we can cool a gas by an adiabatic reduction in pressure. Since the curves of S versus T for different pressures must join at $T = 0$, successive steps involve progressively smaller changes, in S and in T, on approaching zero temperature. Alternatively, the unattainability of zero temperatures implies that $S(T = 0, P)$ is independent of P.

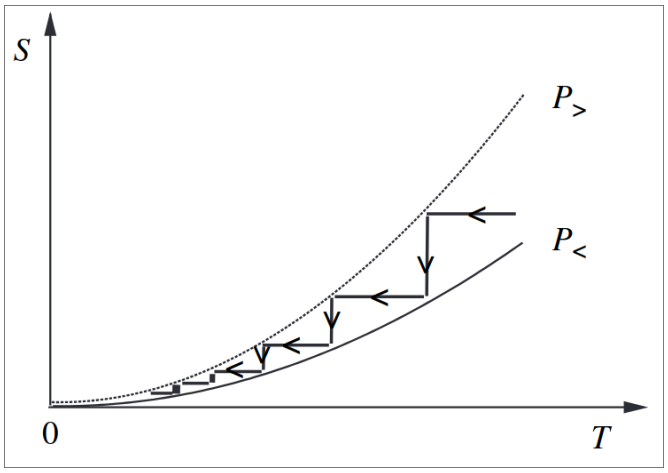


Figure 1.3: Approach to Absolute Zero Temperature in finite steps

2 Lecture 02: (06/01/2025)

Let us start with the classical example of coin tossing. We have a fair coin, and we toss it. The probability of getting a head H (or a tail T) is

$$p_{\text{coin}}(\{H\}) = p_{\text{coin}}(\{T\}) = \frac{1}{2}. \tag{2.1}$$

Say, we start writing down the outcomes of the coin tosses, intuitively, we expect that the number of heads and tails will be equal.

$$\#H = \#T. \tag{2.2}$$

Building upon this, we can consider very famous examples of probability theory, *i.e.* the *random walk*.

Example 2.1 (Random Walk)

Consider a particle at the origin of a one-dimensional lattice. At each time step, the particle moves to the right with probability 1/2 and to the left with probability 1/2.

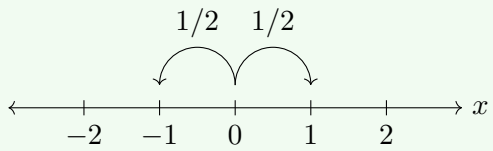


Figure 2.1: Random Walk (in 1-D)

There are many things, which we can study about this system, but one of the most interesting results is that "there is a finite probability of return."

In 2-D: the probability is very small, but it is finite.

In 3-D or higher dimensions: the probability of returning to the origin decreases as the number of dimensions increases (for more details and proof, see [5]).

§2.1 Probability Theory

Definition 2.2 (Sample Space):
The *sample space* Ω is the set of all possible outcomes of an experiment.

Definition 2.3 (Random Variable):
A *random variable* is a function $X : \Omega \to \mathbb{R}$, where Ω is the sample space.

Intuitively, a random variable is an assignment of a real number to each outcome of an experiment.

Definition 2.4 (Event):
An *event* is a subset of the sample space, *i.e.* $A \subseteq \Omega$.

Example 2.5

Some basic examples of an experiment and its sample space are as follows:

1. Tossing a coin: $\Omega = \{H, T\}$.
2. Rolling a die: $\Omega = \{1, 2, 3, 4, 5, 6\}$.
3. Velocity of a particle: $\Omega = \mathbb{R}^3 = \{(v_x, v_y, v_z) \mid -\infty < v_x, v_y, v_z < \infty\}$.

From these examples, we can see that the sample space can be discrete or continuous.

Axiom 2.6 (Axiom's of Probability (Kolmogorov's Axioms)):
Let p be a function defined on the set of all events of a sample space Ω. The function p is called a probability *if it satisfies the following axioms:*

1. *Positivity:* $0 \leq p(A) \leq 1$ *for all events A.*
2. *Normalization:* $p(\Omega) = 1$.
3. *Additivity: If $A_1, A_2, \ldots$ are mutually exclusive events, then*

$$p\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} p(A_i). \tag{2.3}$$

Theorem 2.7
Let A and B be two events in a sample space Ω, then

1. $p(\bar{A}) = 1 - p(A)$.
2. If $A \subseteq B$, then $p(A) \leq p(B)$.

But these axioms doesn't tell us how assign probabilities to events. For that, we have two practical methods:

- Objective Probability: Let a random experiment be repeated N times, in which the frequency of the event A was N_A, then the probability of the event A is

$$p(A) = \lim_{N\to\infty} \frac{N_A}{N}. \tag{2.4}$$

- Subjective Probability: We assign a (theoretical) estimate of the probability based on the uncertainties related to lack of precise knowledge of outcomes.

Remark 2.8 (Probabilities in statistical mechanics). All assignments of probability in statistical mechanics are subjective. The consequences of such subjective assignments of probability have to be checked against measurements, and they may need to be modified as more information about the outcomes becomes available.

The notion of objective probability hides an important concept of *ensemble*.

Definition 2.9 (Ensemble):
An *ensemble* is a collection of systems, each of which is in a different state. The probability of a system being in a particular state is given by the fraction of systems in the ensemble that are in that state.

§2.2 One Random Variable

Consider a random variable x, whose outcomes are real numbers.

Definition 2.10 (Cumulative Distribution Function):
The *cumulative distribution function* (CDF) $P(x)$ of a random variable x is the probability of an outcome with any value less than x.

$$P(x) = \text{prob}(E \subseteq [-\infty, x]) \tag{2.5}$$

CDF must be a monotonically increasing function of x, with

$$P(-\infty) = 0, \qquad P(\infty) = 1. \tag{2.6}$$

Also, the probability of the random variable x lying in the interval $[a, b]$ is

$$\text{prob}(a \leq x \leq b) = P(b) - P(a). \tag{2.7}$$

Definition 2.11 (Probability distribution Function):
The *probability distribution function* (PDF) $p(x)$ of a random variable x is the derivative of the cumulative distribution function.

$$p(x) = \frac{\mathrm{d}P(x)}{\mathrm{d}x} \tag{2.8}$$

Hence, $p(x)\,\mathrm{d}x = \mathrm{prob}(E \in [x, x + \mathrm{d}x])$. And due to the normalization condition, we have

$$\mathrm{prob}(\Omega) = \int_{-\infty}^{\infty} p(x)\,\mathrm{d}x = 1. \tag{2.9}$$

Remark 2.12. Let $p(x)$ be a probability distribution function, then

1. $0 \leq p(x) < \infty$.
2. The dimension of $p(x)$ is same as the dimension of x^{-1}.
3. The probability of x lying in the interval $[a, b]$ is

$$\mathrm{prob}(a \leq x \leq b) = \int_a^b p(x)\,\mathrm{d}x\,. \tag{2.10}$$

4. A PDF can have integrable singularities, but not essential singularities.

Definition 2.13 (Expectation Value):
Let $F(x)$ be a function of the random variable x, then the *expectation value* of $F(x)$ is

$$\langle F(x) \rangle = \int_{-\infty}^{\infty} \mathrm{d}x\, p(x) F(x). \tag{2.11}$$

§2.3 Change of Variables

Consider a random variable x, with a probability distribution function $p(x)$. Let $y = f(x)$ be another random variable, where f is invertible. Then we want to find the probability distribution function $\tilde{p}(y)$.

Let A be an event in the x space, then the probability of A is

$$\mathrm{prob}(A) = \int_A p(x)\,\mathrm{d}x\,. \tag{2.12}$$

Now, let $B = f(A)$ be the event in the y space, then the probability of B is

$$\mathrm{prob}(B) = \int_B \tilde{p}(y)\,\mathrm{d}y\,. \tag{2.13}$$

Since these two probabilities are equal, we have

$$\int_B \tilde{p}(y)\,\mathrm{d}y = \int_A p(x)\,\mathrm{d}x\,. \tag{2.14}$$

Using the inverse of f, we can write

$$\int_B \tilde{p}(y)\,\mathrm{d}y = \int_B p(f^{-1}(y)) \left| \frac{\mathrm{d}f^{-1}}{\mathrm{d}y} \right| \mathrm{d}y\,. \tag{2.15}$$

Hence, we have

$$\tilde{p}(y) = p(f^{-1}(y)) \left| \frac{\mathrm{d}f^{-1}}{\mathrm{d}y} \right|. \tag{2.16}$$

Example 2.14

Let $0 < x < 1$ be a random variable with a probability distribution function $p(x) = 3x^2$. Let $y = x^2$, then we have

$$\tilde{p}(y) = p(\sqrt{y})\left|\frac{\mathrm{d}\sqrt{y}}{\mathrm{d}y}\right| = \frac{3}{2}\sqrt{y}.$$

We can arrive at the same result using, CDF.

$$\begin{aligned}\tilde{P}(y) &= \text{prob}(B < y)\\ &= \text{prob}(x^2 < y)\\ &= \text{prob}(x < \sqrt{y})\\ &= \text{prob}(A < \sqrt{y})\\ &= P(\sqrt{y})\end{aligned}$$

With this, we can find CDF of random variable y.

$$\tilde{P}(y) = \int_0^{\sqrt{y}} 3x^2\,\mathrm{d}x = y^{3/2}.$$

Thus, the probability distribution function is

$$\tilde{p}(y) = \frac{\mathrm{d}\tilde{P}(y)}{\mathrm{d}y} = \frac{3}{2}\sqrt{y}.$$

3 Lecture 03: (08/01/2025)

§3.1 Moments of a Distribution

Definition 3.1 (Moment of a Distribution):
The n-th moment of a distribution $p(x)$ is defined as

$$m_n \equiv \langle x^n \rangle := \int_{-\infty}^{\infty} x^n p(x)\,\mathrm{d}x \tag{3.1}$$

It is cumbersome to calculate the moments of a distribution directly. It would be more convenient to calculate the moments of a distribution from something like "Moment Generating Function" (MGF) or "Characteristic Function" (CF). The MGF and CF are defined as follows:

Definition 3.2 (Moment Generating Function):
The MGF of a distribution $p(x)$ is defined as the Fourier transform of the distribution:

$$\tilde{p}(k) := \int_{-\infty}^{\infty} p(x) e^{-\mathrm{i}kx}\,\mathrm{d}x \equiv \left\langle e^{-\mathrm{i}kx} \right\rangle \tag{3.2}$$

The inverse Fourier transform of the MGF will give us the distribution:

$$p(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \tilde{p}(k) e^{\mathrm{i}kx}\,\mathrm{d}k \tag{3.3}$$

Using Taylor series expansion of the exponential function, we can write the MGF as:

$$\tilde{p}(k) = \left\langle e^{-\mathrm{i}kx} \right\rangle = \left\langle \sum_{n=0}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} x^n \right\rangle = \sum_{n=0}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} \langle x^n \rangle$$

With this, we can have an alternative definition of the moment of a distribution using the MGF:

$$(\mathrm{i})^n \left[\frac{\partial^n \tilde{p}(k)}{\partial k^n} \right|_{k=0} =: \langle x^n \rangle \tag{3.4}$$

§3.2 Cumulants

In order to define the cumulants, we first need to define the cumulant generating function (CGF):

Definition 3.3 (Cumulant Generating Function):
The CGF of a distribution $p(x)$ is defined as the logarithm of the MGF:

$$K(k) = \ln \tilde{p}(k) = \ln \left\langle e^{-\mathrm{i}kx} \right\rangle \tag{3.5}$$

The cumulant κ_n is obtained from power series expansion of the CGF:

$$K(k) = \sum_{n=1}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} \kappa_n \tag{3.6}$$

Definition 3.4 (Cumulant):
The n-th cumulant of a distribution $p(x)$ is defined as

$$\kappa_n := (\mathrm{i})^n \left[\frac{\partial^n K(k)}{\partial k^n} \right|_{k=0} \tag{3.7}$$

In order to relate the cumulants to the moments, we can use the known Taylor series expansion of the logarithm:

$$\ln(1+\varepsilon) = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \epsilon^m. \tag{3.8}$$

We can rewrite the expansion of MGF as:

$$\tilde{p}(k) = 1 + \sum_{n=1}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} \langle x^n \rangle = 1 + \epsilon. \tag{3.9}$$

Using the Taylor series expansion of the logarithm, we can write the CGF as:

$$K(k) = \ln \tilde{p}(k) = \sum_{m=1}^{\infty} \frac{(-1)^{m+1}}{m} \left[\sum_{n=1}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} \langle x^n \rangle \right]^m \tag{3.10}$$

Comparing the coefficients of the powers of $(-\mathrm{i}k)$ in eqns. (3.6) and (3.10), we can write the cumulants in terms of the moments:

$(-\mathrm{i}k)^1$**:** this is possible only for $m = n = 1$:

$$\kappa_1 = \langle x \rangle =: \mu. \tag{3.11}$$

The first cumulant κ_1 is the mean of the distribution.

$(-\mathrm{i}k)^2$**:** this is possible for $m = 1, n = 2$ and $m = 2, n = 1$:

$$\kappa_2 = \langle x^2 \rangle - \langle x \rangle^2 =: \sigma^2. \tag{3.12}$$

The second cumulant κ_2 is the variance of the distribution.

§3.3 Probability Distributions

Now we will define some probability distributions and calculate their moments and cumulants which will be useful in this course.

§3.3.1 Exponential Distribution

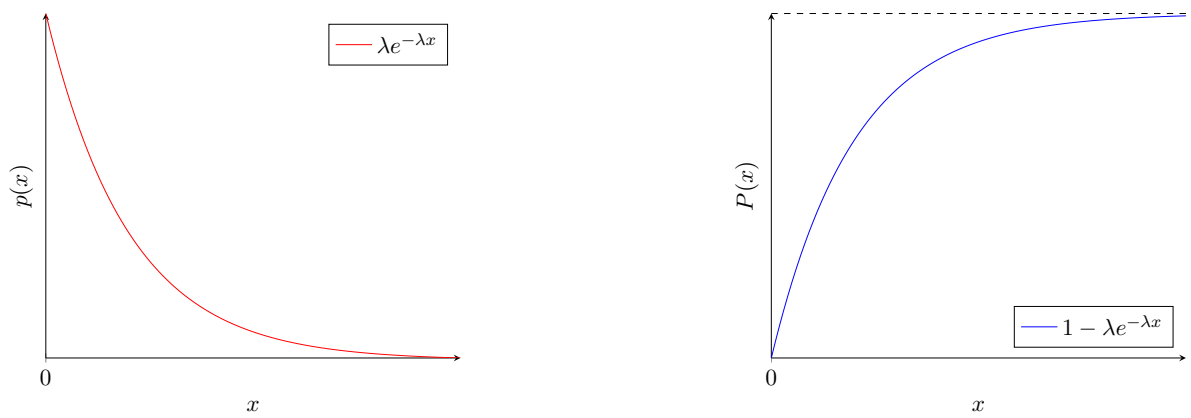
Definition 3.5 (Exponential Distribution):
The exponential distribution is defined as

$$p(x) = \lambda e^{-\lambda x}; \quad x \in [0, \infty) \tag{3.13}$$

where $\lambda > 0$ is a parameter of the distribution.

From this we can calculate the CDF:

$$P(x) = \int_0^x p(x')\,\mathrm{d}x' = 1 - e^{-\lambda x}. \tag{3.14}$$



(a) PDF of the Exponential Distribution **(b)** CDF of the Exponential Distribution

Figure 3.1: Graphs of the Exponential Distribution

And the MGF is given by:

$$\tilde{p}(k) = \int_0^\infty p(x)e^{-\mathrm{i}kx}\,\mathrm{d}x = \frac{\lambda}{\lambda + \mathrm{i}k}. \tag{3.15}$$

The moments of the exponential distribution are:

$$\langle x \rangle = \frac{1}{\lambda}, \quad \langle x^2 \rangle = \frac{2}{\lambda^2}, \quad \langle x^n \rangle = \frac{n!}{\lambda^n}. \tag{3.16}$$

The cumulants of the exponential distribution are:

$$\kappa_1 = \frac{1}{\lambda}, \quad \kappa_2 = \frac{1}{\lambda^2} \tag{3.17}$$

§3.3.2 Bernoulli and Binomial Distributions

Till now, we have discussed continuous probability distributions. Now we will discuss discrete probability distributions which also have applications in statistical mechanics (like the Random Walk problem).

Let us first define the Bernoulli distribution:

§3.3.2.1 Bernoulli Distribution

Let an experiment have only two outcomes: success ($x = 1$) and failure ($x = -1$). The probability of success is p and the probability of failure is $1 - p =: q$. The Bernoulli distribution is defined as:

$$p(x) = \begin{cases} p, & x = 1, \\ q, & x = -1. \end{cases} \tag{3.18}$$

It is the simplest discrete probability distribution. And this distribution give rise to an important distribution called the Binomial distribution.

§3.3.2.2 Binomial Distribution (or Bernoulli Trials)

Let us perform N independent Bernoulli trials. The probability of success in each trial is p and the probability of failure is $1 - p =: q$.

We can understand the Binomial distribution with the help of Random Walk problem: a particle moves on a 1D lattice with probability of moving right as p and left as q. Let the random walker start at the origin and make a total of N steps. Say the walker moves right N_p times and left N_q times. Then the probability of this configuration is:

$$p_N(N_p, N_q) = \binom{N}{N_p} p^{N_p} q^{N_q} = \frac{(N_p + N_q)!}{N_p!\, N_q!} p^{N_p} q^{N_q}. \tag{3.19}$$

Another way to understand eqn. (3.19) is to consider the displacement of the random walker from the origin. The displacement is given by $m := N_p - N_q$. The probability of the walker being at a distance m from the origin is given by:

$$p_N(m) = \frac{N!}{\left(\frac{N+m}{2}\right)!\,\left(\frac{N-m}{2}\right)!} p^{\frac{N+m}{2}} q^{\frac{N-m}{2}}. \tag{3.20}$$

At this point, we can make a connection between these two ways of understanding the Binomial distribution. Observe that $2N_p = N + m \Rightarrow m = 2N_p - N$. Take the expectation value of m:

$$\begin{aligned} \langle m \rangle &= 2\langle N_p \rangle - N \\ &= 2Np - N = N(p - q) \end{aligned} \tag{3.21}$$

This is the mean displacement of the random walker from the origin. As a sanity check, we intuitively expect the mean displacement for an unbiased random walk (*i.e.* $p = q = 1/2$) to be zero.

4 Lecture 04: (09/01/2025)

§4.1 Continuation of Binomial Distribution

In the last lecture, we discussed the binomial distribution. Now, let's calculate the characteristic function, but in this case we will have a finite sum instead of an integral.

$$\tilde{p}(k) = \sum_{N_p=0}^{N} p(N_p)e^{-\mathrm{i}kN_p} = \sum_{N_p=0}^{N} \binom{N}{N_p} p^{N_p} q^{N-N_p} e^{-\mathrm{i}kN_p} = (pe^{-\mathrm{i}k} + q)^N, \tag{4.1}$$
$$\tilde{p}(k) = (pe^{-\mathrm{i}k} + q)^N.$$

With this, the cumulative distribution function is

$$K(k) = \ln \tilde{p}(k) = N \ln\Big(pe^{-\mathrm{i}k} + q\Big). \tag{4.2}$$

The cumulants are then

$$\kappa_1 = \mu = \langle N_p \rangle = Np; \qquad \kappa_2 = \sigma^2 = \langle N_p^2 \rangle - \langle N_p \rangle^2 = Npq. \tag{4.3}$$

From this, we can see calculate the second moment of N_p:

$$\left\langle N_p^2 \right\rangle = Npq + \langle N_p \rangle^2 = Npq + N^2p^2 = Np(q + Np). \tag{4.4}$$

Thus, saying that the mean success in a Binomial distribution is of order N and the standard deviation is of order $\sqrt{N}$.

$$\mu \sim N, \qquad \sigma \sim \sqrt{N}. \tag{4.5}$$

Problem 4.1.1 (Homework).
Prove that the following is true for the Binomial distribution:

$$\langle m \rangle_c = N(p - q), \qquad \left\langle m^2 \right\rangle_c = 4Npq, \tag{4.6}$$

where $m = N_p - N_q$ and $\langle \cdot \rangle_c$ denotes the cumulant.

Solution.
We have shown the mean displacement in the last lecture (see eqn. (3.21)). Now, we need to calculate the second cumulant. We have $m^2 = 4N_p^2 - 4N_pN + N^2$, so the second moment is

$$\begin{aligned}
\left\langle m^2 \right\rangle &= \left\langle 4N_p^2 - 4N_pN + N^2 \right\rangle = 4\left\langle N_p^2 \right\rangle - 4N \langle N_p \rangle + N^2 \\
&= 4(Np(q + Np)) - 4N(Np) + N^2 \\
&= 4Npq + 4N^2p^2 - 4N^2p + N^2 \\
&= 4Npq + N^2(4p^2 - 4p + 1) \\
&= 4Npq + N^2(2p - 1)^2 = 4Npq + [N(p - q)]^2 \\
&= 4Npq + \langle m \rangle^2 .
\end{aligned}$$

Thus, we have $\langle m^2 \rangle_c = \langle m^2 \rangle - \langle m \rangle^2 = 4Npq$. ◀

§4.2 Introduction to Stochastic Processes

The notations and definitions discussed know is not very important in the context of statistical mechanics. However, we will need to understand these terminologies as many books and papers use these terms. We will now introduce the concept of a stochastic process.

Definition 4.1 (Stochastic Process):
A *stochastic process* is a collection of random variables $\{X_t\}_{t\in\mathcal{T}}$ indexed by a parameter t, which can be discrete or continuous.

Usually the parameter t has the interpretation of time. If the parameter is discrete, we have a *discrete-time* stochastic process, and if the parameter is continuous, we have a *continuous-time* stochastic process.

If $X_m = s$, we say that the process is in state s at time m. The set of all possible states of the process is called the *state space* of the process and is denoted by $\mathcal{S}$. The state space can be discrete or continuous.

Example 4.2
The stochastic process is used to model this dynamic system. For example,

- The random walk problem is a stochastic process, with the state space being $\mathbb{Z}$ and the parameter being discrete (can be thought as $t \in \mathbb{N}$).
- The position of a particle in a gas is a stochastic process, with the state space being $\mathbb{R}^3$ and the time being continuous $t \in \mathbb{R}$.

Definition 4.3 (Markov Process):
A stochastic process $\{X_t\}_{t\in\mathcal{T}}$ is said to be a *Markov process* if the conditional probability of the future state of the process depends only on the present state and not on the past states. Mathematically, this is written as (for discrete-time processes):

$$p(X_{t+1} = s' \mid X_t = s, X_{t-1} = s_{t-1}, \ldots, X_0 = s_0) = p(X_{t+1} = s' \mid X_t = s). \tag{4.7}$$

Intuitively, this means that the future is independent of the past given the present. This is a very strong assumption, but it is very useful in many applications.

§4.2.1 Two-State Markov Chain

Let's consider a simple example of a two-state Markov chain. The state space is $\mathcal{S} = \{1, 2\}$, define $P_1(t) := p(X_t = 1)$ as the probability of being in state 1 at time t and $P_2(t) := p(X_t = 2)$ as the probability of being in state 2 at time t. So we have

$$\forall t \in \mathcal{T} : P_1(t) + P_2(t) = 1. \tag{4.8}$$

We can define the rate of transition from state 1 to state 2 as $w(1 \to 2)$ and the rate of transition from state 2 to state 1 as $w(2 \to 1)$. Thus, we can write the probability at time $t + \Delta t$ as

$$P_1(t+\Delta t) = P_1(t) \underbrace{(1 - w(1\to 2)\Delta t)}_{\text{prob. of staying in state 1}} + P_2(t) \underbrace{w(2\to 1)\Delta t}_{\text{prob. of jumping from 2 to 1}} \tag{4.9}$$

$$P_1(t+\Delta t) - P_1(t) = -w(1\to 2)P_1(t)\Delta t + w(2\to 1)P_2(t)\Delta t$$

$$\frac{P_1(t+\Delta t) - P_1(t)}{\Delta t} = -w(1\to 2)P_1(t) + w(2\to 1)P_2(t),$$

and taking the limit $\Delta t \to 0$, we get

$$\frac{\mathrm{d}P_1(t)}{\mathrm{d}t} = -w(1\to 2)P_1(t) + w(2\to 1)P_2(t). \tag{4.10}$$

Similarly, we can write the equation for $P_2(t)$ as

$$\frac{\mathrm{d}P_2(t)}{\mathrm{d}t} = -w(2\to 1)P_2(t) + w(1\to 2)P_1(t). \tag{4.11}$$

For sanity check, we can see that the sum of eqns. (4.10) and (4.11) is zero, as it should be.

§4.2.2 Master Equation

We can generalize the above discussion to a Markov chain with N states. Let $P_i(t) = p(X_t = i)$ be the probability of being in state i at time t and $w(i \to j)$ be the rate of transition from state i to state j. Then, the eqn. (4.10) can be generalized to

$$\frac{\mathrm{d}P_i(t)}{\mathrm{d}t} = \sum_{j \neq i} [w(j \to i)P_j(t) - w(i \to j)P_i(t)]. \tag{4.12}$$

This is called the *master equation* for the Markov process.

With this equation in theory we can study the dynamics of any system which can be modeled as a finite state Markov chain. However, in this course we are interested in the thermodynamic limit, where the system has achieved *equilibrium*. For that we need to define the concept of equilibrium and the conditions for equilibrium.

Definition 4.4 (Steady State):
A Markov process is said to be in a *steady state* if the probabilities of being in the states do not change with time *i.e.*

$$\frac{\mathrm{d}P_i(t)}{\mathrm{d}t} = 0, \quad \forall i \in \mathcal{S}. \tag{4.13}$$

Let's denote the steady state probabilities as P_i^{ss}. Then, the master equation becomes

$$\sum_{j \neq i} \left[w(j \to i)P_j^{\text{ss}} - w(i \to j)P_i^{\text{ss}} \right] = 0. \tag{4.14}$$

For a system, eqn. (4.14) can be satisfied in mainly two ways. But before that, we need to define the concept of *probability current*.

Definition 4.5 (Probability Current):
The *probability current* from state j to state i (given $i \neq j$) at time t is defined as

$$J_{(j \to i)}(t) := w(j \to i)P_j(t) - w(i \to j)P_i(t). \tag{4.15}$$

We can extend this definition to the steady state as well, by replacing $P_i(t)$ with P_i^{ss} in eqn. (4.15).

We can rewrite eqn. (4.14) in terms of probability currents as

$$\sum_{j \neq i} J_{(j \to i)}^{\text{ss}} = 0. \tag{4.16}$$

This equation can be satisfied in two ways:

1. **Detailed Balance Condition:** If the probability current $J_{(j \to i)}^{\text{ss}}$ vanishes for each pair (i, j) of states, then the system is said to satisfy the *detailed balance* condition.

$$J_{(j \to i)}^{\text{ss}} = 0, \quad \Rightarrow \quad w(j \to i)P_j^{\text{ss}} = w(i \to j)P_i^{\text{ss}} \tag{4.17}$$

2. Sum of probability currents into a state is zero. In this case, we say a system is in steady state but hasn't attained detailed balance.

   We can say that the system is in a steady state, but there is a net flow of probability from one state to another. This is called a *non-equilibrium steady state*.

Definition 4.6 (Equilibrium):
A Markov process is said to be in *equilibrium* if it is in a steady state and satisfies the detailed balance condition.

§4.2.3 Chapman-Kolmogorov Equation

Let's consider a particle moving in a gas. The position of the particle at time t is a random variable X_t. We can define the probability of the particle being at position x at time t as $p(x,t) = p(X_t = x)$.

Say the particle moves from (x_0, t_0) to (x, t) in n steps, with intermediate steps as position $x_1, \ldots, x_{n-1}$ at times $t_1, \ldots, t_{n-1}$ respectively. Then, the Markov property will be

$$p(x, t \mid x_{n-1}, t_{n-1}; \cdots; x_1, t_1; x_0, t_0) = p(x, t \mid x_{n-1}, t_{n-1}). \tag{4.18}$$

In general, if the particle starts at position x_0 at time t_0 and ends at position x at time t, then the probability of this happening is

$$p(x, t \mid x_0, t_0) = \int \mathrm{d}x' \, p(x, t \mid x', t') \; p(x', t' \mid x_0, t_0), \tag{4.19}$$

where the integral is over all possible intermediate positions x' at time t'.

This is called the *Chapman-Kolmogorov equation*. This equation is a consequence of the Markov property and basic probability theory (law of total probability).

And one can show that the Chapman-Kolmogorov equation is equivalent to the master equation for the Markov process.

5 Lecture 05: (13/01/2025)

Let's start with rewriting the Master equation (see eqn. (4.12)) for continuous time and continuous state space:[1]

$$\frac{\partial p(x,t)}{\partial t} = \int \mathrm{d}x' \, [w(x \mid x')p(x',t) - w(x' \mid x)p(x,t)]. \tag{5.1}$$

As stated in the last lecture, the above equation can be derived from the Chapman-Kolmogorov equation (see [1]).

§5.1 Kramers-Moyal Expansion

Let's redefine the rate of transition from x' to x to be the rate of transition from x' to the state $x = x' + r$ with a displacement $r = x' - x$. Thus, we have

$$w(x \mid x') = f(x', r) \tag{5.2}$$

and, rewriting the Master equation, we get

$$\frac{\partial p(x,t)}{\partial t} = \int \mathrm{d}r \, [f(x-r,r)p(x-r,t) - f(x,-r)p(x,t)], \tag{5.3}$$

where, $r = x - x'$, and $f(x,r)$ is the rate of transition from x to $x + r$. We can expand the first term in the integrand (about the first argument) in a Taylor series around x,

$$f(x-r,r)p(x-r,t) = f(x,r)p(x,t) + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} r^n \frac{\partial^n}{\partial x^n}[f(x,r)p(x,t)]. \tag{5.4}$$

And the second term can be written as,[2]

$$\int_{-\infty}^{\infty} \mathrm{d}r \, f(x,-r)p(x,t) = p(x,t) \int_{-\infty}^{\infty} \mathrm{d}r \, f(x,-r) = \int_{-\infty}^{\infty} \mathrm{d}r \, f(x,r)p(x,t). \tag{5.5}$$

Substituting eqns. (5.4) and (5.5) back into eqn. (5.3), we get

$$\frac{\partial p(x,t)}{\partial t} = \int \mathrm{d}r \left[f(x,r)p(x,t) + \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} r^n \frac{\partial^n}{\partial x^n}[f(x,r)p(x,t)] - f(x,r)p(x,t) \right]. \tag{5.6}$$

Rearranging the terms, we get

$$\frac{\partial p(x,t)}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{\partial^n}{\partial x^n} \left[\left\{ \int \mathrm{d}r \, r^n f(x,r) \right\} p(x,t) \right]. \tag{5.7}$$

Let's define $R_n(x) := \int \mathrm{d}r \, r^n f(x,r)$,[3] and rewrite the above equation as

$$\frac{\partial p(x,t)}{\partial t} = \sum_{n=1}^{\infty} \frac{(-1)^n}{n!} \frac{\partial^n}{\partial x^n}[R_n(x)p(x,t)]. \tag{5.8}$$

This is the Kramers-Moyal expansion of the Master equation. If the series truncates at $n = 2$ (exactly), we get the Fokker-Planck equation (or drift-diffusion equation).

$$\frac{\partial p(x,t)}{\partial t} = -\frac{\partial}{\partial x}[A(x)p(x,t)] + \frac{1}{2}\frac{\partial^2}{\partial x^2}[B(x)p(x,t)], \tag{5.9}$$

where, $A(x) = R_1(x)$ and $B(x) = R_2(x)$.

Theorem 5.1 (Pawula theorem)
Either the sequence $\{R_n(x)\}_{n=1}^{\infty}$ becomes zero for all $n \geq 3$, or all its even terms are positive.

[1] Here, we change the notation from $w(x \to x')$ to $w(x' \mid x)$.
[2] Here we have used the fact that the position of the particle can be anywhere in the state space, *i.e.* $x \in \mathbb{R}$.
[3] In terms of transition rate, $R_n(x) := \int (x' - x)^n w(x' \mid x) \, \mathrm{d}x'$

§5.2 Revisiting Random Walk Problem: $N \gg 1$

Recall, $P(m, N)$ is the probability of finding the particle at position m after N steps. The probability of finding the particle at position m after $N+1$ steps can be written as

$$P(m, N+1) = p \cdot P(m-1, N) + q \cdot P(m+1, N), \tag{5.10}$$

where, p and q are the probabilities of moving right and left, respectively. Assume that the random walk is unbiased, *i.e.* $p = q = \frac{1}{2}$ and subtracting $P(m, N)$ from the above equation, we get

$$P(m, N+1) - P(m, N) = \frac{1}{2}[P(m-1, N) - 2P(m, N) + P(m+1, N)]. \tag{5.11}$$

Say, we discretize the space such that $x = m \cdot \Delta x$ and $t = N \cdot \Delta t$, where, Δx and Δt are the lattice spacing and time step, respectively. Thus, the above equation can be written as

$$\Delta t \frac{\partial P(x,t)}{\partial t} = \frac{1}{2}(\Delta x)^2 \frac{\partial^2 P(x,t)}{\partial x^2}. \tag{5.12}$$

Define $D := \frac{(\Delta x)^2}{2\Delta t}$, and take the limit $\Delta x \to 0$ and $\Delta t \to 0$, we get

$$\frac{\partial P(x,t)}{\partial t} = D \frac{\partial^2 P(x,t)}{\partial x^2}. \tag{5.13}$$

This is the diffusion equation.

Problem 5.2.1.
Show that the solution of the diffusion equation for the following boundary conditions:

$$P(x, 0) = \delta(x),$$
$$P(x,t) \to 0 \quad \text{as} \quad x \to \pm\infty,$$

is given by

$$P(x,t) = \frac{1}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right). \tag{5.14}$$

Solution.
There are multiple ways to solve the above problem. One way is to use the Fourier transform (space coordinate in this case). Let $\tilde{P}(k,t)$ be the Fourier transform of $P(x,t)$, *i.e.*

$$\tilde{P}(k,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{d}x\, e^{\mathrm{i}kx} P(x,t) \quad \text{and} \quad P(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{d}k\, e^{-\mathrm{i}kx} \tilde{P}(k,t). \tag{5.15}$$

We need to transform the initial condition to the Fourier space, *i.e.*

$$\tilde{P}(k,0) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \mathrm{d}x\, e^{\mathrm{i}kx} \delta(x) = \frac{1}{\sqrt{2\pi}}. \tag{5.16}$$

Now, let's transform the diffusion equation to the Fourier space,

$$\frac{\partial P(x,t)}{\partial t} = D \frac{\partial^2 P(x,t)}{\partial x^2} \quad \Rightarrow \quad \frac{\partial \tilde{P}(k,t)}{\partial t} = -Dk^2 \tilde{P}(k,t). \tag{5.17}$$

The solution of the above equation is given by

$$\left[\ln \tilde{P}(k,s)\right|_{\tilde{P}(k,s=0)}^{\tilde{P}(k,s=t)} = -Dk^2 t \quad \Rightarrow \quad \tilde{P}(k,t) = \tilde{P}(k,0) e^{-Dk^2 t} = \frac{1}{\sqrt{2\pi}} e^{-Dtk^2}. \tag{5.18}$$

Now, taking the inverse Fourier transform of $\tilde{P}(k,t)$, we get

$$P(x,t) = \frac{1}{\sqrt{4\pi D t}} \exp\left(-\frac{x^2}{4Dt}\right) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{x^2}{2\sigma^2}\right), \tag{5.19}$$

where, $\sigma^2 = 2Dt$. ◀

The solution of the diffusion equation (with the above boundary conditions) is a Gaussian distribution.

§5.2.1 Normal Distribution: As a Limiting Case of Binomial Distribution

Now let's try to derive the same result using the binomial distribution. We know the probability of finding the particle at position m after N steps is given by

$$P(m,N) = \frac{N!}{\left(\frac{N+m}{2}\right)!\,\left(\frac{N-m}{2}\right)!} p^{\frac{N+m}{2}} q^{\frac{N-m}{2}}, \tag{5.20}$$

taking logarithm on both sides, we get

$$\ln P(m,N) = \ln N! - \ln\left(\frac{N+m}{2}\right)! - \ln\left(\frac{N-m}{2}\right)! + \left(\frac{N+m}{2}\right)\ln p + \left(\frac{N-m}{2}\right)\ln q. \tag{5.21}$$

Recall the Stirling's approximation,

$$\ln N! = \left(N+\frac{1}{2}\right)\ln N - N + \frac{1}{2}\ln(2\pi) + \mathcal{O}(1/N), \tag{5.22}$$

and using the above approximation, we get

$$\begin{aligned}\ln P(m,N) = \left(N+\frac{1}{2}\right)\ln N - \cancel{N} + \frac{1}{2}\ln(2\pi) - \left(\frac{N+m}{2}+\frac{1}{2}\right)\ln\frac{N+m}{2} + \frac{\cancel{N}+\cancel{m}}{2} - \frac{1}{2}\ln(2\pi)\\ - \left(\frac{N-m}{2}+\frac{1}{2}\right)\ln\frac{N-m}{2} + \frac{\cancel{N}-\cancel{m}}{2} - \frac{1}{2}\ln(2\pi) + \left(\frac{N+m}{2}\right)\ln p + \left(\frac{N-m}{2}\right)\ln q\end{aligned}$$

$$\begin{aligned}\ln P(m,N) = \left(N+\frac{1}{2}\right)\ln N - \left(\frac{N+m}{2}+\frac{1}{2}\right)\ln\frac{N+m}{2} - \left(\frac{N-m}{2}+\frac{1}{2}\right)\ln\frac{N-m}{2}\\ + \left(\frac{N+m}{2}\right)\ln p + \left(\frac{N-m}{2}\right)\ln q - \frac{1}{2}\ln(2\pi)\end{aligned}$$

Define the deviation from the mean as $\delta_m := m - \langle m\rangle = m - N(p-q)$, and rewrite the term $(N+m)/2$ and $(N-m)/2$ as

$$\frac{N+m}{2} = Np\left[1+\frac{\delta_m}{2Np}\right] \qquad \frac{N-m}{2} = Nq\left[1-\frac{\delta_m}{2Nq}\right]$$

$$\ln\frac{N+m}{2} = \ln N + \ln p + \ln\left[1+\frac{\delta_m}{2Np}\right] \qquad \ln\frac{N-m}{2} = \ln N + \ln q + \ln\left[1-\frac{\delta_m}{2Nq}\right].$$

Thus, we have

$$\begin{aligned}-\left(\frac{N+m}{2}+\frac{1}{2}\right)\ln\frac{N+m}{2} + \left(\frac{N+m}{2}\right)\ln p &= -\left(\frac{N+m}{2}+\frac{1}{2}\right)\left[\ln N + \ln\left(1+\frac{\delta_m}{2Np}\right)\right] - \frac{1}{2}\ln p\\ -\left(\frac{N-m}{2}+\frac{1}{2}\right)\ln\frac{N-m}{2} + \left(\frac{N-m}{2}\right)\ln q &= -\left(\frac{N-m}{2}+\frac{1}{2}\right)\left[\ln N + \ln\left(1-\frac{\delta_m}{2Nq}\right)\right] - \frac{1}{2}\ln q.\end{aligned}$$

Substitute the above expressions back into the equation for $\ln P(m,N)$, we get

$$\ln P(m,N) = \ln\left(\frac{1}{\sqrt{2\pi Npq}}\right) - \left(\frac{N+m}{2}+\frac{1}{2}\right)\ln\left(1+\frac{\delta_m}{2Np}\right) - \left(\frac{N-m}{2}+\frac{1}{2}\right)\ln\left(1-\frac{\delta_m}{2Nq}\right)$$

In the limit $N\to\infty$, we have $N+1\approx N$, so

$$\ln P(m,N) = \ln\left(\frac{1}{\sqrt{2\pi Npq}}\right) - \frac{N+m}{2}\ln\left(1+\frac{\delta_m}{2Np}\right) - \frac{N-m}{2}\ln\left(1-\frac{\delta_m}{2Nq}\right)$$

Using the Taylor expansion, $\ln(1+x) = x - \frac{x^2}{2} + \mathcal{O}(x^3)$ and $\ln(1-x) = -x - \frac{x^2}{2} + \mathcal{O}(x^3)$, we get

$$\ln P(m,N) = \ln\left(\frac{1}{\sqrt{2\pi Npq}}\right) - \frac{N+m}{2}\left[\frac{\delta_m}{2Np} - \frac{\delta_m^2}{8N^2p^2}\right] - \frac{N-m}{2}\left[-\frac{\delta_m}{2Nq} - \frac{\delta_m^2}{8N^2q^2}\right]$$

Using the relation $\delta_m = m - N(p-q)$, we get

$$\ln P(m,N) = \ln\left(\frac{1}{\sqrt{2\pi Npq}}\right) - Np\left(1+\frac{\delta_m}{2Np}\right)\left[\frac{\delta_m}{2Np} - \frac{\delta_m^2}{8N^2p^2}\right] + Nq\left(1-\frac{\delta_m}{2Nq}\right)\left[-\frac{\delta_m}{2Nq} - \frac{\delta_m^2}{8N^2q^2}\right]$$

Rearranging the terms and taking the terms up to $\mathcal{O}(1/N)$, we get

$$\ln P(m,N) = \ln\left(\frac{1}{\sqrt{2\pi Npq}}\right) - \frac{\delta_m^2}{2(4Npq)}$$

So we have,

$$\ln P(m,N) = \ln\left(\frac{2}{\sqrt{2\pi(4Npq)}}\right) - \frac{\delta_m^2}{2(4Npq)}. \tag{5.23}$$

Thus, the probability distribution is given by

$$P(m,N) = \frac{2}{\sqrt{2\pi(4Npq)}}\exp\left(-\frac{\delta_m^2}{2(4Npq)}\right),$$

where, $\delta_m = m - \langle m\rangle = m - N(p-q)$. Define $\sigma^2 = 4Npq$, and we get

$$P(m,N) = \frac{2}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{1}{2}\left(\frac{m-\langle m\rangle}{\sigma}\right)^2\right). \tag{5.24}$$

§5.3 Normal Distribution

The normal distribution is given by

$$P(x) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{5.25}$$

where, μ, σ are real constants. Let's calculate the MGF of the normal distribution,

$$\begin{aligned}\tilde{p}(k) &= \frac{1}{\sqrt{2\pi\sigma^2}}\int_{-\infty}^{\infty}\exp\left(-\frac{(x-\mu)^2}{2\sigma^2} - \mathrm{i}kx\right)\mathrm{d}x \\ &= \exp\left(-\mathrm{i}k\mu - \frac{\sigma^2k^2}{2}\right).\end{aligned} \tag{5.26}$$

Now, let's calculate the CGF of the normal distribution,

$$\ln\tilde{p}(k) = -\mathrm{i}k\mu - \frac{\sigma^2k^2}{2} = \sum_{n=1}^{\infty}\frac{(-\mathrm{i}k)^n}{n!}\langle x^n\rangle_c. \tag{5.27}$$

From the above equation, we can identify the cumulants of the normal distribution as

$$\langle x\rangle_c = \langle x\rangle = \mu; \qquad \left\langle x^2\right\rangle_c = \left\langle x^2\right\rangle - \langle x\rangle^2 = \sigma^2; \qquad \langle x^n\rangle_c = 0 \quad \forall n \geq 3. \tag{5.28}$$

So, the normal distribution is completely characterized by the first two cumulants.

Theorem 5.2 (Marcinkiewicz Theorem)
The cumulant generating function cannot be a finite polynomial of degree greater than 2; thus there is no distribution with any mean, any non-zero variance, and any non-zero skew but no non-zero higher cumulants.[a]

[a] For a proof, see [4].

§5.4 Poisson Distribution

Let's understand the Poisson distribution with the help of a biological example.

Example 5.3 (Transcription and Translation)
Consider a DNA that codes for a protein. The DNA is transcribed into mRNA (with rate k_t) and the mRNA is translated into protein (with rate k_p). Let the concentration of DNA at time t be $[D]$, the concentration of mRNA be $[M]$, so the rate law for the concentration of mRNA is given by

$$\frac{\mathrm{d}[M]}{\mathrm{d}t} = k_t[D] - k_p[M]. \tag{5.29}$$

Assuming that the concentration of DNA is constant (and initial concentration of mRNA is 0), the solution of the above equation is given by

$$[M](t) = \frac{k_t[D]}{k_p}(1 - \exp(-k_p t)). \tag{5.30}$$

The steady state concentration of mRNA (*i.e.* $t \to \infty$) is given by

$$[M]_{\mathrm{ss}} = \frac{k_t[D]}{k_p}. \tag{5.31}$$

In the experiment, the result doesn't correspond to the deterministic analytic solution to the rate law. The reason for this discrepancy is the stochastic nature of the process. So, we need to consider the stochastic nature of the process for a better understanding of the experiment.

Let's consider $P_m(t)$ to be the probability of finding m mRNA molecules at time t. Then the master equation becomes,

$$\frac{\partial P_m(t)}{\partial t} = \underbrace{k_t P_{m-1}(t) + k_p(m+1)P_{m+1}(t)}_{\text{gain}} - \underbrace{k_t P_m(t) - k_p m P_m(t)}_{\text{loss}}. \tag{5.32}$$

Assuming that the system is in steady state (*i.e.* $\partial P_m(t)/\partial t = 0$), we get

$$(m+1)P_{m+1} = mP_m + \frac{k_t}{k_p}[P_m - P_{m-1}]. \tag{5.33}$$

For this recurrence relation, we assume $P_{-1} = 0$ and P_0 is some constant. The solution of the above equation is given by

$$\begin{aligned} P_1 &= \frac{k_t}{k_p}P_0 \\ P_2 &= \frac{1}{2}\left(\frac{k_t}{k_p}\right)^2 P_0 \\ &\vdots \\ P_m &= \frac{1}{m!}\left(\frac{k_t}{k_p}\right)^m P_0. \end{aligned}$$

For finding the value of P_0, we use the normalization condition, *i.e.*

$$\sum_{m=0}^{\infty} P_m = 1 \quad \Rightarrow \quad P_0 = \exp\left(-\frac{k_t}{k_p}\right). \tag{5.34}$$

Thus, the probability of finding m mRNA molecules in the steady state is given by

$$P_m = \frac{1}{m!}\left(\frac{k_t}{k_p}\right)^m \exp\left(-\frac{k_t}{k_p}\right). \tag{5.35}$$

This distribution is called the Poisson distribution. For simplicity, we define $\lambda := \frac{k_t}{k_p}$, and the Poisson distribution becomes

$$p(m) = \frac{\lambda^m}{m!} e^{-\lambda}. \tag{5.36}$$

Let's calculate the MGF of the Poisson distribution,

$$\begin{aligned} \tilde{p}(n) &= \sum_{m=0}^{\infty} \frac{\lambda^m}{m!} e^{-\lambda} e^{-\mathrm{i}nm} \\ &= \exp\Big(\lambda\Big[e^{-\mathrm{i}n} - 1\Big]\Big). \end{aligned} \tag{5.37}$$

Now, let's calculate the CGF of the Poisson distribution,

$$\ln \tilde{p}(n) = \lambda\Big[e^{-\mathrm{i}n} - 1\Big] = \lambda \sum_{\nu=1}^{\infty} \frac{(-\mathrm{i}n)^\nu}{\nu!}. \tag{5.38}$$

Comparing the coefficients in the above equation with eqn. (3.6), we get

$$\langle m^\nu \rangle_c = \lambda \quad \forall \nu \geq 1. \tag{5.39}$$

6 Lecture 06: (15/01/2025)

Till now, we have discussed the single random variable. But in real life, the system can't be described by a single random variable (in general). We need to consider multiple random variables. For example describing the position and momentum of a particle, we need six random variables.

§6.1 Many Random Variables

Say we have N random variables $X_1, X_2, \ldots, X_N$ taking values in $\mathbb{R}$. Thus, our sample space is $\Omega = \mathbb{R}^N$.

Definition 6.1 (Joint Probability Density Function):
The joint probability density function (joint PDF) $p(\mathbf{x})$ is the probability density of an outcome in a volume element $\mathrm{d}^N\mathbf{x}$ around $\mathbf{x}$.

$$p(\mathbf{x})\,\mathrm{d}^N\mathbf{x} = \Pr\Big(\mathbf{X} \in \mathrm{d}^N\mathbf{x}\Big). \tag{6.1}$$

Clarifying the notation, $\mathbf{x} = (x_1, x_2, \ldots, x_N)$ and $p(\mathbf{x}) \equiv p(X_1 = x_1, X_2 = x_2, \ldots, X_N = x_N)$. And the volume element $\mathrm{d}^N\mathbf{x} = \prod_{i=1}^N \mathrm{d}x_i$.
The condition that total probability is 1 (or the normalization condition) in this case is

$$\int_{\mathbb{R}^N} p(\mathbf{x})\,\mathrm{d}^N\mathbf{x} = 1. \tag{6.2}$$

In case, all the random variables are independent, then the joint PDF factorizes into the product of individual PDFs.

$$p(\mathbf{x}) = \prod_{i=1}^N p_i(x_i). \tag{6.3}$$

Definition 6.2 (Marginal Probability Density Function):
The marginal probability density function of X_i is given by

$$p_i(x_i) = \int_{\mathbb{R}^{N-1}} p(\mathbf{x}) \prod_{j\neq i} \mathrm{d}x_j\,. \tag{6.4}$$

In other words, the marginal PDF is the PDF of a single random variable obtained by integrating over all other random variables. We can also extend this definition to find the marginal PDF of any subset of random variables.

$$p(x_1, \cdots, x_k) = \int_{\mathbb{R}^{N-k}} p(x_1, \cdots, x_N) \prod_{j=k+1}^{N} \mathrm{d}x_j\,. \tag{6.5}$$

Definition 6.3 (Conditional Probability Density Function):
The conditional probability density function of X_i given X_j is given by

$$p_{i|j}(x_i \mid x_j) = \frac{p(x_i, x_j)}{p_j(x_j)}. \tag{6.6}$$

The conditional PDF describes the behavior of a subset of random variables, for specified values of the others.

Theorem 6.4 (Bayes' Theorem)

$$p_{i|j}(x_i \mid x_j) = \frac{p_{j|i}(x_j \mid x_i)\; p_i(x_i)}{p_j(x_j)}. \tag{6.7}$$

§6.2 Central Limit Theorem

Let $X_1, X_2, \ldots, X_N$ be N independent and identically distributed (i.i.d.) random variables. Define the sum of these random variables as

$$Y = \sum_{i=1}^{N} X_i. \tag{6.8}$$

Then the PDF of Y is given by

$$p_Y(y) = \int \mathrm{d}^N\mathbf{x}\ \delta\left(y - \sum_{i=1}^{N} x_i\right) p(\mathbf{x}), \tag{6.9}$$

where $p(\mathbf{x})$ is the joint PDF of $X_1, X_2, \ldots, X_N$. With this definition, we can calculate the MGF of Y.

$$\begin{aligned}\tilde{p}_Y(k) &= \int \mathrm{d}y\ e^{-\mathrm{i}ky}\ p_Y(y) \\ &= \int_{\mathbb{R}^N} \mathrm{d}^N\mathbf{x}\, p(\mathbf{x}) \int_{-\infty}^{\infty} \mathrm{d}y\ e^{-\mathrm{i}ky}\ \delta\left(y - \sum_{i=1}^{N} x_i\right) \\ &= \int_{\mathbb{R}^N} \mathrm{d}^N\mathbf{x}\, p(\mathbf{x}) \exp\left(-\mathrm{i}k \sum_{i=1}^{N} x_i\right)\end{aligned} \tag{6.10}$$

Since X_i's are independent, we have

$$p(\mathbf{x}) = \prod_{i=1}^{N} p(x_i) \qquad\qquad \exp\left(-\mathrm{i}k \sum_{i=1}^{N} x_i\right) = \prod_{i=1}^{N} \exp(-\mathrm{i}kx_i) \tag{6.11}$$

Therefore,

$$\begin{aligned}\tilde{p}_Y(k) &= \prod_{i=1}^{N} \int_{-\infty}^{\infty} \mathrm{d}x_i\ p(x_i)\ \exp(-\mathrm{i}kx_i) \\ &= [\tilde{p}(k)]^N,\end{aligned} \tag{6.12}$$

where $\tilde{p}(k)$ is the MGF of X_i (since they are identical, they have the same MGF). Let's define the average of X_i as

$$\bar{X} = \frac{\sum_{i=1}^{N} X_i}{N}. \tag{6.13}$$

Then, it is easy to see that the MGF of $\bar{X}$ is

$$\tilde{p}_{\bar{X}}(k) = \left[\tilde{p}\left(\frac{k}{N}\right)\right]^N, \tag{6.14}$$

we can rewrite this as

$$\tilde{p}_{\bar{X}}(k) = \exp\left(N \ln\left[\tilde{p}\left(\frac{k}{N}\right)\right]\right) \tag{6.15}$$

Observe that $\ln[\tilde{p}(k/N)]$ is the CGF of X_i. We know,

$$K\left(\frac{k}{N}\right) = \ln\left[\tilde{p}\left(\frac{k}{N}\right)\right] = \sum_{n=1}^{\infty} \frac{(-\mathrm{i}k)^n}{n!} \frac{\langle x^n \rangle_c}{N^n} \tag{6.16}$$

Therefore,

$$\tilde{p}_{\bar{X}}(k) = \exp\left(-\mathrm{i}k \langle x \rangle_c - \frac{k^2}{2} \frac{\langle x^2 \rangle_c}{N} + \mathcal{O}\left(\frac{1}{N^2}\right)\right). \tag{6.17}$$

In limit $N \to \infty$, we drop all terms of order $1/N^2$ and higher. Then, this is the MGF of a Gaussian distribution (see eqn. (5.26)) with mean $\langle x \rangle_c$ and variance $\langle x^2 \rangle_c / N$.

Theorem 6.5 (Central Limit Theorem)
Let $X_1, X_2, \ldots, X_N$ be N i.i.d. random variables with mean μ and variance σ^2. Define, the average of these random variables as

$$\bar{X} = \frac{\sum_{i=1}^{N} X_i}{N}. \tag{6.18}$$

Then, in the limit $N \to \infty$, the PDF of $\bar{X}$ approaches a Normal distribution with mean μ and variance σ^2/N.

$$\lim_{N\to\infty} p_{\bar{X}}(x) = \frac{1}{\sqrt{2\pi(\sigma^2/N)}} \exp\left(-\frac{(x-\mu)^2}{2(\sigma^2/N)}\right). \tag{6.19}$$

Note: The Central Limit Theorem holds only for random variables with finite mean and variance. From the derivation, we can see that the Central Limit Theorem doesn't hold for dependent random variables.

§6.3 Rule for large numbers

To describe equilibrium properties of macroscopic bodies, statistical mechanics has to deal with the very large number N of microscopic degrees of freedom. Usually, taking the thermodynamic limit $N \to \infty$ is a good approximation and simplifies the calculations.

There are mainly three types of N dependence encountered in statistical mechanics:

(a) *Intensive* quantities: $\sim \mathcal{O}(N^0)$. These are independent of the system size. For example, temperature, pressure, chemical potential, etc.

(b) *Extensive* quantities: $\sim \mathcal{O}(N)$. These are proportional to the system size. For example, energy, volume, entropy, etc.

(c) *Exponential* dependence: $\sim \mathcal{O}(\exp(N\phi))$. These are the most interesting quantities. For example, the number of microstates, available volume in phase space, etc.

In statistical mechanics we frequently encounter sums or integrals of exponential variables. Performing such sums in the thermodynamic limit is considerably simplified due to the following results.

1. *Summation of exponential quantities*: Consider a sum of n exponential variables

$$S_n = \sum_{i=1}^{n} \mathcal{E}_i \tag{6.20}$$

where each term is positive, with an exponential dependence on N, that is

$$0 \leq \mathcal{E}_i \sim \mathcal{O}(\exp(N\phi_i)), \tag{6.21}$$

and the number of terms n is proportional to some power of N. Let $\mathcal{E}_{\max}$ be the maximum of all $\mathcal{E}_i$'s. Ignoring trivial cases (*i.e.* when $\mathcal{E}_i = 0$ for all i), then the sum S_n is bounded between

$$\mathcal{E}_{\max} \leq S_n \leq n\mathcal{E}_{\max}. \tag{6.22}$$

An intensive quantity can be constructed from S_n as $\ln S_n/N$. This quantity is bounded between

$$\frac{\ln \mathcal{E}_{\max}}{N} \leq \frac{\ln S_n}{N} \leq \frac{\ln \mathcal{E}_{\max}}{N} + \frac{\ln n}{N}. \tag{6.23}$$

For $n \propto N^\alpha$, where $\alpha > 0$, the ratio $\ln n/N \to 0$ as $N \to \infty$ and,

$$\lim_{N\to\infty} \frac{\ln S_n}{N} = \lim_{N\to\infty} \frac{\ln \mathcal{E}_{\max}}{N} = \phi_{\max}. \tag{6.24}$$

2. *Saddle point integration*: Consider the integral

$$I = \int_{-\infty}^{\infty} \mathrm{d}x \exp(N\phi(x)), \tag{6.25}$$

where $\phi(x)$ is a smooth function of x. Let ϕ attains global maximum at $x = x_{\max}$. Expand $\phi(x)$ around $x_{\max}$, give us

$$I = \int \mathrm{d}x \exp\left(N\left[\phi(x_{\max}) - \frac{1}{2}|\phi''(x_{\max})|(x - x_{\max})^2 + \cdots\right]\right). \tag{6.26}$$

Note that at the maximum, $\phi'(x_{\max}) = 0$ and $\phi''(x_{\max}) < 0$. Approximating the integral as a Gaussian integral (terminating the expansion at the quadratic term), we get

$$I \approx e^{N\phi(x_{\max})} \int \mathrm{d}x \exp\left(-\frac{N}{2}|\phi''(x_{\max})|(x - x_{\max})^2\right) = \sqrt{\frac{2\pi}{N|\phi''(x_{\max})|}} e^{N\phi(x_{\max})}. \tag{6.27}$$

Using this approximation, we can calculate $\ln I/N$ in thermodynamic limit as

$$\lim_{N\to\infty} \frac{\ln I}{N} = \lim_{N\to\infty} \left[\phi(x_{\max}) - \frac{1}{2N} \ln\left(\frac{N|\phi''(x_{\max})|}{2\pi}\right)\right] = \phi(x_{\max}). \tag{6.28}$$

7 Lecture 07: (20/01/2025)

Recalling the last lecture, we discussed the concept of 'Saddle point approximation.'

Theorem 7.1 (Laplace's Method)
Let $I = \int_a^b \exp(Mf(x))\,\mathrm{d}x$, where $M \gg 1$ and $f(x)$ is twice differentiable in the interval $[a, b]$ with a global maximum at $x_0 \in (a, b)$. Then, the integral I can be approximated as

$$I \approx \sqrt{\frac{2\pi}{M|f''(x_0)|}}\exp(Mf(x_0)). \tag{7.1}$$

§7.1 Stirling's Approximation

Stirling's approximation is a method used to approximate factorials for large numbers. Let's start with analytic continuation of the factorial function *i.e.* the Gamma function ($\Gamma(n)$).

$$\Gamma(n+1) \equiv n! = \int_0^\infty \mathrm{d}x\, x^n e^{-x}. \tag{7.2}$$

Now, we can rewrite the integrand as $x^n e^{-x} = e^{n\ln x}e^{-x} = \exp(n[\ln x - x/n])$, comparing with the form of the integrand in Laplace's method, we can write $f(x) = \ln x - x/n$. The maximum of this function is at $x = n$, and the second derivative of $f(x)$ is $-1/n^2$. Substituting these values in eqn. (7.1), we get

$$n! = e^{n[\ln n - 1]}\sqrt{\frac{2\pi}{n(1/n^2)}} = n^n e^{-n}\sqrt{2\pi n}. \tag{7.3}$$

This is the Stirling's approximation for the factorial function. Another form of Stirling's approximation is

$$\ln n! \approx n\ln n - n + \frac{1}{2}\ln(2\pi n). \tag{7.4}$$

Problem 7.1.1.
Using Laplace's method, find the approximate value of the integral in limit $n \gg 1$

$$I = \int_0^\infty x^{2n} e^{-ax^2 - bx^4/2}\,\mathrm{d}x\,, \tag{7.5}$$

where $a, b > 0$.

Solution.
Let's start with a change of variables $u = x^2$, then the integral becomes

$$I = \frac{1}{2}\int_0^\infty u^m e^{-au - bu^2/2}\,\mathrm{d}u\,,$$

where $m = n - 1/2$. Define $f(u) = au + bu^2/2 - m\ln u$, then the maximum of $f(u)$ is at $u = (-a + \sqrt{a^2 + 4bm})/2b$. The second derivative of $f(u)$ is $b - m/u_{\max}^2 = -a/u_{\max}$, substituting these values we get

$$I \approx e^{f(u_{\max})}\frac{1}{2}\int_0^\infty \exp\left(-\frac{1}{2}\left(\frac{2ab}{a - \sqrt{a^2 + 4bm}}\right)(u - u_{\max})^2\right)\mathrm{d}u\,.$$

◀

§7.2 Entropy

Boltzmann introduced the formula for entropy in statistical mechanics depending on the number of microstates Ω of a system. The entropy is defined as

$$S = k_B \ln \Omega, \tag{7.6}$$

where k_B is the Boltzmann constant. Let each microstate is equal probable with probability $p = 1/\Omega$, then the entropy can be written as

$$S = k_B \ln \Omega = -k_B \ln p. \tag{7.7}$$

Example 7.2 (Coin Toss and Bit Entropy)

Consider a biased coin with probability p of heads and $1-p$ of tails. Recall eqn. (4.3), if we have N coin tosses, then on average we get Np heads and $N(1-p)$ tails. Let this head and tail corresponds to 0 and 1 respectively, then we have Ω different (binary) messages of length N.

$$\Omega = \frac{N!}{(Np)!(N(1-p))!}. \tag{7.8}$$

Using (crude) Stirling's approximation, $N! = N^N$, we get

$$\Omega \approx \frac{N^N}{(Np)^{Np}(N(1-p))^{N(1-p)}} = \frac{\cancel{N^N}}{\cancel{N^{Np}}\cancel{N^{N(1-p)}}p^{Np}(1-p)^{N(1-p)}} = \left[\frac{1}{p^p(1-p)^{1-p}}\right]^N. \tag{7.9}$$

Therefore, the entropy of this message is

$$S = k_B \ln \Omega = -N[p \ln p + (1-p)\ln(1-p)]. \tag{7.10}$$

So if we calculate the entropy per bit (*i.e.* $S_{\text{bit}} = S/N$) we get

$$S_{\text{bit}} = -[p \ln p + (1-p)\ln(1-p)]. \tag{7.11}$$

The last example is a special case of the Shannon entropy, which is defined as

$$S = -\sum_i p_i \ln p_i, \tag{7.12}$$

where p_i is the probability of the i-th microstate.

§7.2.1 Application of Maximum Entropy Principle

At this stage, this new quantity S is just a mathematical construct, but we can estimate the probability of a microstate using the principle of maximum entropy (if system is subjected to some constraints). In case, the system is not subjected to any constraints, then the 'best guess' is that all microstates are equally probable.

Let us consider a system with energy E, let the i-th microstate has energy E_i having probability p_i. So we want to maximize the entropy,

$$S = -\sum_i p_i \ln p_i, \tag{7.13}$$

subject to the constraints

$$\sum_i p_i = 1 \qquad \sum_i p_i E_i = E. \tag{7.14}$$

Using the method of Lagrange multipliers, we can write the Lagrangian as

$$\mathcal{L}(p_i, \alpha, \beta) = -\sum_i p_i \ln p_i - \alpha\left(\sum_i p_i - 1\right) - \beta\left(\sum_i p_i E_i - E\right). \tag{7.15}$$

Taking the derivative of $\mathcal{L}$ with respect to p_i and setting it to zero, we get

$$\frac{\partial \mathcal{L}}{\partial p_i} = -\ln p_i - 1 - \alpha - \beta E_i = 0. \tag{7.16}$$

Solving this equation, we get

$$p_i = \exp(-1-\alpha)\exp(-\beta E_i). \tag{7.17}$$

Using the constraint (normalizability), we get

$$1 = \sum_i p_i = \exp(-1-\alpha)\sum_i \exp(-\beta E_i) \quad \Rightarrow \quad \exp(-1-\alpha) = \frac{1}{\sum_i e^{-\beta E_i}}. \tag{7.18}$$

We define the partition function as

$$Z = \sum_i e^{-\beta E_i}, \tag{7.19}$$

then the probability of the i-th microstate is

$$p_i = \frac{1}{Z} e^{-\beta E_i}. \tag{7.20}$$

Intuitively, we know that $\beta \propto T^{-1}$. But, let's see how we can derive this for the gas of non-interacting particles.

Remark 7.3 (Kinetic Theory of Gases). Let P_x be the momentum of a particle, then the energy of the particle is $E = P_x^2/2m$. The probability of a particle having momentum P_x is

$$p(P_x) = \frac{1}{Z} e^{-\beta P_x^2/2m}, \tag{7.21}$$

where $Z = \int_{-\infty}^{\infty} e^{-\beta P_x^2/2m}\, \mathrm{d}P_x$. This is a Gaussian integral, and we can write

$$Z = \sqrt{\frac{2\pi m}{\beta}}. \tag{7.22}$$

Therefore, the average energy of the particle is

$$\langle E \rangle = \int_{-\infty}^{\infty} \frac{P_x^2}{2m}\, p(P_x)\ \mathrm{d}P_x = \frac{1}{Z}\int_{-\infty}^{\infty} \frac{P_x^2}{2m} e^{-\beta P_x^2/2m}\, \mathrm{d}P_x\,, \tag{7.23}$$

which is a Gaussian integral, and we can write

$$\langle E \rangle = \frac{1}{2\beta}, \tag{7.24}$$

which is the average kinetic energy of the particle. Therefore, the temperature of the gas is

$$\langle E \rangle = \frac{1}{2\beta} = \frac{1}{2} k_B T \quad \Rightarrow \quad \beta = \frac{1}{k_B T}. \tag{7.25}$$

§7.2.2 Alternative Method to Arrive at formulation of Canonical Ensemble

Consider an atom in contact with a bath at temperature T. The system (atom + bath) is isolated, so the total energy is conserved. Let's assume that the atom can only be in two microstates, S_1 and S_2, with energies $E(S_1)$ and $E(S_2)$ respectively with probabilities $p(S_1)$ and $p(S_2)$. Let $\Omega_R(S_1)$ and $\Omega_R(S_2)$ be the number of microstates of the bath corresponding to atom in microstate S_1 and S_2 respectively. Thus, entropy of reservoir is,

$$S_R(S_1) = k_B \ln \Omega_R(S_1) \quad \text{and} \quad S_R(S_2) = k_B \ln \Omega_R(S_2). \tag{7.26}$$

Consider the ratio of probabilities of the atom being in microstate S_1 and S_2,

$$\frac{p(S_1)}{p(S_2)} = \frac{\Omega_R(S_2)}{\Omega_R(S_1)} = \exp\left(\frac{S_R(S_2) - S_R(S_1)}{k_B}\right). \tag{7.27}$$

Using a result from thermodynamics, $T\mathrm{d}S_R = \mathrm{d}U_R + P\mathrm{d}V_R - \mu\,\mathrm{d}N_R$, where U_R is the internal energy of the reservoir, P is the pressure, V_R is the volume of the reservoir, and μ is the chemical potential. Here, $\mathrm{d}V_R$ and $\mathrm{d}N_R$ are zero, so we get

$$T\mathrm{d}S_R = \mathrm{d}U_R \quad \Rightarrow \quad \mathrm{d}S_R = \frac{\mathrm{d}U_R}{T}. \tag{7.28}$$

And since the total energy of the system is conserved, we have $\mathrm{d}U_R = -\,\mathrm{d}U$, where $\mathrm{d}U$ is the change in energy of the atom. Therefore, we get

$$S_R(S_2) - S_R(S_1) = -\frac{1}{T}[E(S_2) - E(S_1)]. \tag{7.29}$$

Substituting this in the ratio of probabilities, we get

$$\frac{p(S_1)}{p(S_2)} = \frac{\exp(-E(S_1)/k_BT)}{\exp(-E(S_2)/k_BT)}. \tag{7.30}$$

Therefore, the probability of the atom being in microstate S is

$$p(S) = \frac{1}{Z}\exp\left(-\frac{E(S)}{k_BT}\right), \tag{7.31}$$

where Z is the partition function.

$$Z = \sum_S \exp\left(-\frac{E(S)}{k_BT}\right). \tag{7.32}$$

8 Lecture 08: (22/01/2025)

At this stage, we know enough probability theory to start talking about the most statistical mechanics. Historically, one of the first problems that statistical mechanics was developed to solve was the problem of the ideal gas by formulating "kinetic theory of gases".

§8.1 Kinetic Theory of Gases

Kinetic theory studies the macroscopic properties of large numbers of particles, starting from their (classical) equations of motion.
Thermodynamics describes the macroscopic properties of a system in terms of a few variables (pressure, volume, temperature, etc.) with help of empirical laws. The goal of kinetic theory is to derive these laws from the microscopic information about the particles (atoms or molecules) that make up the system. In particular, we shall try to answer the following questions:

1. How we can define "equilibrium" for a system of moving particles?
2. Do all systems naturally evolve towards an equilibrium state?
3. What is the time evolution of a system that is not quite in equilibrium?

The simplest system to study, the veritable workhorse of thermodynamics, is the dilute (nearly ideal) gas. A typical volume of gas contains of the order of 10^{23} particles, and in kinetic theory we try to deduce the macroscopic properties of the gas from the time evolution of the set of atomic coordinates.

Definition 8.1 (Microstate and Macrostate):
At any time t, the *microstate* of a system of N particles is described by specifying the positions $\mathbf{q}_i(t)$, and momenta $\mathbf{p}_i(t)$, of all particles. The microstate thus corresponds to a point $\mu(t)$, in the $6N$-dimensional phase space $\Gamma = \Pi_{i=1}^{N}\{\mathbf{q}_i, \mathbf{p}_i\}$.
A *macrostate* is a set of macroscopic variables that are assigned to each microstate.

For example, in the case of an ideal gas, the macrostate is described by the volume V, the number of particles N, and the total energy E of the gas.
The space of macrostates is considerably smaller than the phase space spanned by microstates. Therefore, there must be a very large number of microstates μ corresponding to the same macrostate M.
Let us consider a system of N particles evolving in time under the influence of some Hamiltonian $\mathcal{H}(\mathbf{q}_i, \mathbf{p}_i)$. Then the time evolution of a microstate $\mu(t)$ is given by the Hamiltonian equations of motion (canonical equations):

$$\begin{aligned} \dot{\mathbf{q}}_i &= \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} \\ \dot{\mathbf{p}}_i &= -\frac{\partial \mathcal{H}}{\partial \mathbf{q}_i} \end{aligned} \tag{8.1}$$

At this point, we need to formalize the concept of an ensemble of systems. Since we know that different microstates can correspond to the same macrostate.

Definition 8.2 (Statistical Ensemble):
A *statistical ensemble* is a collection of identical systems, each of which is in a different microstate.

Consider an ensemble of systems, which contains $\mathcal{N}$ copies of the system, each described by a different microstate (or representative point) in the phase space. Let $\mathrm{d}\mathcal{N}(\mathbf{q}, \mathbf{p}, t)$ denote the number of microstates in the ensemble that lie in an infinitesimal volume element $\mathrm{d}\Gamma = \prod_{i=1}^{N} \mathrm{d}^3\mathbf{q}_i\, \mathrm{d}^3\mathbf{p}_i$ around the point $(\mathbf{q}, \mathbf{p})$ in phase space at time t. Then the probability density of finding a system in the ensemble in the microstate $(\mathbf{q}, \mathbf{p})$ at time t is given by

$$\rho(\mathbf{q}, \mathbf{p}, t) = \lim_{\substack{\mathcal{N}\to\infty \\ \mathrm{d}\Gamma\to 0}} \frac{\mathrm{d}\mathcal{N}(\mathbf{q}, \mathbf{p}, t)}{\mathcal{N}\,\mathrm{d}\Gamma}. \tag{8.2}$$

It is evident that $\rho(\mathbf{q}, \mathbf{p}, t)$ is a normalized probability density, i.e., $\int \mathrm{d}\Gamma\, \rho(\mathbf{q}, \mathbf{p}, t) = 1$. To compute macroscopic properties of the system for various function of canonical variables $\mathcal{O}(\mathbf{q}, \mathbf{p})$, we need to compute the *ensemble average* of $\mathcal{O}$:

$$\langle \mathcal{O} \rangle = \int \mathrm{d}\Gamma\, \mathcal{O}(\mathbf{q}, \mathbf{p}) \rho(\mathbf{q}, \mathbf{p}, t). \tag{8.3}$$

When the exact microstate of the system is known, we say that the system is in a *pure state*. On the other hand, when our knowledge of the system is probabilistic, we say that the system is in a *mixed state*.

§8.1.1 Liouville's Theorem

We must observe that if a system is evolving under Hamiltonian dynamics, then the system has time-reversal symmetry. This implies that trajectory of the system in phase space is either a simple closed curve or a curve that never intersects itself. Furthermore, the trajectory of two different representative points in phase space never intersect.

Theorem 8.3 (Liouville's Theorem)
Consider a statistical ensemble of systems evolving under Hamiltonian dynamics. Then the probability density $\rho(\mathbf{q}, \mathbf{p}, t)$ in phase space is conserved along the trajectories of the system. That is,

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = 0. \tag{8.4}$$

Proof:
Since the total number of systems in an ensemble is conserved, the number of representative points leaving any volume in space per second must be equal to the rate of decrease of the number of representative points in the same volume. Let $\mathbf{v} = (\dot{\mathbf{q}}, \dot{\mathbf{p}})$ be the velocity of the representative point in phase space. Let U be an arbitrary volume in phase space with boundary ∂U. The flow of density of states is defined as $\mathbf{v}\rho$, then

$$\frac{\mathrm{d}}{\mathrm{d}t} \int_U \mathrm{d}\Gamma\, \rho = -\oint_{\partial U} \mathrm{d}\mathbf{S} \cdot \mathbf{v}\rho. \tag{8.5}$$

Using the divergence theorem, we can write

$$\int_{\partial U} \mathrm{d}\mathbf{S} \cdot \mathbf{v}\rho = \int_U \mathrm{d}\Gamma\, \boldsymbol{\nabla} \cdot (\mathbf{v}\rho). \tag{8.6}$$

Substituting eqn. (8.6) into eqn. (8.5), we get

$$\begin{aligned} \frac{\mathrm{d}}{\mathrm{d}t} \int_U \mathrm{d}\Gamma\, \rho &= -\int_U \mathrm{d}\Gamma\, \boldsymbol{\nabla} \cdot (\mathbf{v}\rho) \\ \Rightarrow \int_U \mathrm{d}\Gamma \left[\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\mathbf{v}\rho) \right] &= 0. \end{aligned} \tag{8.7}$$

Since the volume U is arbitrary, we must have

$$\frac{\partial \rho}{\partial t} + \boldsymbol{\nabla} \cdot (\mathbf{v}\rho) = 0. \tag{8.8}$$

Now, let us write the divergence of the flow velocity $\mathbf{v}$ in phase space:

$$\boldsymbol{\nabla} \cdot \mathbf{v}\rho = \sum_{i=1}^{N} \left(\frac{\partial}{\partial \mathbf{q}_i} (\dot{\mathbf{q}}_i \rho) + \frac{\partial}{\partial \mathbf{p}_i} (\dot{\mathbf{p}}_i \rho) \right) \tag{8.9}$$

$$= \sum_{i=1}^{N} \left[\frac{\partial \rho}{\partial \mathbf{q}_i} \dot{\mathbf{q}}_i + \frac{\partial \rho}{\partial \mathbf{p}_i} \dot{\mathbf{p}}_i \right] + \rho \sum_{i=1}^{N} \left[\frac{\partial \dot{\mathbf{q}}_i}{\partial \mathbf{q}_i} + \frac{\partial \dot{\mathbf{p}}_i}{\partial \mathbf{p}_i} \right] \tag{8.10}$$

Using the Hamiltonian equations of motion, we can write

$$= \sum_{i=1}^{N} \left[\frac{\partial \rho}{\partial \mathbf{q}_i} \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} - \frac{\partial \rho}{\partial \mathbf{p}_i} \frac{\partial \mathcal{H}}{\partial \mathbf{q}_i} \right] + \rho \sum_{i=1}^{N} \left[\cancelto{0}{\frac{\partial^2 \mathcal{H}}{\partial \mathbf{q}_i \partial \mathbf{p}_i} - \frac{\partial^2 \mathcal{H}}{\partial \mathbf{p}_i \partial \mathbf{q}_i}} \right] \tag{8.11}$$

$$= \sum_{i=1}^{N} \left[\frac{\partial \rho}{\partial \mathbf{q}_i} \frac{\partial \mathcal{H}}{\partial \mathbf{p}_i} - \frac{\partial \rho}{\partial \mathbf{p}_i} \frac{\partial \mathcal{H}}{\partial \mathbf{q}_i} \right] = \{\rho, \mathcal{H}\}. \tag{8.12}$$

Therefore, we have

$$\frac{\partial \rho}{\partial t} + \{\rho, \mathcal{H}\} = 0. \tag{8.13}$$

From classical mechanics, we know

$$\frac{\mathrm{d}\rho}{\mathrm{d}t} = \frac{\partial \rho}{\partial t} + \{\rho, \mathcal{H}\} = 0. \tag{8.14}$$

Q.E.D.

Definition 8.4 (Equilibrium):
A system is said to be in *equilibrium* if the probability density $\rho(\mathbf{q}, \mathbf{p}, t)$ is time-independent. Mathematically, this implies that

$$\frac{\partial \rho}{\partial t} = 0 \quad \Rightarrow \quad \{\rho, \mathcal{H}\} = 0. \tag{8.15}$$

Let's denote the equilibrium probability density by ρ_{eq}. Then, we have

$$\frac{\partial \rho_{\text{eq}}}{\partial \mathbf{q}_i} \dot{\mathbf{q}}_i + \frac{\partial \rho_{\text{eq}}}{\partial \mathbf{p}_i} \dot{\mathbf{p}}_i = 0. \tag{8.16}$$

We have following possible solutions:

1. Microcanonical Ensemble: $\rho_{\text{eq}} = \text{constant}$. This corresponds to a probability density that is uniform over whole phase space. This case says that each microstate have equal a priori probability.

2. Canonical Ensemble: $\rho_{\text{eq}} = \rho(\mathcal{H}(\mathbf{q}, \mathbf{p}))$. This corresponds to a probability density that is constant on the energy surface $\mathcal{H}(\mathbf{q}, \mathbf{p}) = E$.

   Say the system have n different conserved quantities $L_1, L_2, \ldots, L_n$. In this case, probability density of the form

   $$\rho_{\text{eq}} = \rho(\mathcal{H}(\mathbf{q}, \mathbf{p}), L_1, L_2, \ldots, L_n) \tag{8.17}$$

   is also a valid equilibrium probability density. Specifically, the probability density is constant on the surface $\mathcal{H}(\mathbf{q}, \mathbf{p}) = E$, $L_1 = l_1$, $L_2 = l_2, \ldots, L_n = l_n$.

With this, we have defined the concept of equilibrium for a system of moving particles. Two answers the second question posed at the beginning of the lecture, we need to show that non-stationary density $\rho(\mathbf{q}, \mathbf{p}, t)$ converges to the equilibrium density ρ_{eq}. In case of Hamiltonian dynamics, we have time reversal symmetry *i.e.* for any solution $\rho(t)$ converging to ρ_{eq}, there is a time-reversed solution $\rho(-t)$ that diverges away from ρ_{eq}. This implies that the system can never reach equilibrium.

One way to resolve this paradox is to hope that our solution $\rho(t)$ is in neighborhood of ρ_{eq} the majority of the time, so that the time average of the system is dominated by the stationary solution. This is the essence of the *ergodic hypothesis*. The ergodic hypothesis states that the time average of a system is equal to the ensemble average. This hypothesis is the foundation of statistical mechanics.

Theorem 8.5 (Ergodic Theorem)
If we wait long enough, the representative point of a system will visit every accessible point in phase space.

If we are able to prove the ergodic theorem for a system (under some conditions on system we can do this for some physical systems), then we can say that the time average of a system is equal to the ensemble average.

9 Lecture 09: (20/01/2025)

§9.1 Bogoliubov-Born-Green-Kirkwood-Yvon (BBGKY) hierarchy

The BBGKY hierarchy is a set of equations that describe the time evolution of the n-particle distribution functions. Let's consider a system of N particles, evolving according to the Hamiltonian dynamics with

$$\mathcal{H}(\mathbf{q},\mathbf{p}) = \sum_{i=1}^{N}\left[\frac{\mathbf{p}_i^2}{2m} + V(\mathbf{q}_i)\right] + \frac{1}{2}\sum_{i,j=1}^{N} U_{ij} \tag{9.1}$$

where $U_{ij} = U(|\mathbf{q}_i - \mathbf{q}_j|)$ and $U_{ii} = 0$. Observe the interaction potential is symmetric, i.e., $U_{ij} = U_{ji}$.

Notationally, $f_s(\mathbf{q}_1, \mathbf{p}_1; \cdots; \mathbf{q}_s, \mathbf{p}_s; t)$ is the s-particle distribution function, which is the probability density of finding s particles at positions $\mathbf{q}_1, \ldots, \mathbf{q}_s$ with momenta $\mathbf{p}_1, \ldots, \mathbf{p}_s$ respectively at time t. Let $\rho(\mathbf{q}, \mathbf{p}, t)$ be the probability density of finding the system at $\mu(t) = (\mathbf{q}, \mathbf{p})$ at time t. Then, the s-particle distribution function is given by

$$f_s(\mathbf{q}_1, \mathbf{p}_1; \ldots; \mathbf{q}_s, \mathbf{p}_s; t) = \frac{N!}{(N-s)!}\int \prod_{i=s+1}^{N} \mathrm{d}V_i\, \rho(\mathbf{q}_1, \mathbf{p}_1; \ldots; \mathbf{q}_s, \mathbf{p}_s; \mathbf{q}_{s+1}, \mathbf{p}_{s+1}; \ldots; \mathbf{q}_N, \mathbf{p}_N; t), \tag{9.2}$$

where $\mathrm{d}V_i = \mathrm{d}^3\mathbf{q}_i\, \mathrm{d}^3\mathbf{p}_i$. Let's define the s-particle coordinates probability density as

$$\rho_s(\mathbf{q}_1, \mathbf{p}_1; \ldots; \mathbf{q}_s, \mathbf{p}_s; t) = \frac{(N-s)!}{N!} f_s(\mathbf{q}_1, \mathbf{p}_1; \ldots; \mathbf{q}_s, \mathbf{p}_s; t). \tag{9.3}$$

For easier calculations, let's divide the Hamiltonian into three parts: $\mathcal{H} = \mathcal{H}_s + \mathcal{H}_{N-s} + \mathcal{H}'$, where

$$\mathcal{H}_s = \sum_{n=1}^{s}\left[\frac{\mathbf{p}_n^2}{2m} + V(\mathbf{q}_n)\right] + \frac{1}{2}\sum_{n,m=1}^{s} U(|\mathbf{q}_n - \mathbf{q}_m|), \tag{9.4}$$

$$\mathcal{H}_{N-s} = \sum_{i=s+1}^{N}\left[\frac{\mathbf{p}_i^2}{2m} + V(\mathbf{q}_i)\right] + \frac{1}{2}\sum_{i,j=s+1}^{N} U(|\mathbf{q}_i - \mathbf{q}_j|), \tag{9.5}$$

$$\mathcal{H}' = \sum_{n=1}^{s}\sum_{i=s+1}^{N} U(|\mathbf{q}_n - \mathbf{q}_i|). \tag{9.6}$$

Now we can write the time evolution of the s-particle distribution function as

$$\frac{\partial f_s}{\partial t} = \frac{N!}{(N-s)!}\int \prod_{i=s+1}^{N} \mathrm{d}V_i\, \frac{\partial \rho}{\partial t} \tag{9.7}$$

from the Liouville Theorem (see theorem 8.3), we have

$$\frac{\partial f_s}{\partial t} = -\frac{N!}{(N-s)!}\int \prod_{i=s+1}^{N} \mathrm{d}V_i\, \{\rho, \mathcal{H}_s + \mathcal{H}_{N-s} + \mathcal{H}'\}. \tag{9.8}$$

In this equation, we have to compute three Poisson brackets. For the first one, the integral is over the coordinates and momenta of particles $s+1, \ldots, N$, and the Poisson bracket is with respect to the Hamiltonian $\mathcal{H}_s$. Thus, we can change the order of integration and derivatives for Poisson brackets, and we get

$$\{\rho, \mathcal{H}_s\} = \left\{\left(\int \prod_{i=s+1}^{N} \mathrm{d}V_i\, \rho\right), \mathcal{H}_s\right\} = \{\rho_s, \mathcal{H}_s\}. \tag{9.9}$$

For the second term, let's write the Poisson bracket explicitly:

$$\int \prod_{i=s+1}^{N} \mathrm{d}V_i\, \{\rho, \mathcal{H}_{N-s}\} = \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{j=1}^{N}\left[\frac{\partial \rho}{\partial \mathbf{q}_j}\frac{\partial \mathcal{H}_{N-s}}{\partial \mathbf{p}_j} - \frac{\partial \rho}{\partial \mathbf{p}_j}\frac{\partial \mathcal{H}_{N-s}}{\partial \mathbf{q}_j}\right] \tag{9.10}$$

$$= \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{j=s+1}^{N} \underbrace{\frac{\partial \rho}{\partial \mathbf{q}_j} \frac{\mathbf{p}_j}{m}}_{(a)} - \underbrace{\frac{\partial \rho}{\partial \mathbf{p}_j} \left(\frac{\partial V}{\partial \mathbf{q}_j} + \sum_{k=s+1}^{N} \frac{\partial}{\partial \mathbf{q}_j} U(|\mathbf{q}_j - \mathbf{q}_k|) \right)}_{(b)} = 0. \quad (9.11)$$

The last equality is obtained by integrating by parts and using the fact that the probability density is normalized (i.e., at the boundaries of the integration domain, the probability density vanishes). In term (a): $\mathbf{p}_j/m$ is independent of $\mathbf{q}_j$, and in term (b): $\partial \mathcal{H}_{N-s}/\partial \mathbf{q}_j$ is independent of $\mathbf{p}_j$. Finally, for the third term, we have

$$\int \prod_{i=s+1}^{N} \mathrm{d}V_i \, \{\rho, \mathcal{H}'\} = \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{j=1}^{N} \left[\frac{\partial \rho}{\partial \mathbf{q}_j} \frac{\partial \mathcal{H}'}{\partial \mathbf{p}_j} - \frac{\partial \rho}{\partial \mathbf{p}_j} \frac{\partial \mathcal{H}'}{\partial \mathbf{q}_j} \right] \quad (9.12)$$

Since $\mathcal{H}'$ is independent of $\mathbf{p}_j$, the first term vanishes. For the second term, we have

$$= - \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{j=1}^{N} \frac{\partial \rho}{\partial \mathbf{p}_j} \left(\sum_{n=1}^{s} \sum_{k=s+1}^{N} \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_k|)}{\partial \mathbf{q}_j} \right) \quad (9.13)$$

Breaking the sum into two parts, first in which derivatives are taken with respect to $q_n, 1 \leq n \leq s$, and the second in which derivatives are taken with respect to $q_k, s+1 \leq k \leq N$, we get

$$= - \int \prod_{i=s+1}^{N} \mathrm{d}V_i \left[\sum_{n=1}^{s} \frac{\partial \rho}{\partial \mathbf{p}_n} \sum_{k=s+1}^{N} \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_k|)}{\partial \mathbf{q}_n} + \sum_{k=s+1}^{N} \frac{\partial \rho}{\partial \mathbf{p}_k} \sum_{n=1}^{s} \frac{\partial U(|\mathbf{q}_k - \mathbf{q}_n|)}{\partial \mathbf{q}_k} \right] \quad (9.14)$$

The second term vanishes using the same argument as before (since the variables which are getting integrated are the same variables which are differentiated).

$$= - \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{n=1}^{s} \frac{\partial \rho}{\partial \mathbf{p}_n} \sum_{k=s+1}^{N} \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_k|)}{\partial \mathbf{q}_n} \quad (9.15)$$

The terms in the sum (over k) are equal, so we can write

$$= -(N-s) \int \prod_{i=s+1}^{N} \mathrm{d}V_i \sum_{n=1}^{s} \frac{\partial \rho}{\partial \mathbf{p}_n} \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \quad (9.16)$$

After rearranging the terms, we get

$$= -(N-s) \sum_{n=1}^{s} \int \prod_{i=s+1}^{N} \mathrm{d}V_i \, \frac{\partial \rho}{\partial \mathbf{p}_n} \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \quad (9.17)$$

$$= -(N-s) \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \int \prod_{i=s+2}^{N} \mathrm{d}V_i \, \frac{\partial \rho}{\partial \mathbf{p}_n} \quad (9.18)$$

The second integral is over the coordinates and momenta of particles $s+1, \ldots, N$, and the derivative is with respect to the coordinates of particles $1, \ldots, s$. Thus, we can change the order of integration and derivatives

$$= -(N-s) \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \frac{\partial}{\partial \mathbf{p}_n} \left(\int \prod_{i=s+2}^{N} \mathrm{d}V_i \, \rho \right) \quad (9.19)$$

$$= -(N-s) \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \frac{\partial \rho_{s+1}}{\partial \mathbf{p}_n}. \quad (9.20)$$

Substituting the results of all three Poisson brackets back into the equation for the time evolution of the s-particle distribution function, we get

$$\frac{\partial f_s}{\partial t} = - \frac{N!}{(N-s)!} \left[\{\rho_s, \mathcal{H}_s\} - (N-s) \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \frac{\partial \rho_{s+1}}{\partial \mathbf{p}_n} \right] \quad (9.21)$$

Using the equation eqn. (9.3), we can write

$$\frac{\partial f_s}{\partial t} = -\{f_s, \mathcal{H}_s\} + \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \frac{\partial f_{s+1}}{\partial \mathbf{p}_n}. \tag{9.22}$$

Thus, we get the BBGKY hierarchy for the s-particle distribution functions:

$$\frac{\partial f_s}{\partial t} + \{f_s, \mathcal{H}_s\} = \sum_{n=1}^{s} \int \mathrm{d}V_{s+1} \, \frac{\partial U(|\mathbf{q}_n - \mathbf{q}_{s+1}|)}{\partial \mathbf{q}_n} \cdot \frac{\partial f_{s+1}}{\partial \mathbf{p}_n}. \tag{9.23}$$

Let's consider the first equation of the BBGKY hierarchy, i.e., the equation for the one-particle distribution function.

$$\frac{\partial f_1}{\partial t} + \{f_1, \mathcal{H}_1\} = \int \mathrm{d}V_2 \, \frac{\partial U(|\mathbf{q}_1 - \mathbf{q}_2|)}{\partial \mathbf{q}_1} \cdot \frac{\partial f_2}{\partial \mathbf{p}_1}. \tag{9.24}$$

10 Lecture 10: (27/01/2025)

In the previous lecture, we have derived the BBGKY hierarchy.

$$\frac{\partial f_n}{\partial t} = \{\mathcal{H}_n, f_n\} + \sum_{i=1}^{n} \int \mathrm{d}^3\mathbf{p}_{n+1}\, \mathrm{d}^3\mathbf{r}_{n+1}\, \frac{\partial U(|\mathbf{r}_i - \mathbf{r}_{n+1}|)}{\partial \mathbf{r}_i} \cdot \frac{\partial f_{n+1}}{\partial \mathbf{p}_n}, \tag{10.1}$$

We use following notations:

$$\left(\frac{\partial f_n}{\partial t}\right)_{\text{collision}} = \sum_{i=1}^{n} \int \mathrm{d}^3\mathbf{p}_{n+1}\, \mathrm{d}^3\mathbf{r}_{n+1}\, \frac{\partial U(|\mathbf{r}_i - \mathbf{r}_{n+1}|)}{\partial \mathbf{r}_i} \cdot \frac{\partial f_{n+1}}{\partial \mathbf{p}_n},$$

$$\left(\frac{\partial f_n}{\partial t}\right)_{\text{streaming}} = \{\mathcal{H}_n, f_n\}.$$

With the help of the BBGKY hierarchy, we are able to describe the evolution of the n-particle distribution function in terms of the $(n+1)$-particle distribution function. This is a great achievement as initially we had to solve the N-particle distribution function, which is a very difficult task. However, the BBGKY hierarchy is still not closed, as it involves the $(n+1)$-particle distribution function. We need to find a way to close the BBGKY hierarchy or to find a way to express rate of collision in terms of n-particle distribution function only.

§10.1 Boltzmann Equation

For simplicity, we will consider $n = 1$ case. Say we have a system of particles, and we are interested in the evolution of the one-particle distribution function $f_1(\mathbf{r}, \mathbf{p}, t)$. We can write the BBGKY hierarchy for $n = 1$ as

$$\frac{\partial f_1}{\partial t}(\mathbf{r}, \mathbf{p}, t) = \{\mathcal{H}_1, f_1\} + \int \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{r}_2\, \frac{\partial U(|\mathbf{r} - \mathbf{r}_2|)}{\partial \mathbf{r}} \cdot \frac{\partial f_2}{\partial \mathbf{p}}. \tag{10.2}$$

Specifically, we are interested in the collision term. Say two particles collide at point $\mathbf{r}$ at time t, each particle having momentum $\mathbf{p}_1$ and $\mathbf{p}_2$. And after the collision, the particles have momentum $\mathbf{p}_1'$ and $\mathbf{p}_2'$. This whole process can be thought as transition of the two-particle distribution function $f_2(\mathbf{r}, \mathbf{p}_1; \mathbf{r}, \mathbf{p}_2, t)$ to $f_2(\mathbf{r}, \mathbf{p}_1'; \mathbf{r}, \mathbf{p}_2', t)$. The collision term $\left(\frac{\partial f_1}{\partial t}\right)_{\text{col.}}(\mathbf{r}, \mathbf{p}, t)$ can be written as

$$= \int \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{p}_1'\, \mathrm{d}^3\mathbf{p}_2'\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) f_2(\mathbf{r}, \mathbf{p}_1'; \mathbf{r}, \mathbf{p}_2', t) - w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2') f_2(\mathbf{r}, \mathbf{p}_1; \mathbf{r}, \mathbf{p}_2, t), \tag{10.3}$$

where $w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2)$ is the transition rate from state $(\mathbf{p}_1', \mathbf{p}_2')$ to $(\mathbf{p}_1, \mathbf{p}_2)$ (and referred as *scattering function* or *differential cross-section*). And form of this equation is very similar to Master equation in stochastic processes (see eqn. (4.12)).

Till this point, we have not made any assumption about $\mathbf{p}_1'$ and $\mathbf{p}_2'$. Let's make an assumption that the system is *dilute* and *weakly interacting* with *time translational invariance* as well as *spatial translational invariance*. This means that the system is not very dense, and the particles are not interacting very strongly, and linear momentum and energy are conserved in the system. However, there is a problem with this assumption. The problem is the presence of the external potential $U(|\mathbf{r}_1 - \mathbf{r}_2|)$. This potential breaks the translational invariance of the system, for example, the potential can pump energy into the system. To overcome this problem, we will make another assumption that the potential in the system is locally flat.

Since the system is time reversible (at microscopic level), and classical mechanics is parity invariant, we have the following property for the scattering function:

$$w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) = w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2'). \tag{10.4}$$

This property can also be understood from the fact that the system has exchange symmetry.

Remark 10.1 (Exchange Symmetry as a Consequence of Time Reversibility and Parity Invariance). Consider the same scenario as above, where two particles collide at point $\mathbf{r}$ at time t, each particle having momentum $\mathbf{p}_1$ and $\mathbf{p}_2$. And after the collision, the particles have momentum $\mathbf{p}_1'$ and $\mathbf{p}_2'$.

(i) Time Reversibility: If we reverse the time, the particles will collide with momentum $-\mathbf{p}_1'$ and $-\mathbf{p}_2'$ and after the collision, the particles will have momentum $-\mathbf{p}_1$ and $-\mathbf{p}_2$. Thus, the scattering function satisfies

$$w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2') = w(-\mathbf{p}_1', -\mathbf{p}_2' \to -\mathbf{p}_1, -\mathbf{p}_2). \tag{10.5}$$

(ii) Parity Invariance: If we reverse the spatial coordinates, the particles will collide at point $-\mathbf{r}$ at time t, each particle having momentum $-\mathbf{p}_1$ and $-\mathbf{p}_2$. And after the collision, the particles will have momentum $-\mathbf{p}_1'$ and $-\mathbf{p}_2'$. Thus, the scattering function satisfies

$$w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2') = w(-\mathbf{p}_1, -\mathbf{p}_2 \to -\mathbf{p}_1', -\mathbf{p}_2'). \tag{10.6}$$

From the above two properties, combining them, we get

$$w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2') = w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2). \tag{10.7}$$

Still we are not able to close the BBGKY hierarchy. For the solution for this, Boltzmann made an assumption that the two-particle distribution function is factorizable. This means that the two-particle distribution function can be written as a product of two one-particle distribution functions. This assumption is known as *molecular chaos assumption*. With this assumption, the two-particle distribution function can be written as

$$f_2(\mathbf{r}, \mathbf{p}_1; \mathbf{r}, \mathbf{p}_2, t) = f_1(\mathbf{r}, \mathbf{p}_1, t) \cdot f_1(\mathbf{r}, \mathbf{p}_2, t). \tag{10.8}$$

Till this point, the momenta of the particles are correlated by conservation of linear momentum and energy. However, the molecular chaos assumption breaks this correlation. Thus, we have a problem with the molecular chaos assumption. To overcome this problem, Boltzmann explains that gas is dilute, so this particle before another collision will have a lot of time to interact with other particles and thus the correlation will be destroyed.

Another big consequence of the molecular chaos assumption is that we have introduced a notion of arrow of time in the system. This is because the molecular chaos assumption breaks the time-reversibility of the system. This is a very important point, as this is the reason why we have the second law of thermodynamics. Now, we can write the collision term as

$$\left(\frac{\partial f_1}{\partial t}\right)_{\text{col.}}(\mathbf{r}, \mathbf{p}, t) = \int \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{p}_1'\, \mathrm{d}^3\mathbf{p}_2'\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \left[f_1(\mathbf{r}, \mathbf{p}_1', t) f_1(\mathbf{r}, \mathbf{p}_2', t) - f_1(\mathbf{r}, \mathbf{p}_1, t) f_1(\mathbf{r}, \mathbf{p}_2, t)\right]. \tag{10.9}$$

Using this expression, we can write the evolution equation for the one-particle distribution function as

$$\frac{\partial f_1}{\partial t} = \{\mathcal{H}_1, f_1\} + \int \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{p}_1'\, \mathrm{d}^3\mathbf{p}_2'\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \left[f_1(\mathbf{r}, \mathbf{p}_1', t) f_1(\mathbf{r}, \mathbf{p}_2', t) - f_1(\mathbf{r}, \mathbf{p}_1, t) f_1(\mathbf{r}, \mathbf{p}_2, t)\right]. \tag{10.10}$$

This equation is known as the *Boltzmann equation*. For equilibrium, the Boltzmann equation reduces to the Liouville equation. Thus, we have the following condition for equilibrium

$$f_1^{\text{eq}}(\mathbf{r}, \mathbf{p}_1) f_1^{\text{eq}}(\mathbf{r}, \mathbf{p}_2) = f_1^{\text{eq}}(\mathbf{r}, \mathbf{p}_1') f_1^{\text{eq}}(\mathbf{r}, \mathbf{p}_2'). \tag{10.11}$$

This equation is satisfied for the Maxwell-Boltzmann distribution function.

§10.2 Quantum Boltzmann Equation

The Boltzmann equation can be generalized to quantum mechanics. The collision term in the quantum Boltzmann equation is given by

$$\begin{aligned}\left(\frac{\partial f_1}{\partial t}\right)_{\text{col.}} = \int \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{p}_1'\, \mathrm{d}^3\mathbf{p}_2'\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \big[&f_1(\mathbf{p}_1', t) f_1(\mathbf{p}_2', t)\{1 \pm f_1(\mathbf{p}_1, t)\}\{1 \pm f_1(\mathbf{p}_2, t)\} \\ &- f_1(\mathbf{p}_1, t) f_1(\mathbf{p}_2, t)\{1 \pm f_1(\mathbf{p}_1', t)\}\{1 \pm f_1(\mathbf{p}_2', t)\}\big]. \end{aligned} \tag{10.12}$$

Here, the $+$ sign is for bosons and the $-$ sign is for fermions.

Similar to the classical case, we can write the condition for equilibrium as

$$f_1^{\mathrm{eq}}(\mathbf{p}_1')f_1^{\mathrm{eq}}(\mathbf{p}_2')\{1 \pm f_1^{\mathrm{eq}}(\mathbf{p}_1)\}\{1 \pm f_1^{\mathrm{eq}}(\mathbf{p}_2)\} = f_1^{\mathrm{eq}}(\mathbf{p}_1)f_1^{\mathrm{eq}}(\mathbf{p}_2)\{1 \pm f_1^{\mathrm{eq}}(\mathbf{p}_1')\}\{1 \pm f_1^{\mathrm{eq}}(\mathbf{p}_2')\}. \tag{10.13}$$

The distribution function which satisfies this equilibrium condition in general given as the Fermi-Dirac distribution function for fermions and Bose-Einstein distribution function for bosons.

$$f_1^{\mathrm{eq}}(\mathbf{p}) = \frac{1}{\exp\{\beta(E(\mathbf{p}) - \mu + \mathbf{u}\cdot\mathbf{p})\} \pm 1}. \tag{10.14}$$

11 Lecture 11: (29/01/2025)

§11.1 H-Theorem

Theorem 11.1 (H-Theorem)
If $f_1(\mathbf{r}, \mathbf{p}, t)$ satisfies the Boltzmann equation. Define $H(t)$ as

$$H(t) = \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, f_1(\mathbf{r}, \mathbf{p}, t) \ln f_1(\mathbf{r}, \mathbf{p}, t). \tag{11.1}$$

Then $\frac{\mathrm{d}H}{\mathrm{d}t} \leq 0$.

This quantity $H(t)$ has a similar form to the entropy for a probability distribution (*i.e.* $S = -k_B p \ln p$).
Proof:

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \frac{\partial}{\partial t}(f_1 \ln f_1) = \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, (\ln f_1 + 1)\frac{\partial f_1}{\partial t} \tag{11.2}$$

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p} \ln f_1 \frac{\partial f_1}{\partial t} + \frac{\mathrm{d}}{\mathrm{d}t}\int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, f_1 \tag{11.3}$$

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p} \ln f_1 \frac{\partial f_1}{\partial t} + 0 \tag{11.4}$$

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p} \ln f_1 \left[\{\mathcal{H}_1, f_1\} + \left(\frac{\partial f_1}{\partial t}\right)_{\mathrm{col}}\right] \tag{11.5}$$

Using integration by parts, for streaming term, we can show that the integral vanishes.

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p} \ln f_1 \left(\frac{\partial f_1}{\partial t}\right)_{\mathrm{col}} \tag{11.6}$$

Using the Boltzmann equation, (and relabeling the dummy variable $\mathbf{p} \to \mathbf{p}_1$), also suppressing the spatial coordinates $\mathbf{r}$ and the time t,

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \int \mathrm{d}\tau\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \ln f_1(\mathbf{p}_1)\left[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') - f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)\right] \tag{11.7}$$

where $\mathrm{d}\tau = \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}_1\, \mathrm{d}^3\mathbf{p}_2\, \mathrm{d}^3\mathbf{p}_1'\, \mathrm{d}^3\mathbf{p}_2'$. At this stage, all momenta are integrated over so they are really nothing but dummy variables. Let's relabel $1 \leftrightarrow 2$ on the momenta.

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \int \mathrm{d}\tau\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \ln f_1(\mathbf{p}_2)\left[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') - f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)\right] \tag{11.8}$$

Adding eqns. (11.7) and (11.8), we get

$$\frac{\mathrm{d}H}{\mathrm{d}t} = \frac{1}{2}\int \mathrm{d}\tau\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \ln[f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)]\left[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') - f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)\right] \tag{11.9}$$

With similar arguments as before, we now swap the dummy variables $\mathbf{p} \leftrightarrow \mathbf{p}'$. And using the symmetry of the scattering function, we have (see eqn. (10.4), $w(\mathbf{p}_1, \mathbf{p}_2 \to \mathbf{p}_1', \mathbf{p}_2') = w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2)$) we get

$$\frac{\mathrm{d}H}{\mathrm{d}t} = -\frac{1}{2}\int \mathrm{d}\tau\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2) \ln[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2')]\left[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') - f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)\right] \tag{11.10}$$

Adding eqns. (11.9) and (11.10), we get

$$\begin{aligned}\frac{\mathrm{d}H}{\mathrm{d}t} = -\frac{1}{4}\int \mathrm{d}\tau\, w(\mathbf{p}_1', \mathbf{p}_2' \to \mathbf{p}_1, \mathbf{p}_2)&\left[\ln(f_1(\mathbf{p}_1')f_1(\mathbf{p}_2')) - \ln(f_1(\mathbf{p}_1)f_1(\mathbf{p}_2))\right] \\ &\times \left[f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') - f_1(\mathbf{p}_1)f_1(\mathbf{p}_2)\right]\end{aligned} \tag{11.11}$$

Identify $f_1(\mathbf{p}_1')f_1(\mathbf{p}_2') = x$ and $f_1(\mathbf{p}_1)f_1(\mathbf{p}_2) = y$. Then the last line is an integral of function $(\ln x - \ln y)(x - y)$. And we know, this function is always positive. Thus, we have

$$\frac{\mathrm{d}H}{\mathrm{d}t} \leq 0 \tag{11.12}$$

Q.E.D.

We define the entropy as

$$S = -k_B H(t). \tag{11.13}$$

Thus, the H-Theorem states that the entropy of the system always increases or remains constant. And there we see the arrow of time seemingly emerging from time-invariant Hamiltonian mechanics. Everything hinges on the assumption of molecular chaos (see eqn. (10.8)). This was where we broke time-reversal symmetry, ultimately ensuring that entropy increases only in the future.

§11.2 Ensemble Averages

Let's consider a function of canonical variables $A(\mathbf{r}, \mathbf{p}, t)$. We can define the ensemble average of this function as

$$\langle A \rangle (t) = \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho(\mathbf{r}, \mathbf{p}, t) A(\mathbf{r}, \mathbf{p}). \tag{11.14}$$

Where $\rho(\mathbf{r}, \mathbf{p}, t)$ is probability density function. Now, let's consider the time evolution of this ensemble average. We have

$$\frac{\mathrm{d}\langle A \rangle}{\mathrm{d}t} = \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p} \left[\frac{\mathrm{d}\rho}{\mathrm{d}t} A + \rho \frac{\mathrm{d}A}{\mathrm{d}t} \right] \tag{11.15}$$

Using the Liouville equation, we have

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \frac{\mathrm{d}A}{\mathrm{d}t} \tag{11.16}$$

Using property of partial derivative, we can write

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \left[\frac{\partial A}{\partial \mathbf{r}} \cdot \dot{\mathbf{r}} + \frac{\partial A}{\partial \mathbf{p}} \cdot \dot{\mathbf{p}} + \frac{\partial A}{\partial t} \right] \tag{11.17}$$

Using Hamilton's equations, we have

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \left[\frac{\partial A}{\partial \mathbf{r}} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{p}} - \frac{\partial A}{\partial \mathbf{p}} \cdot \frac{\partial \mathcal{H}}{\partial \mathbf{r}} \right] + \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \frac{\partial A}{\partial t} \tag{11.18}$$

$$= \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \{A, \mathcal{H}\} + \int \mathrm{d}^3\mathbf{r}\, \mathrm{d}^3\mathbf{p}\, \rho \frac{\partial A}{\partial t} \tag{11.19}$$

$$= \langle \{A, \mathcal{H}\} \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle \tag{11.20}$$

Thus, we have

$$\frac{\mathrm{d}\langle A \rangle}{\mathrm{d}t} = \langle \{A, \mathcal{H}\} \rangle + \left\langle \frac{\partial A}{\partial t} \right\rangle. \tag{11.21}$$

This is the equation of motion for ensemble averages. This equation is very similar to the Ehrenfest theorem in quantum mechanics.

12 Lecture 12: (03/02/2025)

Definition 12.1 (Statistical Mechanics):
Statistical mechanics is a probabilistic approach to equilibrium macroscopic properties of large number of degree of freedom.

We started by reviewing thermodynamics (*i.e.* studying equilibrium properties of macroscopic systems). Then we developed the notion of macrostates and microstates. The macrostate M depends on a relatively small number of thermodynamic coordinates. To arrive at the same thermodynamic laws, we will examine the dynamics of the many degrees of freedom comprising a macroscopic body.

Description of each microstate μ requires an enormous amount of information and the corresponding time evolution, governed by Hamilton's equations. Rather than studying the evolution of a microstate, statistical mechanics examines an ensemble of microstates corresponding to a given macrostate. It aims to provide the probabilities $p_M(\mu)$ for the equilibrium ensemble. Liouville's theorem justifies the assumption that all accessible microstates are equally likely in an equilibrium ensemble.

§12.1 Microcanonical Ensemble

Our starting point in thermodynamics is a mechanically and adiabatically isolated system *i.e.* a system that does not exchange heat or work with its surroundings. Thus, the total energy E and the generalized coordinates $\mathbf{x}$ are fixed, specifying a macrostate $M \equiv (E, \mathbf{x})$. The set of all microstates μ that correspond to the macrostate M form the *microcanonical ensemble*.

Since the time evolution of the microstates is governed by Hamilton's equations, so if the Hamiltonian H is time-independent, the total energy E is conserved. Thus, all the microstates in this ensemble are confined to the surface of constant energy ($\mathcal{H}(\mu) = E$) in the phase space.

Let us consider that system does not any other conserved quantities, so that all microstates are mutually accessible. The central postulate of statistical mechanics is that the equilibrium probability distribution is given by

$$p_M(\mu) = \begin{cases} \dfrac{1}{\Omega(E, \mathbf{x})} & \text{if } \mathcal{H}(\mu) = E, \\ 0 & \text{otherwise} \end{cases} \tag{12.1}$$

where $\Omega(E)$ is the area of the surface of constant energy E in the phase space. This assumption was introduced by Boltzmann and is known as the *equal a priori probability postulate*. This assignment of probabilities is consistent with Liouville's theorem, as Under a canonical change of variables, $\mu \to \mu'$, volumes in phase space are left invariant (see theorem 8.3 and problem A.2.2). Thus, the transformed probability is given by

$$p_M(\mu') = p_M(\mu)\left|\frac{\partial \mu}{\partial \mu'}\right| = p_M(\mu), \tag{12.2}$$

which is the same as the original probability.

To avoid subtleties, we allow all the microstates lying in the phase space volume $E - \Delta \leq \mathcal{H}(\mu) \leq E + \Delta$ to correspond to the macrostate M, where Δ is uncertainity in the energy. Now, the normalization factor becomes $\Omega' \approx 2\Delta\Omega$. Since Ω typically depends exponentially on E, as long as $\Delta \sim \mathcal{O}(E^0)$ (or even $\mathcal{O}(E^1)$), the difference between the surface and volume of the shell is negligible in the $E \propto N \to \infty$ limit, and we can use Ω and Ω' interchangeably.

The entropy of this uniform probability distribution is given by

$$S(E, \mathbf{x}) = k_B \ln \Omega(E, \mathbf{x}), \tag{12.3}$$

where k_B is the Boltzmann constant, used to make the dimension of entropy consistent with the thermodynamic entropy. For a collection of independent systems, the overall allowed phase space is the product of individual ones, that is, $\Omega_{\text{Total}} = \prod_i \Omega_i$. The resulting entropy is thus additive, as expected for an extensive quantity.

§12.1.1 Deriving Laws of Thermodynamics

§12.1.1.1 Zeroth Law

Consider two microcanonical systems, and allow them to exchange energy, but not work. If the original systems have energies E_1 and E_2 respectively, the total energy is $E = E_1 + E_2$. Assuming that the two systems are weakly interacting, each microstate of the joint system can be written as $\mu = \mu_1 \otimes \mu_2$, where μ_1 and μ_2 are the microstates of the two systems and $\mathcal{H}(\mu_1 \otimes \mu_2) = \mathcal{H}_1(\mu_1) + \mathcal{H}_2(\mu_2)$. In equilibrium, the probability distribution is given by

$$p_E(\mu_1 \otimes \mu_2) = \frac{1}{\Omega(E)} \cdot \begin{cases} 1 & \text{if } \mathcal{H}(\mu_1 \otimes \mu_2) = \mathcal{H}_1(\mu_1) + \mathcal{H}_2(\mu_2) = E, \\ 0 & \text{otherwise} \end{cases}. \tag{12.4}$$

The entropy of the joint system is given by (using the extensivity of entropy along with weak interaction assumption)

$$S(E) = k_B \ln \Omega(E) = k_B \ln \Omega_1(E_1) + k_B \ln \Omega_2(E_2) = S_1(E_1) + S_2(E_2). \tag{12.5}$$

Since only the overall energy is fixed, the total allowed phase space is computed from

$$\Omega(E) = \int \mathrm{d}E_1\, \Omega_1(E_1)\Omega_2(E - E_1) = \int \mathrm{d}E_1 \exp\left[\frac{S_1(E_1)}{k_B}\right] \exp\left[\frac{S_2(E - E_1)}{k_B}\right], \tag{12.6}$$

where we have used the fact that $\Omega_i = \exp[S_i/k_B]$. Extensivity of entropy implies that S_1 and S_2 are proportional to the numbers of particles in each system, making the integrand an exponentially large quantity. Hence, the integral can be equated by the saddle point method to the maximum value of the integrand, obtained for energies E_1^* and $E_2^* = E - E_1^*$, that is,

$$S(E) = k_B \ln \Omega(E) \approx S_1(E_1^*) + S_2(E_2^*). \tag{12.7}$$

To find position of the maximum, we extermize the exponnet in the integrand with respect to E_1, that is,

$$\frac{\partial S(E)}{\partial E_1} = 0 \quad \Rightarrow \quad \frac{\partial S_1(E_1)}{\partial E_1} + \frac{\partial S_2(E - E_1)}{\partial E_1} = 0. \tag{12.8}$$

This gives us

$$\left(\frac{\partial S_1}{\partial E_1}\right)_{\mathbf{x}_1}\bigg|_{E_1 = E_1^*} = \left(\frac{\partial S_2}{\partial E_2}\right)_{\mathbf{x}_2}\bigg|_{E_2 = E_2^*}. \tag{12.9}$$

Observe that all the microstates of the joint system are equally likely, but there is an exponentially large number of microstates in the vaccinity of (E_1^*, E_2^*). This probabilistic arguments provide no information on the dynamics of evolution amongst these microstates, or on the amount of time needed to establish equilibrium.

At the equilibrium point, condition eqn. (12.9) is satisfied. With this condition, we can define a *temperature* for a *system in equilibrium*, as

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{\mathbf{x}}. \tag{12.10}$$

§12.1.1.2 First Law

Let's study the variation of $S(E, \mathbf{x})$ with $\mathbf{x}$, by changing the generalized coordinates reversibly by $\delta\mathbf{x}$. This results in doing work on the system by an amount $đW = \mathbf{J} \cdot \delta\mathbf{x}$, where $\mathbf{J} = \left(\frac{\partial H}{\partial \mathbf{x}}\right)_E$ is the generalized force. And this work changes the internal energy of the system to $E + \mathbf{J} \cdot \delta\mathbf{x}$. The first order change in entropy is given by

$$\delta S = S(E + \mathbf{J} \cdot \delta\mathbf{x},\ \mathbf{x} + \delta\mathbf{x}) - S(E, \mathbf{x}) = \left[\left(\frac{\partial S}{\partial E}\right)_{\mathbf{x}} \mathbf{J} + \left(\frac{\partial S}{\partial \mathbf{x}}\right)_E\right] \cdot \delta\mathbf{x}. \tag{12.11}$$

At equilibrium, the variation of entropy is zero (*i.e.* $\delta S = 0$), which gives us

$$\left(\frac{\partial S}{\partial E}\right)_{\mathbf{x}} \mathbf{J} + \left(\frac{\partial S}{\partial \mathbf{x}}\right)_E = 0 \quad \Rightarrow \quad \left(\frac{\partial S}{\partial \mathbf{x}}\right)_E = -\frac{\mathbf{J}}{T}. \tag{12.12}$$

With this relation, we can write the total change in entropy as

$$\mathrm{d}S = \left(\frac{\partial S}{\partial E}\right)_{\mathbf{x}} \mathrm{d}E + \left(\frac{\partial S}{\partial \mathbf{x}}\right)_{E} \cdot \mathrm{d}\mathbf{x} = \frac{\mathrm{d}E}{T} - \frac{\mathbf{J}\cdot\mathrm{d}\mathbf{x}}{T}. \tag{12.13}$$

After rearranging the terms, we get

$$\mathrm{d}E = T\,\mathrm{d}S + \mathbf{J}\cdot\mathrm{d}\mathbf{x}\,. \tag{12.14}$$

Now, identifying the heat input as $đQ = T\,\mathrm{d}S$, we get the first law of thermodynamics as

$$\mathrm{d}E = đQ + đW. \tag{12.15}$$

§12.1.1.3 Second Law

Our definition of equilibrium is based on the fact that the system has many degrees of freedom $N \gg 1$, which make it exponentially unlikely in N that the combined systems are found with component energies different from (E_1^*, E_2^*). By this argument, the equilibrium point has a larger number of accessible states than any starting point

$$\Omega(E_1^*, \mathbf{x}_1)\Omega(E_2^*, \mathbf{x}_2) \geq \Omega(E_1, \mathbf{x}_1)\Omega(E_2, \mathbf{x}_2). \tag{12.16}$$

This inequality can be written in terms of the entropy as

$$S(E_1^*, \mathbf{x}_1) + S(E_2^*, \mathbf{x}_2) \geq S(E_1, \mathbf{x}_1) + S(E_2, \mathbf{x}_2). \tag{12.17}$$

Thus, the change in entropy of the system evolving to more densely populated microstates is always positive,

$$\delta S = S(E_1^*, \mathbf{x}_1) + S(E_2^*, \mathbf{x}_2) - S(E_1, \mathbf{x}_1) - S(E_2, \mathbf{x}_2) \geq 0. \tag{12.18}$$

When the system is not in equilibrium, the equality in eqn. (12.9) is not satisfied, we get following inequality

$$\delta S = \left[\left(\frac{\partial S_1}{\partial E_1}\right)_{\mathbf{x}_1} - \left(\frac{\partial S_2}{\partial E_2}\right)_{\mathbf{x}_2}\right]\delta E_1 = \left(\frac{1}{T_1} - \frac{1}{T_2}\right)\delta E_1 \geq 0. \tag{12.19}$$

This gives us the Clausius inequality, which states that heat flows from a hot body to a cold body, and not the other way around.

§12.2 Ideal Gas

The ideal gas is a collection of N non-interacting particles, each of mass m, moving in a volume V with total energy E. The Hamiltonian of the system is given by

$$H = \sum_{i=1}^{N} \frac{\mathbf{p}_i^2}{2m} + U(\mathbf{q}_i), \tag{12.20}$$

where $U(\mathbf{q}_i)$ is the potential imposed by the walls of the container. Here the microcanonical ensemble is specified by the total energy E, the volume V, and the number of particles N. The PDF of the microstates is given by

$$p_M(\mu) = \frac{1}{\Omega(E,V,N)} \cdot \begin{cases} 1 & \text{if } \mathbf{q}_i \in V,\ \sum_{i=1}^{N} \mathbf{p}_i^2/2m = E \quad (\pm\Delta_E), \\ 0 & \text{otherwise} \end{cases}. \tag{12.21}$$

The total number of microstates is given by

$$\Omega(E,V,N) = \int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{q}_i\,\mathrm{d}^3\mathbf{p}_i\,. \tag{12.22}$$

Since the potential only constrains the particles to the volume V, thus the integral over the coordinates is simply V^N. Then, the integral over the momenta is basically the volume a hyperspherical shell of radius $R = \sqrt{2mE}$ with thickness $\Delta_R = \sqrt{2m/E}\Delta_E$.

Let $\mathcal{S}_d$ be the solid angle in d dimensions, then the volume $\mathscr{V}_d$ and surface area $\mathscr{A}_d$ of d-dimensional hypersphere of radius R is given by[1]

$$\mathscr{V}_d = \frac{\mathcal{S}_d}{d} R^d, \qquad \mathscr{A}_d = \mathcal{S}_d R^{d-1}. \tag{12.23}$$

So the volume of the shell of thickness Δ is $\mathscr{A}_d \Delta$. Thus, we need to compute the solid angle $\mathcal{S}_d$. Consider a hypersphere in d dimensions, and following integral

$$I_d = \int \prod_{i=1}^{d} \mathrm{d}x_i \exp\left[-\sum_{i=1}^{d} x_i^2\right] \tag{12.24}$$

$$= \left(\int_{-\infty}^{\infty} \mathrm{d}x_1 \exp\left[-x_1^2\right]\right)^d \tag{12.25}$$

$$= \pi^{d/2}. \tag{12.26}$$

Alternatively, we can write the integral in generalized spherical coordinates as

$$I_d = \int_0^\infty \mathrm{d}r\, r^{d-1} \exp\left[-r^2\right] \int \mathrm{d}\Omega_d\,, \tag{12.27}$$

$$= \mathcal{S}_d \int_0^\infty \mathrm{d}r\, r^{d-1} e^{-r^2}, \tag{12.28}$$

subsitiuting $u = r^2$, we get

$$I_d = \frac{1}{2}\mathcal{S}_d \int_0^\infty \mathrm{d}u\, u^{d/2-1} e^{-u} \tag{12.29}$$

$$= \frac{1}{2}\mathcal{S}_d \Gamma\left(\frac{d}{2}\right). \tag{12.30}$$

Thus, we get the solid angle in d dimensions as

$$\mathcal{S}_d = \frac{2\pi^{d/2}}{\Gamma\left(\frac{d}{2}\right)}. \tag{12.31}$$

For our case, $d = 3N$, and the volume of the shell for generalized momenta is

$$\int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{p}_i = \mathcal{S}_{3N}\left(\sqrt{2mE}\right)^{3N-1} \sqrt{\frac{2m}{E}}\Delta_E. \tag{12.32}$$

Thus, the total number of microstates is given by

$$\Omega(E,V,N) = V^N \frac{2\pi^{3N/2}}{\Gamma\left(\frac{3N}{2}\right)} \left(\sqrt{2mE}\right)^{3N-1} \sqrt{\frac{2m}{E}}\Delta_E. \tag{12.33}$$

Cleaning up the expression, we get

$$\Omega(E,V,N) = V^N \frac{(2\pi mE)^{3N/2}}{\Gamma\left(\frac{3N}{2}\right)} \frac{\Delta_E}{E}. \tag{12.34}$$

The entropy of the ideal gas is given by

$$S(E,V,N) = k_B \ln \Omega(E,V,N) = k_B\left[N \ln V + \frac{3N}{2}\ln(2\pi mE) - \ln\Gamma\left(\frac{3N}{2}\right) + \ln\frac{\Delta_E}{E}\right]. \tag{12.35}$$

Using Stirling's approximation for the gamma function ($\ln\Gamma(x) \approx x\ln x - x$), we get

$$S(E,V,N) = Nk_B\left[\ln V + \frac{3}{2}\ln(2\pi mE) - \frac{3}{2}\ln\frac{3N}{2} + \frac{3}{2}\right]. \tag{12.36}$$

Thus, the entropy of the ideal gas is

$$S(E,V,N) = Nk_B \ln\left[V\left(\frac{4\pi emE}{3N}\right)^{3/2}\right]. \tag{12.37}$$

[1]For better understanding, refer to notes The volume and surface area of an n-dimensional hypersphere.

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In the last class we have computed the entropy of ideal gas. Now let's compute the temperature and pressure of the ideal gas.

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_{V,N} \qquad \frac{P}{T} = \left(\frac{\partial S}{\partial V}\right)_{E,N} \tag{13.1}$$

$$= \frac{3}{2}\frac{Nk_B}{E} \qquad = \frac{Nk_B}{V} \tag{13.2}$$

$$\Rightarrow \boxed{E = \frac{3}{2}Nk_BT} \qquad \Rightarrow \boxed{PV = Nk_BT} \tag{13.3}$$

Extensivity of Entropy: We want entropy to scale linearly with the size of the system. Let's scale the system by a factor λ (i.e., $E \to \lambda E$, $V \to \lambda V$, $N \to \lambda N$). Then,

$$S(\lambda E, \lambda V, \lambda N) = \lambda N k_B \ln\left[\lambda V\left(\frac{4\pi emE}{3N}\right)^{3/2}\right] \tag{13.4}$$

$$= \lambda S(E, V, N) + \lambda N k_B \ln \lambda \tag{13.5}$$

Thus, entropy is not scaled linearly with the size of the system. The problem is with our calculation of number of states $\Omega(E, V, N)$, we have not taken into account the fact that the particles are indistinguishable *i.e.* we have overcounted the number of states. We need to divide by the number of ways we can permute the particles in the states (*i.e.* $N!$). Thus, the correct expression for the number of states is

$$\Omega(E, V, N) = \frac{V^N}{N!}\frac{(2\pi mE)^{3N/2}}{\Gamma(3N/2)}\frac{\Delta_E}{E}. \tag{13.6}$$

Now, the correct expression for entropy is

$$S(E, V, N) = Nk_B \ln\left[e\frac{V}{N}\left(\frac{4\pi emE}{3N}\right)^{3/2}\right] \tag{13.7}$$

§13.1 'N' Classical Harmonic Oscillators

Consider a system of N classical harmonic oscillators. The Hamiltonian of total system is

$$\mathcal{H} = \sum_{i=1}^{3N}\frac{p_i^2}{2m} + \frac{1}{2}m\omega^2 r_i^2. \tag{13.8}$$

Thus, our phase space is $6N$-dimensional. Let's calculate the number of states $\Omega(E, V, N)$ for this system.

$$\Omega(E, N) = \int\limits_{E-\Delta_E \le \mathcal{H}(\mathbf{p},\mathbf{r}) \le E+\Delta_E} \prod_{i=1}^{3N}\frac{\mathrm{d}p_i\,\mathrm{d}r_i}{h_\circ}, \tag{13.9}$$

where $h_\circ$ is a constant with dimensions of angular momentum, used to make Ω dimensionless. To calculate the integral, let's do the following change of variables:

$$p_i = \sqrt{2m}\,x_i \qquad i = 1, 2, \ldots, 3N \tag{13.10}$$

$$r_i = \sqrt{\frac{2}{m\omega^2}}\,x_{3N+i} \qquad i = 1, 2, \ldots, 3N \tag{13.11}$$

Now, the number of states is the volume of the hyperspherical shell in the $6N$-dimensional phase space with radius $\sqrt{E}$ and thickness $\Delta_R = \Delta_E/\sqrt{E}$. The number of states is

$$\Omega(E, N) = \frac{1}{h_\circ^{3N}}\left(\sqrt{\frac{2}{m\omega^2}}\right)^{3N}\left(\sqrt{2m}\right)^{3N}\frac{2(\pi E)^{3N}}{(3N-1)!}\frac{\Delta_E}{E}. \tag{13.12}$$

Simplifying, we get

$$\Omega(E,N) = \left(\frac{2\pi E}{h_\circ \omega}\right)^{3N} \frac{2}{(3N-1)!} \frac{\Delta_E}{E}. \tag{13.13}$$

Now, the entropy is

$$S(E,N) = 3Nk_B \ln\left(\frac{2\pi E}{h_\circ \omega}\right) + k_B\left[\ln(2) - \ln(3N-1)! + \ln\frac{\Delta_E}{E}\right]. \tag{13.14}$$

Note, $\ln \Delta_E/E \sim -\ln N$, and using Stirling's approximation, we get

$$-\ln(3N-1)! = -(3N-1)\ln(3N-1) + (3N-1) \tag{13.15}$$

$$= -(3N-1)\left[\ln(3N) + \ln\left(1 - \frac{1}{3N}\right)\right] + (3N-1) \tag{13.16}$$

Using $\ln(1-x) \approx -x$, we get

$$-\ln(3N-1)! \approx -3N\ln(3N) + \ln(3N) + 3N - \frac{1}{3N} \tag{13.17}$$

In the limit $N \to \infty$, we get

$$-\ln(3N-1)! \approx -3N\ln(3N) + \ln(3N) + 3N \tag{13.18}$$

Thus, the entropy is

$$S(E,N) = 3Nk_B\left[\ln\left(\frac{2\pi E}{3Nh_\circ \omega}\right) + 1\right] + k_B\left[\ln(2) + \ln\left(3N\frac{\Delta_E}{E}\right)\right] \tag{13.19}$$

The second term is of order N^0, thus in the thermodynamic limit, the entropy is

$$S(E,N) = 3Nk_B\left[\ln\left(\frac{2\pi E}{3Nh_\circ \omega}\right) + 1\right]. \tag{13.20}$$

Thus, the temperature is

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N = \frac{3Nk_B}{E} \Rightarrow E = 3Nk_BT. \tag{13.21}$$

So the energy per degree of freedom is $1/2\ k_BT$, which is the equipartition theorem.

§13.2 Mixing Entropy and Gibbs Paradox

Consider two distinct gases, contained in two separate containers with volumes V_1 and V_2, and number of particles N_1 and N_2 at same temperature T. The gas 1 has energy E_1 and gas 2 has energy E_2. The partition between them is removed, and they are allowed to expand and occupy the combined volume $V = V_1 + V_2$. The final number of particles in the combined volume is $N = N_1 + N_2$ with energy $E = E_1 + E_2$ at the same temperature T. Since the gases are at same temperature, we have

$$\frac{3}{2}k_BT = \frac{E_1}{N_1} = \frac{E_2}{N_2} = \frac{E_1+E_2}{N_1+N_2}. \tag{13.22}$$

Now, let's calculate the entropy of the system.

$$S_{\text{initial}} = S_1(E_1,V_1,N_1) + S_2(E_2,V_2,N_2) = N_1k_B(\ln V_1 + \sigma_1) + N_2k_B(\ln V_2 + \sigma_2), \tag{13.23}$$

where

$$\sigma_i = \frac{3}{2}\ln\left(\frac{4\pi e m_i}{3} \cdot \frac{E_i}{N_i}\right). \tag{13.24}$$

We can rewrite this as a fucntion of T:

$$\sigma_i(T) = \frac{3}{2}\ln\left(2\pi e m_i k_B T\right). \tag{13.25}$$

Now, the final entropy is

$$S_{\text{final}} = N_1 k_B \ln(V_1 + V_2) + N_2 k_B \ln(V_1 + V_2) + k_B(N_1\sigma_1 + N_2\sigma_2). \tag{13.26}$$

Thus, the change in entropy is (or the mixing entropy)

$$\Delta S = S_{\text{final}} - S_{\text{initial}} = N_1 k_B \ln \frac{V}{V_1} + N_2 k_B \ln \frac{V}{V_2}. \tag{13.27}$$

Generalizing this to n gases, we get

$$\Delta S = -N k_B \sum_{i=1}^{n} \frac{N_i}{N} \ln \frac{V_i}{V} \geq 0, \tag{13.28}$$

where $N = \sum_{i=1}^{n} N_i$ and $V = \sum_{i=1}^{n} V_i$.

Consider the case where in both the containers, we have the same gas with same density $n = N_1/V_1 = N_2/V_2$. Then removing or inserting the partition does not change the state of the system, there should be no entropy of mixing, but our formula eqn. (13.28) gives a non-zero value. This is the Gibbs paradox. Observe after removing and reinserting the partition, the macroscopic system does return to its initial configuration, the actual particles that occupy the two chambers are not the same. But as the particles are by assumption *identical*, these configurations cannot be distinguished.

Therefore, we have over-counted the phase space associated with N identical particles by the number of possible permutations. To correct this, we need to divide by the number of ways we can permute the particles in the states (*i.e.* $N!$). Thus, the correct expression for the number of states is

$$\Omega(E, V, N) = \frac{V^N}{N!} \frac{(2\pi m E)^{3N/2}}{\Gamma(3N/2)} \frac{\Delta_E}{E}. \tag{13.29}$$

Now, the correct expression for entropy is

$$S(E, V, N) = N k_B \ln \left[e \frac{V}{N} \left(\frac{4\pi e m E}{3N} \right)^{3/2} \right] \tag{13.30}$$

Rewriting the entropy in terms of σ,

$$S(E, V, N) = N k_B \left[\ln \frac{eV}{N} + \sigma \right]. \tag{13.31}$$

Now the mixing entropy can be calculated for both the cases,

1. When the gases are distinguishable:
   The number of microstates for final configuration is

$$\Omega_{\text{final}} = \frac{V^{N_1+N_2}}{N_1! N_2!} \cdots \tag{13.32}$$

   Thus, the change in entropy is

$$\begin{aligned} \Delta S = S_{\text{final}} - S_{\text{initial}} &= N_1 k_B \ln \frac{V}{N_1} + N_2 k_B \ln \frac{V}{N_2} - N_1 k_B \ln \frac{V_1}{N_1} - N_2 k_B \ln \frac{V_2}{N_2} && (13.33) \\ &= N_1 k_B \ln \frac{V}{V_1} + N_2 k_B \ln \frac{V}{V_2} && (13.34) \\ &= -N k_B \left[\frac{N_1}{N} \ln \frac{V_1}{V} + \frac{N_2}{N} \ln \frac{V_2}{V} \right] \geq 0 && (13.35) \end{aligned}$$

2. When the gases are indistinguishable, with density $n = N_1/V_1 = N_2/V_2 = N/V$:
   The number of microstates for final configuration is

$$\Omega_{\text{final}} = \frac{V^{N_1+N_2}}{(N_1 + N_2)!} \cdots \tag{13.36}$$

   Thus, the change in entropy is

$$\Delta S = S_{\text{final}} - S_{\text{initial}} = N k_B \ln \frac{V}{N} - N_1 k_B \ln \frac{V_1}{N_1} - N_2 k_B \ln \frac{V_2}{N_2} = 0 \tag{13.37}$$

14 Lecture 14: (06/02/2025)

§14.1 Ultra-Relativistic Gas (Non-Interacting)

Consider N indistinguishable massless particles in a volume V, moving at the speed of light. Let E be the total energy of the system. The Hamiltonian is given by

$$\mathcal{H} = \sum_{i=1}^{N} |\mathbf{p}_i| c + U(\mathbf{q}_i), \tag{14.1}$$

where $U(\mathbf{q}_i)$ is the potential due to the walls of the container. The number of states in the phase space is given by

$$\Omega(E, V, N) = \frac{1}{N!} \frac{1}{h^{3N}} \int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{q}_i \, \mathrm{d}^3\mathbf{p}_i \,. \tag{14.2}$$

Since the energy of the system is independent of the position of the particles, the integral over the position variables can be done separately.

$$\Omega(E, V, N) = \frac{V^N}{N!} \frac{1}{h^{3N}} \int\limits_{\mathcal{H}=E} \prod_{i=1}^{N} \mathrm{d}^3\mathbf{p}_i \,. \tag{14.3}$$

Converting each momentum integral in spherical coordinates, we get

$$\Omega(E, V, N) = \frac{V^N}{N!} \frac{1}{h^{3N}} \int\limits_{\mathcal{H}=E} \prod_{i=1}^{N} 4\pi p_i^2 \, \mathrm{d}p_i \tag{14.4}$$

The integral over the angles gives 4π. Since limits of the integral are evident, we can write

$$= \left(\frac{4\pi V}{h^3}\right)^N \frac{1}{N!} \underbrace{\int \prod_{i=1}^{N} p_i^2 \, \mathrm{d}p_i}_{I_N(E)} \,. \tag{14.5}$$

The integral $I_N(E)$ need to be evaluated keep in mind the constraint $\mathcal{H} = E$. Observe that each momentum can maximally be E/c. Therefore, we guess that the integral is of the form

$$I_N(E) = A_N \left(\frac{E}{c}\right)^{3N} . \tag{14.6}$$

Let us fix the momentum of the first particle to be p_1. The remaining $N-1$ particles have a total energy of $E - p_1 c$. Therefore, the integral for $N-1$ particles is

$$I_{N-1}(E - p_1 c) = A_{N-1} \left(\frac{E - p_1 c}{c}\right)^{3(N-1)} . \tag{14.7}$$

The integral for N particles is

$$I_N(E) = \int_0^{E/c} p_1^2 \, \mathrm{d}p_1 \, I_{N-1}(E - p_1 c). \tag{14.8}$$

Substituting the expression for $I_{N-1}(E - p_1 c)$ and $I_N(E)$ in the above equation, we get

$$A_N = \left(\frac{c}{E}\right)^{3N} A_{N-1} \int_0^{E/c} p_1^2 \left(\frac{E}{c} - p_1\right)^{3(N-1)} \mathrm{d}p_1 \,. \tag{14.9}$$

Substitute $x = p_1/(E/c)$ to get

$$A_N = A_{N-1} \int_0^1 x^2 (1-x)^{3(N-1)} \, \mathrm{d}x \,. \tag{14.10}$$

Make the substitution $x = 1 - x$ to get

$$A_N = A_{N-1} \int_0^1 (1-x)^2 x^{3(N-1)} \, \mathrm{d}x \,. \tag{14.11}$$

The integral evaluates to

$$A_N = \frac{2}{(3N)(3N-1)(3N-2)} A_{N-1}. \tag{14.12}$$

Solvin the recursion relation, we get

$$A_N = \frac{2^N}{(3N)!} A_0. \tag{14.13}$$

Setting $N = 1$, we get $A_1 = 1/3$ thus, $A_0 = 1$ and $A_N = \frac{2^N}{(3N)!}$. Therefore, the number of states is

$$\Omega(E, V, N) = \frac{1}{N!(3N)!} \left(\frac{8\pi V E^3}{h^3 c^3} \right)^N. \tag{14.14}$$

The entropy is given by

$$S = k_B \ln \Omega(E, V, N) = N k_B \ln \left(\frac{8\pi V E^3}{h^3 c^3} \right) - k_B [\ln N! + \ln(3N)!]. \tag{14.15}$$

Using Stirling's approximation, we simplify the second term

$$\begin{aligned} \ln N! + \ln(3N)! &= N \ln N - N + (3N) \ln(3N) - 3N \\ &= N \ln N + 3N \ln 3N - 4N \\ &= N \ln N + 3N \ln N + 3N \ln 3 - 4N \\ &= 4N \ln N + 3N \ln 3 - 4N \\ &= N \ln \left(\frac{27 N^4}{e^4} \right). \end{aligned}$$

Therefore, the entropy is

$$S(E, V, N) = N k_B \ln \left(\frac{8\pi e^4}{27 h^3 c^3} \cdot \frac{V E^3}{N^4} \right). \tag{14.16}$$

Thus, the temperature is

$$\frac{1}{T} = \left(\frac{\partial S}{\partial E} \right)_{V,N} = \frac{3 N k_B}{E} \quad \Rightarrow \quad E = 3 N k_B T. \tag{14.17}$$

The pressure is

$$\frac{P}{T} = \left(\frac{\partial S}{\partial V} \right)_{E,N} = \frac{N k_B}{V} \quad \Rightarrow \quad PV = N k_B T. \tag{14.18}$$

Heat capacity at constant volume C_V is

$$C_V = \left(\frac{\partial E}{\partial T} \right)_{V,N} = 3 N k_B. \tag{14.19}$$

For heat capacity at constant pressure C_P, we need to express entropy as $S = S(N, T, P)$, so we have

$$S = S(N, T, P) = N k_B \ln \left(\frac{8\pi e^4 k_B^4}{h^3 c^3} \cdot \frac{T^4}{P} \right). \tag{14.20}$$

Thus, the heat capacity at constant pressure is

$$C_P = T \left(\frac{\partial S}{\partial T} \right)_{P,N} = 4 N k_B. \tag{14.21}$$

Thus, the adiaiabatic index $\gamma = C_P / C_V = 4/3$.

15 Lecture 15: (10/02/2025)

§15.1 Two Level System

Consider N impurities in a crystal lattice. Each impurity can be in one of two states, with energies 0 and ϵ. Define occupation numbers n_i as

$$n_i = \begin{cases} 1 & \text{if impurity } i \text{ is in the excited state} \\ 0 & \text{if impurity } i \text{ is in the ground state} \end{cases}. \tag{15.1}$$

So $\{n_i\}_{i=1}^N$ specifies the microstate of the system. The total energy of the system is

$$\mathcal{H}(\{n_i\}) = \sum_{i=1}^N \epsilon n_i = \epsilon N_1, \tag{15.2}$$

where $N_1 = \sum_{i=1}^N n_i$ is the number of impurities in the excited state. The macrostate of the system is specified by E and N_1. The number of microstates $\Omega(E, N)$ is the number of ways to distribute N_1 excitations among N impurities. This is given by

$$\Omega(E, N) = \binom{N}{N_1} = \frac{N!}{N_1!(N - N_1)!}. \tag{15.3}$$

The entropy of the system is (using Stirling's approximation)

$$S(E, N) = k_B \ln \Omega(E, N) = k_B[N \ln N - N_1 \ln N_1 - (N - N_1) \ln(N - N_1)]. \tag{15.4}$$

Rewrite first term as $N \ln N = [N_1 + (N - N_1)] \ln N = N_1 \ln N + (N - N_1) \ln N$. Then

$$S(E, N) = -Nk_B \left[\frac{N_1}{N} \ln \frac{N_1}{N} + \frac{N - N_1}{N} \ln \frac{N - N_1}{N} \right]. \tag{15.5}$$

Now substitute $N_1 = E/\epsilon$, then the entropy becomes

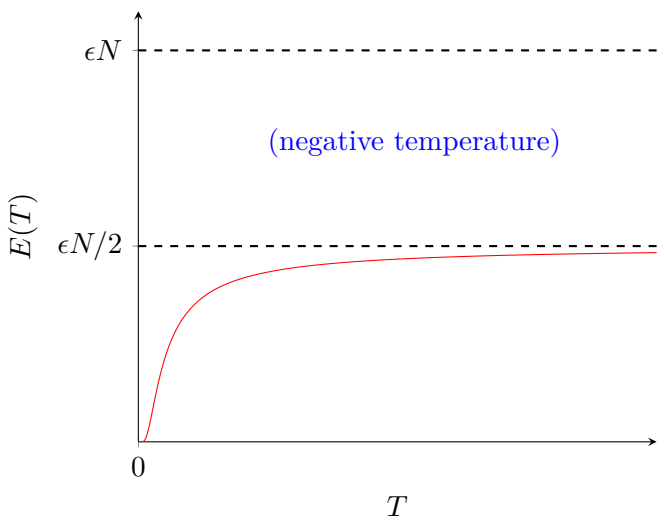
$$S(E, N) = -Nk_B \left[\frac{E}{N\epsilon} \ln \frac{E}{N\epsilon} + \left(1 - \frac{E}{N\epsilon}\right) \ln \left(1 - \frac{E}{N\epsilon}\right) \right]. \tag{15.6}$$

Let's calculate the temperature of the system. The temperature is given by

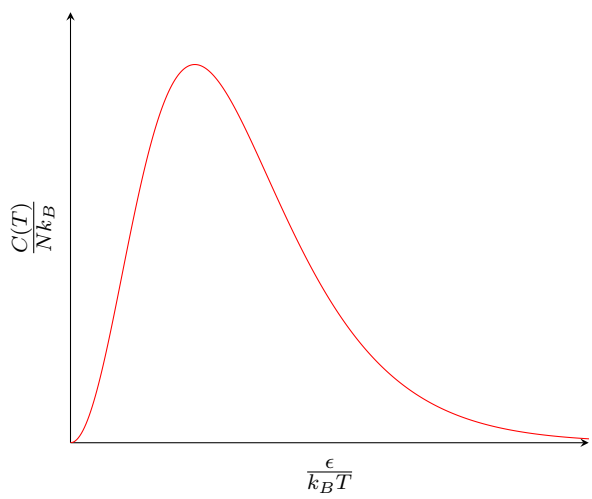
$$\frac{1}{T} = \left(\frac{\partial S}{\partial E}\right)_N = -\frac{k_B}{\epsilon} \ln \left(\frac{E}{N\epsilon - E}\right), \tag{15.7}$$

which gives

$$E(T) = \frac{N\epsilon}{1 + \exp(\epsilon/k_B T)}. \tag{15.8}$$



(a) Energy as a function of temperature



(b) Heat capacity as a function of inverse temperature

Negative Temperature: From eqn. (15.7), we see that the internal energy of the system is a monotonically increasing function of temperature (see fig. 15.1a). Going from minimum of $E = 0$ at $T = 0$ to a maximum of $E = N\epsilon$ at $T = \infty$. But if we start with $E > N\epsilon/2$, from eqn. (15.7), we see that the temperature of the system is negative. The origin of the negative temperature is the decrease in the number of microstates with increasing energy, the opposite of what happens in most systems.

Two-level systems have an upper bound on their energy, and very few microstates close to this maximal energy. Hence, increased energy leads to more order in the system. However, once a negative temperature system is brought into contact with the rest of the Universe (or any portion of it without an upper bound in energy), it loses its excess energy and comes to equilibrium at a positive temperature. The world of negative temperatures is quite unusual in that systems can be cooled by adding heat, and heated by removing it. There are physical examples of systems temporarily prepared at a metastable equilibrium of negative temperature in lasers, and for magnetic spins.

Heat Capacity: The heat capacity is given by

$$C(T) = \left(\frac{\partial E}{\partial T}\right)_N = Nk_B\left(\frac{\epsilon}{k_BT}\right)^2 \exp\left(\frac{\epsilon}{k_BT}\right)\left(1+\exp\left(\frac{\epsilon}{k_BT}\right)\right)^{-2}. \tag{15.9}$$

The vanishing of C as $\exp(-\epsilon/k_BT)$ at low temperatures is characteristic of all systems with an energy gap separating the ground state and lowest excited states. The vanishing of C at high temperatures is a saturation effect, common to systems with a maximum in the number of states as a function of energy. The vanishing of C at high temperatures is a *saturation* effect, common to systems with a maximum in the number of states as a function of energy.

Marginal Probability: The marginal probability of an impurity at site i being in the state n_i is given by

$$P(n_i) = \sum_{j\neq i} p(\{n_j\}) = \frac{\Omega(E-\epsilon n_i, N-1)}{\Omega(E,N)}. \tag{15.10}$$

Observing that $P(n_i = 1) = 1 - P(n_i = 0)$, so it is sufficient to calculate $P(n_i = 0)$. We have

$$P(n_i = 0) = \frac{\Omega(E, N-1)}{\Omega(E,N)} = \frac{N-N_1}{N} = 1 - \frac{E}{N\epsilon}. \tag{15.11}$$

Using eqn. (15.8), we can write

$$P(n_i = 0) = \frac{1}{1+\exp(-\epsilon/k_BT)} \qquad P(n_i = 1) = \frac{\exp(-\epsilon/k_BT)}{1+\exp(-\epsilon/k_BT)}. \tag{15.12}$$

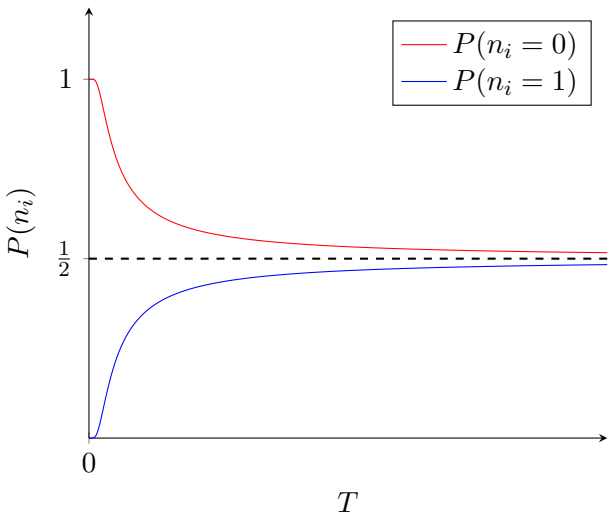


Figure 15.2: Marginal probability of an impurity being in the ground and excited states

§15.2 N-Hard Core Particles

Consider N hard core particles (*i.e.* no two particles can occupy the same site) on a lattice with L sites. The pair (L, N) specifies the macrostate of the system. The number of microstates $\Omega(L, N)$ is the number of ways to distribute N particles among L sites such that no two particles occupy the same site. This is given by

$$\Omega(L, N) = \binom{L}{N} = \frac{L!}{N!(L-N)!}. \tag{15.13}$$

Define the occupation number n_i as

$$n_i = \begin{cases} 0 & \text{if site } i \text{ is empty} \\ 1 & \text{if site } i \text{ is occupied} \end{cases}. \tag{15.14}$$

With this, we can define the microstate of the system as $\{n_i\}_{i=1}^{L}$. Define the denisty of particles as $\rho := N/L$. The (dimensionless-)entropy of the system is

$$S(L, N) = \ln \Omega(L, N) = L \ln L - N \ln N - (L - N) \ln(L - N)$$
$$\boxed{S(L, N) = -L[\rho \ln \rho + (1 - \rho) \ln(1 - \rho)]}. \tag{15.15}$$

Thus, we can find entropy per lattice site as

$$\frac{S}{L} := s(\rho) = -\rho \ln \rho - (1 - \rho) \ln(1 - \rho). \tag{15.16}$$

Let's calculate the probability that site i and j are both occupied ($i \neq j$). This is given by

$$P(i, j) = \frac{\Omega(L-2, N-2)}{\Omega(L, N)} = \frac{N(N-1)}{L(L-1)} = \rho^2 \frac{(1 - 1/N)}{(1 - 1/L)}. \tag{15.17}$$

Since $N < L$, we have $1 - 1/N < 1 - 1/L$, so $P(i, j) < \rho^2$. Individually, the probability that site i is occupied is ρ. Thus, we can find the covariance as

$$C(i, j) = P(i, j) - P(i)P(j) < \rho^2 - \rho^2 = 0. \tag{15.18}$$

Thus, the particles are anti-correlated *i.e.* if one site is occupied, the probability of another site being occupied decreases.

16 Lecture 16: (12/02/2025)

§16.1 Hard Sphere Gas

Consider N indistinguishable non-interacting particles in a volume V with each particle having a volume ω. The total energy of the system is E. Thus, (E, V, N) specifies the macrostate of the system. Let's calculate the number of microstates $\Omega(E, V, N)$ corresponding to this macrostate.

$$\Omega(E,V,N) = \frac{1}{N!(h)^{3N}} \int\limits_{E-\Delta_E/2 \leq \mathcal{H} \leq E+\Delta_E/2} \prod_{i=1}^{N} \mathrm{d}^3\mathbf{q}_i \, \mathrm{d}^3\mathbf{p}_i \,, \tag{16.1}$$

since momenta and positions are independent in this case, we can use previous results to write the momentum integral as

$$\int\limits_{E-\Delta_E/2 \leq \mathcal{H} \leq E+\Delta_E/2} \prod_{i=1}^{N} \mathrm{d}^3\mathbf{p}_i = \frac{(2\pi m E)^{3N/2}}{\left(\frac{3N}{2}-1\right)!} \frac{\Delta_E}{E}. \tag{16.2}$$

Observe that the accessible volume for j^{th} particle is $V - \omega(j-1)$. Thus, the position integral can be written as

$$\int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{q}_i = \prod_{j=0}^{N-1} (V - j\omega). \tag{16.3}$$

We can simplify the position integral as,

$$\begin{aligned}
\prod_{j=0}^{N-1} (V - j\omega) &= V[(V-\omega)(V-(N-1)\omega)][(V-2\omega)(V-(N-2)\omega)] \cdots (\sim N/2 \text{ terms}) \\
&= V^N \left[\left(1 - \frac{\omega}{V}\right)\left(1 - \frac{(N-1)\omega}{V}\right)\right]\left[\left(1 - \frac{2\omega}{V}\right)\left(1 - \frac{(N-2)\omega}{V}\right)\right] \cdots \\
&= V^N \left[1 - \frac{N\omega}{V} + \frac{(N-1)\omega^2}{V^2}\right]\left[1 - \frac{N\omega}{V} + \frac{2(N-2)\omega^2}{V^2}\right] \cdots
\end{aligned}$$

since $\omega \lll V$ we can neglect the second order terms in the above expression,

$$\begin{aligned}
&\approx V^N \left[1 - \frac{N\omega}{V}\right]\left[1 - \frac{N\omega}{V}\right] \cdots (\sim N/2 \text{ terms}) \\
&\prod_{j=0}^{N-1} (V - j\omega) \approx V^N \left[1 - \frac{N\omega}{V}\right]^{N/2} \approx V^N \left[1 - \frac{N\omega}{2V}\right]^{N}.
\end{aligned} \tag{16.4}$$

Substituting the above results in $\Omega(E, V, N)$ we get,

$$\Omega(E,V,N) = \left(V - \frac{N\omega}{2}\right)^N \frac{1}{N!(\frac{3N}{2}-1)!} \left(\frac{2\pi m E}{h^2}\right)^{3N/2} \frac{\Delta_E}{E}. \tag{16.5}$$

The entropy of the system is given by,

$$S(E,V,N) = k_B \ln \Omega(E,V,N) = N k_B \ln \left[\left(V - \frac{N\omega}{2}\right)\left(\frac{2\pi m E}{h^2}\right)^{3/2}\right] - k_B \ln \left[N!\left(\frac{3N}{2}-1\right)!\right] + k_B \ln \frac{\Delta_E}{E}. \tag{16.6}$$

Let's expand the second term using Stirling's approximation,

$$\begin{aligned}
\ln \left[N!\left(\frac{3N}{2}-1\right)!\right] &= N \ln N - N + \left(\frac{3N}{2}-1\right) \ln \left(\frac{3N}{2}-1\right) - \left(\frac{3N}{2}-1\right) \\
&= N \ln N + \left(\frac{3N}{2}-1\right) \ln \left[\left(\frac{3N}{2}\right)\left(1 - \frac{2}{3N}\right)\right] - \frac{5N}{2} + 1 \\
&= N \ln N + \left(\frac{3N}{2}-1\right) \left[\ln \left(\frac{3N}{2}\right) + \ln \left(1 - \frac{2}{3N}\right)\right] - \frac{5N}{2} + 1
\end{aligned}$$

using $\ln(1-x) \approx -x$, we get

$$\approx N\ln N + \frac{3N}{2}\ln\frac{3N}{2} - \ln\frac{3N}{2} + \frac{2}{3N} - \frac{5N}{2}$$

since $N \gg 1$, we can neglect the terms of order $1/N$,

$$\approx N\ln N + \frac{3N}{2}\ln\frac{3N}{2} - \ln\frac{3N}{2} - \frac{5N}{2}. \tag{16.7}$$

Substituting the above result in the expression for entropy, we get,

$$S(E,V,N) = Nk_B\left\{\ln\left[\left(\frac{V}{N}-\frac{\omega}{2}\right)\left(\frac{4\pi mE}{3Nh^2}\right)^{3/2}\right] + \frac{5}{2}\right\} + k_B\ln\left(\frac{3N}{2}\frac{\Delta_E}{E}\right). \tag{16.8}$$

Recall $\ln(\Delta_E/E) \sim -\ln N$, thus the second term in the above expression is of order N^0 so in the thermodynamic limit it is negligible. The entropy of the system in the thermodynamic limit is given by,

$$S(E,V,N) = Nk_B\left\{\ln\left[\left(\frac{V}{N}-\frac{\omega}{2}\right)\left(\frac{4\pi mE}{3Nh^2}\right)^{3/2}\right] + \frac{5}{2}\right\}. \tag{16.9}$$

Let's compute the temperature and pressure of the system,

$$\begin{aligned} \frac{1}{T} &= \left(\frac{\partial S}{\partial E}\right)_{V,N} & \frac{P}{T} &= \left(\frac{\partial S}{\partial V}\right)_{E,N} \\ &= \frac{3Nk_B}{2E} & &= \frac{Nk_B}{V-N\omega/2} \\ \Rightarrow &\boxed{E = \frac{3}{2}Nk_BT} & &\boxed{P\left(V-\frac{N\omega}{2}\right) = Nk_BT}. \end{aligned} \tag{16.10}$$

§16.2 Cauchy Distribution

Till now, we have been dealing with distribution functions which have well-defined mean and variance. However, there are many physical systems where the distribution function does not have a well-defined mean and variance. One such distribution is the Cauchy distribution. The probability density function of the Cauchy distribution is given by,

$$p(x) = \frac{1}{\pi}\frac{1}{1+x^2}. \tag{16.11}$$

Let's calculate the moment generating function (MGF) of the Cauchy distribution,

$$M(t) = \left\langle e^{tx}\right\rangle = \frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{d}x\,\frac{e^{tx}}{1+x^2}. \tag{16.12}$$

The above integral is not convergent, thus the MGF does not exist for the Cauchy distribution. Let's try to calculate characteristic function (CF) (Fourier transform of the probability density function) of the Cauchy distribution,

$$\tilde{p}(k) = \left\langle e^{\mathrm{i}kx}\right\rangle = \frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{d}x\,\frac{e^{\mathrm{i}kx}}{1+x^2}. \tag{16.13}$$

To calculate the above integral, we will use Jordan's lemma. The integrand has poles at $x = \pm\mathrm{i}$, thus the integral can be written as,

$$\tilde{p}(k) = \frac{1}{\pi}\int_{-\infty}^{\infty}\mathrm{d}x\,\frac{e^{\mathrm{i}kx}}{1+x^2} = \begin{cases} \frac{1}{\pi}\left(2\pi\mathrm{i}\,\mathrm{Res}\left(e^{\mathrm{i}kx}/(1+x^2),\mathrm{i}\right)\right) & \text{for } k>0 \\ \frac{1}{\pi}\left(2\pi\mathrm{i}\,\mathrm{Res}(1/(1+x^2),\mathrm{i})\right) & \text{for } k=0 \\ -\frac{1}{\pi}\left(2\pi\mathrm{i}\,\mathrm{Res}\left(e^{\mathrm{i}kx}/(1+x^2),-\mathrm{i}\right)\right) & \text{for } k<0 \end{cases}. \tag{16.14}$$

Thus, the characteristic function of the Cauchy distribution is given by,

$$\tilde{p}(k) = e^{-|k|}. \tag{16.15}$$

The characteristic function of the Cauchy distribution is well-defined, but not differentiable at $k=0$. Thus, the moments of the Cauchy distribution do not exist. The Cauchy distribution is an example of a distribution which does not have a well-defined mean and variance.

§16.3 Example: Saddle Point Approximation

Consider the integral,

$$I(\lambda) = \int_0^{\pi/2} \frac{e^{\lambda \cos x}}{4 + x^2} \, \mathrm{d}x \,, \tag{16.16}$$

where $\lambda > 0$. Let's calculate the leading term of the asymptotics of the given integral for $\lambda \gg 1$.

Let's start by rewriting the integrand as mentioned in theorem 7.1,

$$I(\lambda) = \int_0^{\pi/2} \exp\left(\lambda \left[\cos x - \frac{\ln(x^2+4)}{\lambda}\right]\right) \mathrm{d}x \,. \tag{16.17}$$

The function $\cos x - \ln(x^2+4)/\lambda$ has a global maximum at $x = 0$. Let's compute the second derivative of the function at $x = 0$,

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\cos x - \frac{\ln(x^2+4)}{\lambda}\right)\bigg|_{x=0} = \left[-\cos x - \frac{8-2x^2}{\lambda(x^2+4)^2}\right.\bigg|_{x=0} = -1 - \frac{1}{2\lambda}. \tag{16.18}$$

Thus, the integral $I(\lambda)$ can be approximated as,

$$I(\lambda) \approx \int_0^{\pi/2} \exp\left(\lambda\left[\left(1 - \frac{\ln 4}{\lambda}\right) - \frac{1}{2}\left(1 + \frac{1}{2\lambda}\right)x^2\right]\right) \mathrm{d}x = \frac{e^\lambda}{4} \int_0^{\pi/2} \exp\left(-\frac{\lambda}{2}\left(1 + \frac{1}{2\lambda}\right)x^2\right) \mathrm{d}x \,. \tag{16.19}$$

Since the integrand is Gaussian, the maximum contribution to the integral comes from the region around $x = 0$. Thus, the integral can be approximated as,

$$I(\lambda) \approx \frac{e^\lambda}{4} \int_0^{\infty} \exp\left(-\frac{\lambda}{2}\left(1 + \frac{1}{2\lambda}\right)x^2\right) \mathrm{d}x = \frac{e^\lambda}{8}\sqrt{\frac{4\pi}{2\lambda+1}}. \tag{16.20}$$

Thus, the leading term of the asymptotics of the integral $I(\lambda)$ for $\lambda \gg 1$ is given by,

$$I(\lambda) \approx \sqrt{\frac{\pi}{32\lambda}} e^\lambda. \tag{16.21}$$

17 Lecture 17: (17/02/2025)

In the microcanonical ensemble, the energy E of a large macroscopic system is precisely specified, and its equilibrium temperature T emerges as a consequence (Eqn. (12.10)). However, from a thermodynamic perspective, E and T are both functions of state and on the same footing. It is possible to construct a statistical mechanical formulation in which the temperature of the system is specified, and its internal energy is then deduced. This is the *canonical ensemble*.

§17.1 Canonical Ensemble

In the canonical ensemble, the ensemble is specified by the macrostate $M \equiv (T, \mathbf{x})$. This ensemble allows the input of heat into the system, but no external work. The system S is maintained at a constant temperature T through contact with a reservoir R.

> Remark 17.1. The *reservoir* is another macroscopic system that is sufficiently large so that its temperature is not changed due to interactions with system.

Here, the combined system $S \oplus R$ belongs to the microcanonical ensemble of energy $E_{\text{tot}} \gg E_S$, where E_S is the energy of the system S. So we write the joint probability distribution of microstate $(\mu_S \otimes \mu_R)$ as

$$p(\mu_S \otimes \mu_R) = \frac{1}{\Omega_{S\oplus R}(E_{\text{tot}})} \cdot \delta(E_{\text{tot}} - \mathcal{H}_S(\mu_S) - \mathcal{H}_R(\mu_R)). \tag{17.1}$$

Let's find the probability distribution of the system S being in a microstate μ_S at $(T, \mathbf{x})$. This is given by

$$p(\mu_S) = \int \mathrm{d}\mu_R \;\; p(\mu_S \otimes \mu_R) = \frac{1}{\Omega_{S\oplus R}(E)} \int \mathrm{d}\mu_R \, \delta(E_{\text{tot}} - \mathcal{H}_S(\mu_S) - \mathcal{H}_R(\mu_R)). \tag{17.2}$$

Once μ_S is fixed, the integral is restricted to microstates μ_R such that the reservoir has energy $E_R = E_{\text{tot}} - \mathcal{H}_S(\mu_S)$. Thus, the integral is equal to the number of microstates of the reservoir at energy E_R, which is $\Omega_R(E_R)$. Therefore, we have

$$p(\mu_S) = \frac{\Omega_R(E_{\text{tot}} - \mathcal{H}_S(\mu_S))}{\Omega_{S\oplus R}(E_{\text{tot}})}. \tag{17.3}$$

And using Boltzmann's formula, we can write

$$p(\mu_S) = \frac{\Omega_R(E_{\text{tot}} - \mathcal{H}_S(\mu_S))}{\Omega_{S\oplus R}(E_{\text{tot}})} \propto \exp\left[\frac{1}{k_B} S_R(E_{\text{tot}} - \mathcal{H}_S(\mu_S))\right], \tag{17.4}$$

since the energy of the combined system is fixed $\Omega_{S\oplus R}(E_{\text{tot}})$ is constant. And S_R is the entropy of the reservoir. Using the assumption that the reservoir is large and $E_{\text{tot}} \gg \mathcal{H}_S(\mu_S)$, we can Taylor expand the entropy of the reservoir to get

$$S_R(E_{\text{tot}} - \mathcal{H}_S(\mu_S)) = S_R(E_{\text{tot}}) - \mathcal{H}_S(\mu_S)\left[\frac{\partial S_R}{\partial E_R}\bigg|_{E_R = E_{\text{tot}}} + \mathcal{O}\left(\mathcal{H}_S^2\right)\right]. \tag{17.5}$$

The first term is the entropy of the reservoir at energy E_{tot}, which is a constant. The second term is the inverse temperature of the reservoir, which is the temperature of the system T. Therefore, we have

$$p(\mu_S) \propto \exp\left(-\frac{1}{k_B T}\mathcal{H}_S(\mu_S)\right) = \exp(-\beta \mathcal{H}_S(\mu_S)), \tag{17.6}$$

where $\beta = 1/k_B T$. At this point, we can drop the subscript S from the above equation, so the normalized probability distribution of the system being in a microstate μ at temperature T is

$$p_{(T,\mathbf{x})}(\mu) = \frac{e^{-\beta \mathcal{H}(\mu)}}{Z(T, \mathbf{x})}, \tag{17.7}$$

where $Z(T, \mathbf{x})$ is the *partition function* of the system, given by

$$Z(T, \mathbf{x}) = \int \mathrm{d}\mu \, e^{-\beta \mathcal{H}(\mu)}, \tag{17.8}$$

here the integral is over all microstates of the system. The partition function is a function of the temperature T and the generalized displacements $\mathbf{x}$ of the system.

Now we ask: is energy E of the system well-defined in the canonical ensemble? Unlike the microcanonical ensemble, the energy of the system is a random variable in the canonical ensemble. Since the energy of the system is a random variable, we can calculate what is the probability distribution of the energy of the system. The probability of the system with energy $\mathcal{E}$ is given by

$$p(\mathcal{E}) = \int \mathrm{d}\mu \, \delta(\mathcal{E} - \mathcal{H}(\mu)) p(\mu) = \frac{e^{-\beta \mathcal{E}}}{Z} \int \mathrm{d}\mu \, \delta(\mathcal{E} - \mathcal{H}(\mu)). \tag{17.9}$$

The integral is equal to the number of microstates of where the energy of the system is $\mathcal{E}$, which is $\Omega(\mathcal{E})$. Therefore, we have

$$p(\mathcal{E}) = \frac{e^{-\beta \mathcal{E}}}{Z} \Omega(\mathcal{E}). \tag{17.10}$$

Using normalizability of the probability distribution, we have

$$\int \mathrm{d}\mathcal{E} \;\; p(\mathcal{E}) = 1 \quad \Rightarrow \quad \frac{1}{Z} \int \mathrm{d}\mathcal{E} \;\; \Omega(\mathcal{E}) e^{-\beta \mathcal{E}} = 1. \tag{17.11}$$

The above equation provides another way to calculate the partition function of the system.

$$Z = \int \mathrm{d}\mathcal{E} \, e^{-\beta \mathcal{E}} \Omega(\mathcal{E}). \tag{17.12}$$

If we look closely at the above equation, we can see that the partition function is the Laplace transform of the density of states of the system. The partition function is a generating function for the density of states of the system.

Now we can calculate the average energy of the system in the canonical ensemble, which is given by

$$\langle \mathcal{H} \rangle = \int \mathrm{d}\mathcal{E} \;\; \mathcal{E} p(\mathcal{E}) = \frac{1}{Z} \int \mathrm{d}\mathcal{E} \;\; \mathcal{E} \exp(-\beta \mathcal{E}) \Omega(\mathcal{E}) = \frac{1}{Z} \int \mathrm{d}\mathcal{E} \, \mathcal{E} e^{-\beta \mathcal{E}} \int \mathrm{d}\mu \, \delta(\mathcal{E} - \mathcal{H}(\mu)). \tag{17.13}$$

Changing the order of integration, we get

$$\begin{aligned} \langle \mathcal{H} \rangle &= \frac{1}{Z} \int \mathrm{d}\mu \, \mathcal{H}(\mu) e^{-\beta \mathcal{H}(\mu)} = -\frac{1}{Z} \int \mathrm{d}\mu \, \frac{\partial}{\partial \beta} e^{-\beta \mathcal{H}(\mu)} = -\frac{1}{Z} \frac{\partial}{\partial \beta} \int \mathrm{d}\mu \, e^{-\beta \mathcal{H}(\mu)} \\ &= -\frac{1}{Z} \frac{\partial Z}{\partial \beta} = \boxed{-\frac{\partial \ln Z}{\partial \beta}}. \end{aligned} \tag{17.14}$$

We can reexpress the probability $p(\mathcal{E})$ in terms of free energy, using Boltzmann's formula, we can write

$$p(\mathcal{E}) = \frac{1}{Z} \exp\left(\frac{S(\mathcal{E})}{k_B} - \frac{\mathcal{E}}{k_B T} \right) = \frac{1}{Z} \exp\left(-\frac{F(\mathcal{E})}{k_B T} \right), \tag{17.15}$$

where $F(\mathcal{E}) = \mathcal{E} - TS(\mathcal{E})$ is the *Helmholtz free energy* of the system. In thermodynamics, we have a similar expression

$$U = F + TS = F - T \left[\frac{\partial F}{\partial T} \right]_{\mathbf{x}} = -T^2 \frac{\partial}{\partial T} \left(\frac{F}{T} \right) = \frac{\partial (\beta F)}{\partial \beta}. \tag{17.16}$$

Comparing eqns. (17.14) and (17.16), we get

$$\ln Z = -\beta F \qquad \Rightarrow \qquad F(T, \mathbf{x}) = -k_B T \ln Z(T, \mathbf{x}). \tag{17.17}$$

Let's calculate the second cumulant of the energy of the system in the canonical ensemble. The second cumulant is given by

$$\left\langle \mathcal{H}^2 \right\rangle_c = \left\langle \mathcal{H}^2 \right\rangle - \langle \mathcal{H} \rangle^2 = \frac{1}{Z} \int \mathrm{d}\mu \, \mathcal{H}^2(\mu) e^{-\beta \mathcal{H}(\mu)} - \left(\frac{\partial \ln Z}{\partial \beta} \right)^2 = \frac{\partial^2 \ln Z}{\partial \beta^2} = -\frac{\partial \langle \mathcal{H} \rangle}{\partial \beta}. \tag{17.18}$$

More generally, we can calculate the n-th cumulant of the energy by

$$\langle \mathcal{H}^n \rangle_c = (-1)^n \frac{\partial^n \ln Z}{\partial \beta^n}. \tag{17.19}$$

Also observe,

$$\left\langle \mathcal{H}^2 \right\rangle_c = -\frac{\partial \langle \mathcal{H} \rangle}{\partial 1/k_B T} = k_B T^2 \left. \frac{\partial \langle \mathcal{H} \rangle}{\partial T} \right|_{\mathbf{x}} = k_B T^2 C_{\mathbf{x}}, \tag{17.20}$$

where $C_{\mathbf{x}}$ is the heat capacity of the system at constant generalized displacements $\mathbf{x}$.

§17.2 Two Level System

Here we will revisit the two-level system (see section 15.1) but in the canonical ensemble. The Hamiltonian of the system is given by

$$\mathcal{H} = \epsilon \sum_{i=1}^{N} n_i, \tag{17.21}$$

where $n_i = 0, 1$ is the occupation number of the i-th level. The partition function of the system is given by

$$\begin{aligned} Z = \sum_{\{n_i\}} \exp\left(-\beta\epsilon \sum_{i=1}^{N} n_i\right) &= \sum_{\{n_i\}} \prod_{i=1}^{N} e^{-\beta\epsilon n_i} \\ &= \prod_{i=1}^{N} \sum_{n_i=0}^{1} e^{-\beta\epsilon n_i} \\ &= \prod_{i=1}^{N} \left(1 + e^{-\beta\epsilon}\right) \\ &= \left(1 + e^{-\beta\epsilon}\right)^N. \end{aligned} \tag{17.22}$$

The average energy of the system is given by

$$\langle \mathcal{H} \rangle = -\frac{\partial \ln Z}{\partial \beta} = -N \frac{-\epsilon e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}} = N\epsilon \frac{e^{-\beta\epsilon}}{1 + e^{-\beta\epsilon}} = \frac{N\epsilon}{e^{\beta\epsilon} + 1}. \tag{17.23}$$

The free energy of the system is given by

$$F(T, N) = -k_B T \ln Z = -N k_B T \ln\left(1 + e^{-\epsilon/(k_B T)}\right). \tag{17.24}$$

The entropy of the system is given by

$$S = -\left. \frac{\partial F}{\partial T} \right|_N = -N k_B \ln\left(1 + e^{-\epsilon/(k_B T)}\right) + N k_B T \left(\frac{\epsilon}{k_B T^2}\right) \frac{e^{-\epsilon/(k_B T)}}{1 + e^{-\epsilon/(k_B T)}}. \tag{17.25}$$

The probability of the system being in one of the microstates $\{n_i\}$ is given by

$$p(\{n_i\}) = \frac{1}{Z} \exp\left(-\beta\epsilon \sum_{i=1}^{N} n_i\right) = \prod_{i=1}^{N} \frac{e^{-\beta\epsilon n_i}}{1 + e^{-\beta\epsilon}}. \tag{17.26}$$

Thus, the probability that the i-th particle is in the n_i state is given by

$$p_i(n_i) = \frac{e^{-\beta\epsilon n_i}}{1 + e^{-\beta\epsilon}}. \tag{17.27}$$

And this probability is same as what we obtained in the microcanonical ensemble.

18 Lecture 18: (19/02/2025)

§18.1 Approximate Distribution of Energy for a Canonical Ensemble

We expect the average energy of a system to be increase with number of particles *i.e.* $\langle \mathcal{H} \rangle \propto N$. Thus, we expect

$$\ln Z \propto N. \tag{18.1}$$

So variance of energy is expected to be of order N (see eqn. (17.18)). Thus, we can treat the mean and most likely energies interchangeably, since the width of the distribution $p(\mathcal{E})$ only scales as $\sqrt{\langle \mathcal{H}^2 \rangle_c} \propto \sqrt{N}$. The relative error, $\sqrt{\langle \mathcal{H}^2 \rangle_c} / \langle \mathcal{H} \rangle$, vanishes in the thermodynamic limit as $1/\sqrt{N}$. The PDF for the energy of the system in the canonical ensemble can be approximated by a Gaussian distribution (using the central limit theorem) with mean $\langle \mathcal{H} \rangle$ and variance $\langle \mathcal{H}^2 \rangle_c$.

$$p(\mathcal{E}) \approx \frac{1}{\sqrt{2\pi(k_B T^2 C_{\mathbf{x}})}} \exp\left(-\frac{(\mathcal{E} - \langle \mathcal{H} \rangle)^2}{2k_B T^2 C_{\mathbf{x}}}\right). \tag{18.2}$$

We need to be careful when using this approximation, if the heat capacity is divergent, the central limit theorem does not apply (in case of phase transitions).

Entropy: Using the probability distribution $p_{(T,\mathbf{x})}(\mu)$ (see eqn. (17.7)), we can write the entropy of the system as

$$S = -k_B \left\langle \ln p_{(T,\mathbf{x})}(\mu) \right\rangle = -k_B[-\beta \langle \mathcal{H} \rangle - \ln Z] = \frac{E - F}{T}, \tag{18.3}$$

where $E = \langle \mathcal{H} \rangle$ is the average energy of the system and $F = -k_B T \ln Z$ is the Helmholtz free energy of the system.

§18.2 Ideal Gas

Consider a system of N non-interacting particles in a volume V at temperature T. The Hamiltonian of the system is

$$\mathcal{H} = \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m} + U(\mathbf{q}_i), \tag{18.4}$$

where $U(\mathbf{q}_i)$ is the potential imposed by the walls of the container. For canonical ensemble, the macrostate $M \equiv (T, V, N)$ is fixed. The joint PDF for the microstates $\mu = \{\mathbf{q}_i, \mathbf{p}_i\}$ is given by

$$p_{(T,V,N)}(\{\mathbf{q}_i, \mathbf{p}_i\}) = \frac{1}{Z(T,V,N)} \exp\left(-\beta \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m}\right) \cdot \begin{cases} 1 & \text{if } \mathbf{q}_i \in V, \\ 0 & \text{otherwise.} \end{cases} \tag{18.5}$$

The partition function is given by

$$Z(T,V,N) = \frac{1}{N!} \int \prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{q}_i \, \mathrm{d}^3\mathbf{p}_i}{h^3} \exp\left(-\beta \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m}\right) \tag{18.6}$$

where the factor of $1/N!$ accounts for the indistinguishability of the particles. The integral over the positions $\mathbf{q}_i$ is equal to the volume V. Thus, the partition function can be written as

$$Z(T,V,N) = \frac{V^N}{h^{3N} N!} \int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{p}_i \exp\left(-\beta \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m}\right). \tag{18.7}$$

The integral over the momenta can be done by writing the integrand as a product of N integrals over the momenta of individual particles.

$$\int \prod_{i=1}^{N} \mathrm{d}^3\mathbf{p}_i \exp\left(-\beta \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m}\right) = \left(\int \mathrm{d}^3\mathbf{p} \exp\left(-\beta \frac{|\mathbf{p}|^2}{2m}\right)\right)^N \tag{18.8}$$

The integral over the momentum of a single particle can be written a multiple of three Gaussian integrals. Thus,

$$= \left(\int \mathrm{d}p \exp\left(-\beta \frac{p^2}{2m} \right) \right)^{3N} \tag{18.9}$$

$$= \left(\sqrt{2\pi m k_B T} \right)^{3N} \tag{18.10}$$

Thus, the partition function for an ideal gas is given by

$$Z(T,V,N) = \frac{V^N}{N!} \left(\frac{2\pi m k_B T}{h^2} \right)^{3N/2} = \frac{1}{N!} \left(\frac{V}{\lambda^3(T)} \right)^N, \tag{18.11}$$

where

$$\lambda(T) = \frac{h}{\sqrt{2\pi m k_B T}} \tag{18.12}$$

is the thermal de Broglie wavelength of the particles. The Helmholtz free energy of the system is given by

$$F(T,V,N) = -k_B T \ln Z = -N k_B T \left\{ -\ln N + 1 + \ln \left(\frac{V}{\lambda^3(T)} \right) \right\} \tag{18.13}$$

$$= -N k_B T \ln \left[\frac{Ve}{N} \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \right]. \tag{18.14}$$

Since the Helmholtz free energy is a function of (T,V,N), the exact differential of the Helmholtz free energy is given by

$$\mathrm{d}F = \left(\frac{\partial F}{\partial T} \right)_{V,N} \mathrm{d}T + \left(\frac{\partial F}{\partial V} \right)_{T,N} \mathrm{d}V + \left(\frac{\partial F}{\partial N} \right)_{T,V} \mathrm{d}N\,. \tag{18.15}$$

Using the thermodynamic identity $\mathrm{d}F = -S\,\mathrm{d}T - P\,\mathrm{d}V + \boldsymbol{\mu}\,\mathrm{d}N$, we can write the entropy of the system as

$$S = -\left(\frac{\partial F}{\partial T} \right)_{V,N} = N k_B \ln \left[\frac{Ve}{N} \left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \right] + \frac{3}{2} N k_B = -\frac{F}{T} + \frac{3}{2} N k_B. \tag{18.16}$$

After comparing eqn. (18.16) with eqn. (18.3), we can write the average energy of the system as

$$E = \frac{3}{2} N k_B T. \tag{18.17}$$

The pressure of the system is given by

$$P = -\left(\frac{\partial F}{\partial V} \right)_{T,N} = \frac{N k_B T}{V} \qquad \Rightarrow \qquad PV = N k_B T. \tag{18.18}$$

The chemical potential of the system is given by

$$\boldsymbol{\mu} = \left(\frac{\partial F}{\partial N} \right)_{T,V} = \frac{F}{N} + k_B T = k_B T \ln \left(n \lambda^3(T) \right), \tag{18.19}$$

where $n = N/V$ is the number density of the particles.

19 Lecture 19: (17/03/2025)

In the canonical ensemble, the system is allowed to exchange energy with the reservoir in form of heat, but there is no external work done on the system. In the Gibb's canonical ensemble, the energy of system changes by the addition of both heat and work.

§19.1 Gibb's Canonical Ensemble

In Gibb's canonical ensemble, the ensemble is specified by the macrostate $M \equiv (N, T, \mathbf{J})$. Notice that the macrostate is specified by the number of particles N *i.e.* we are not considering chemical work as a method of changing the energy of the system. Including the work done against the generalised force $\mathbf{J}$ in the ensemble, the energy of the combined system (comprising the system and the external elements that keep the force $\mathbf{J}$ constant) is given by $\mathcal{H} - \mathbf{J} \cdot \mathbf{x}$. The probability of the combined system being in a microstate $(\mu_S, \mathbf{x})$ is given by

$$p(\mu_S, \mathbf{x}) = \frac{1}{\mathcal{Z}(N, T, \mathbf{J})} e^{-\beta(\mathcal{H}(\mu_S) - \mathbf{J}\cdot\mathbf{x})}, \tag{19.1}$$

where $\mathcal{Z}(N, T, \mathbf{J})$ is the Gibb's partition function given by

$$\mathcal{Z}(N, T, \mathbf{J}) = \int \mathrm{d}\mu_S \int \mathrm{d}\mathbf{x}\, e^{-\beta(\mathcal{H}(\mu_S) - \mathbf{J}\cdot\mathbf{x})}. \tag{19.2}$$

In this ensemble, the expectation value of the generalised displacement $\mathbf{x}$ is given by

$$\langle \mathbf{x} \rangle = \frac{1}{\mathcal{Z}(N, T, \mathbf{J})} \int \mathrm{d}\mu_S \int \mathrm{d}\mathbf{x}\, \mathbf{x}\, e^{-\beta(\mathcal{H}(\mu_S) - \mathbf{J}\cdot\mathbf{x})} = k_B T \frac{\partial \ln \mathcal{Z}}{\partial \mathbf{J}}. \tag{19.3}$$

From thermodynamics, we know $G = E - TS - \mathbf{J} \cdot \mathbf{x}$, thus the average displacement is given by $-\frac{\partial G}{\partial \mathbf{J}}$. Now, comparing the two expressions for the average displacement, we get

$$G = -k_B T \ln \mathcal{Z}. \tag{19.4}$$

The enthalpy H is given by $H = E - \mathbf{J} \cdot \mathbf{x}$, thus we have

$$H = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \frac{\partial(\beta G)}{\partial \beta}. \tag{19.5}$$

Note that heat capacity at constant force (which include work done against the external forces) are obtained from the enthalpy as $C_{\mathbf{J}} = \frac{\partial H}{\partial T}$

§19.1.1 Ideal Gas: Isobaric Ensemble

Consider old example of an ideal gas in a box with a piston, where the volume of the box is allowed to change. The total energy of the combined system is given by

$$E(\mu_S, V) = \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m} + PV \tag{19.6}$$

where P is the pressure of the gas and V is the volume of the box. The probability of the combined system being in a microstate (μ_S, V) is given by

$$p(\mu_S, V) = \begin{cases} \frac{1}{\mathcal{Z}(N,T,P)} e^{-\beta E(\mu_S, V)} & \text{if } \{\mathbf{q}_i\} \in V \\ 0 & \text{otherwise} \end{cases} \tag{19.7}$$

where $\mathcal{Z}(N, T, P)$ is the partition function given by

$$\mathcal{Z}(N, T, P) = \int_0^\infty \mathrm{d}V\, e^{-\beta PV} \frac{1}{N!} \int \prod_{i=1}^{N} \frac{\mathrm{d}^3\mathbf{q}_i\, \mathrm{d}^3\mathbf{p}_i}{h^3} \exp\left\{-\beta \sum_{i=1}^{N} \frac{|\mathbf{p}_i|^2}{2m}\right\}. \tag{19.8}$$

Recall that the second integral is the partition function of the ideal gas in the canonical ensemble (see eqn. (18.6)). Thus the Gibb's partition function is given by

$$\mathcal{Z}(N,T,P) = \int_0^\infty \mathrm{d}V\, e^{-\beta PV} \frac{V^N}{N!} \frac{1}{\lambda^{3N}(T)} \tag{19.9}$$

where $\lambda(T)$ is the thermal de Broglie wavelength of the particles. Thus rewriting the above integral, we get

$$\mathcal{Z}(N,T,P) = \frac{1}{N!} \frac{1}{(\beta P)^{N+1}} \frac{1}{\lambda^{3N}(T)} \int_0^\infty \mathrm{d}u\, u^N e^{-u} \tag{19.10}$$

where $u = \beta PV$. The integral over u is given by $\Gamma(N+1)$, thus the partition function is given by

$$\mathcal{Z}(N,T,P) = \frac{1}{N!} \frac{1}{(\beta P)^{N+1}} \frac{1}{\lambda^{3N}(T)} \Gamma(N+1) = \frac{1}{(\beta P)^{N+1} \lambda^{3N}(T)}. \tag{19.11}$$

Let's calculate the Gibb's free energy of the systemm, which is given by

$$G = -k_B T \ln \mathcal{Z} = k_B T[(N+1)\ln(\beta P) + 3N \ln(\lambda(T))] \approx N k_B T[\ln(\beta P) + 3\ln(\lambda(T))]. \tag{19.12}$$

§19.1.2 Non-interacting Spin Chain in an external magnetic field

Consider a non-interacting spin-1/2 chain in an external magnetic field. The Hamiltonian of the system is given by

$$\mathcal{H} = -\sum_{i=1}^N \mathbf{B} \cdot \mathbf{S}_i \tag{19.13}$$

where $\mathbf{B}$ is the external magnetic field and $\mathbf{S}_i$ is the spin of the i-th particle. Assumnig that the magnetic field is in the z-direction, the Hamiltonian can be written as

$$\mathcal{H} = -B\mu_\circ \sum_{i=1}^N \sigma_i^z \tag{19.14}$$

where σ_i^z is the z-component of the Pauli matrix σ_i. The probability of the system being in a microstate $\{\sigma_i^z\}$ is given by

$$p(\{\sigma_i^z\}) = \frac{1}{\mathcal{Z}(N,T,B)} e^{\beta B \mu_\circ \sum_{i=1}^N \sigma_i^z} \tag{19.15}$$

where $\mathcal{Z}(N,T,B)$ is the partition function given by

$$\mathcal{Z}(N,T,B) = \sum_{\{\sigma_i^z\}} e^{\beta B \mu_\circ \sum_{i=1}^N \sigma_i^z} = \left(e^{\beta B \mu_\circ} + e^{-\beta B \mu_\circ}\right)^N = [2\cosh(\beta B \mu_\circ)]^N. \tag{19.16}$$

The Gibb's free energy of the system is given by

$$G = -k_B T \ln \mathcal{Z} = -N k_B T \ln[2\cosh(\beta B \mu_\circ)]. \tag{19.17}$$

The magnetisation of the system is defined as $M = \mu_\circ \sum_{i=1}^N \sigma_i^z$, thus the average magnetisation of the system is given by

$$\langle M \rangle = \frac{1}{\mathcal{Z}(N,T,B)} \sum_{\{\sigma_i^z\}} M e^{\beta B \mu_\circ \sum_{i=1}^N \sigma_i^z} = \frac{1}{\mathcal{Z}(N,T,B)} \sum_{\{\sigma_i^z\}} M e^{\beta B M} = -\frac{\partial G}{\partial B} \tag{19.18}$$

which gives

$$\langle M \rangle = N \mu_\circ \tanh(\beta B \mu_\circ). \tag{19.19}$$

Let's calculate magnetic susceptibility of the system, which is given by

$$\chi(T) = \left[\frac{\partial M}{\partial B}\right|_{B=0} = \frac{N \mu_\circ^2}{k_B T}. \tag{19.20}$$

This is the Curie's law for magnetic susceptibility.

§19.2 Equipartition Theorem

Theorem 19.1 (Equipartition Theorem)
For a system in thermal equilibrium at temperature T, the average energy of each degree of freedom is $k_B T/2$.

To prove the equipartition theorem, consider a system with N particles in 3 dimensions. Many physical systems have Hamiltonians that, through a canonical transformation, can be cast in the form

$$\mathcal{H} = \sum_{i=1}^{3N} A_i P_i^2 + B_i Q_i^2. \tag{19.21}$$

Let's calculate the expectation value of $x_i\left(\frac{\partial \mathcal{H}}{\partial x_j}\right)$ where x_i, x_j are the canonical coordinates. The expectation value is given by

$$\begin{aligned}
\left\langle x_i \frac{\partial \mathcal{H}}{\partial x_j} \right\rangle &= \frac{1}{Z}\int \mathrm{d}\tau\, x_i \frac{\partial \mathcal{H}}{\partial x_j} e^{-\beta\mathcal{H}} && \text{where } \mathrm{d}\tau = \prod_{i=1}^{3N} \frac{\mathrm{d}q_i\,\mathrm{d}p_i}{h} && (19.22)\\
&= \frac{1}{Z}\left(-\frac{1}{\beta}\right)\int \mathrm{d}\tau_{(j)} \underbrace{\int \mathrm{d}x_j\, x_i \frac{\partial}{\partial x_j}\left(e^{-\beta\mathcal{H}}\right)}_{\text{Integration by parts}} && \text{where } \mathrm{d}\tau_{(j)} = \frac{\mathrm{d}\tau}{\mathrm{d}x_j} \\
&= -\frac{1}{\beta}\frac{1}{Z}\int \left[\left(x_i e^{-\beta\mathcal{H}}\Big|_{x_j^{\min}}^{x_j^{\max}} - \int \mathrm{d}x_j \frac{\partial x_i}{\partial x_j} e^{-\beta\mathcal{H}}\right] \mathrm{d}\tau_{(j)} && (19.23)
\end{aligned}$$

The first term in the $\mathrm{d}\tau_{(j)}$ integral vanishes because whenever any of the coordinates takes an "extreme" value the Hamiltonian of the system becomes infinite[1], and $\frac{\partial x_i}{\partial x_j} = \delta_{ij}$, thus the above expression simplifies to

$$\begin{aligned}
&= \frac{1}{Z}\frac{1}{\beta}\delta_{ij}\int \mathrm{d}\tau\, e^{-\beta\mathcal{H}} \\
&= \frac{1}{\beta}\delta_{ij} = \delta_{ij} k_B T. && (19.24)
\end{aligned}$$

Observe that the following relation holds

$$\sum_{i=1}^{3N} P_i \frac{\partial \mathcal{H}}{\partial P_i} + Q_i \frac{\partial \mathcal{H}}{\partial Q_i} = 2\mathcal{H}. \tag{19.25}$$

Thus, the average energy of the system is given by

$$2\langle \mathcal{H}\rangle = 2E = \sum_{i=1}^{3N}\left\langle P_i \frac{\partial \mathcal{H}}{\partial P_i}\right\rangle + \left\langle Q_i \frac{\partial \mathcal{H}}{\partial Q_i}\right\rangle = 6Nk_BT \qquad \Rightarrow \qquad \varepsilon = \frac{E}{6N} = \frac{k_BT}{2}. \tag{19.26}$$

Here, ε is the average energy of each degree of freedom.

§19.2.1 Generalised Equipartition Theorem

The equipartition theorem that we have proved is for systems with quadratic Hamiltonians in 3 dimensions. However, the theorem can be generalised to systems of N particles in d-dimensions with Hamiltonians of the form

$$\mathcal{H} = \sum_{i=1}^{N} A_i |\mathbf{P}_i|^s + B_i |\mathbf{Q}_i|^t \tag{19.27}$$

[1] For instance, if x_j is a space coordinate, then its extreme values will correspond to "locations at the walls of the container"; accordingly, the potential energy of the system would become infinite. If, on the other hand, x_j is a momentum coordinate, then its extreme values will themselves be $\pm\infty$, in which case the kinetic energy of the system would become infinite.

where s, t are positive integers and A_i, B_i are constants. Let's compute the partition function of the system, which is given by

$$Z = \int \mathrm{d}\tau\, e^{-\beta\mathcal{H}} = \int \prod_{i=1}^{N} \frac{\mathrm{d}^d\mathbf{Q}_i\, \mathrm{d}^d\mathbf{P}_i}{h^d} e^{-\beta\mathcal{H}} \tag{19.28}$$

$$= \frac{1}{h^{dN}} \prod_{i=1}^{N} \left[\left(\int \mathrm{d}^d\mathbf{P}_i \exp(-\beta A_i |\mathbf{P}_i|^s) \right) \left(\int \mathrm{d}^d\mathbf{Q}_i \exp(-\beta B_i |\mathbf{Q}_i|^s) \right) \right] \tag{19.29}$$

This is a product of $2N$ integrals, each of which is of the form

$$I(\lambda, r, d) = \int \mathrm{d}^d\mathbf{x} \exp(-\lambda |\mathbf{x}|^r). \tag{19.30}$$

Making the substitution $\mathbf{u} = \lambda^{1/r}\mathbf{x}$, the integral becomes

$$I(\lambda, r, d) = \lambda^{-d/r} \int \mathrm{d}^d\mathbf{u} \exp(-|\mathbf{u}|^r) = \lambda^{-d/r} I(1, r, d). \tag{19.31}$$

Define the function $C(r, d)$ as

$$C(r, d) = \int \mathrm{d}^d\mathbf{u} \exp(-|\mathbf{u}|^r) = I(1, r, d). \tag{19.32}$$

The partition function of the system is given by

$$Z(T, N) = \frac{1}{h^{dN}} \prod_{i=1}^{N} \left[\left\{ (\beta A_i)^{-d/s} C(s, d) \right\} \left\{ (\beta B_i)^{-d/t} C(t, d) \right\} \right]$$

$$= \left(\frac{C(s, d)\ C(t, d)}{h^d} \right)^N \left(\prod_{i=1}^{N} A_i^{-d/s} B_i^{-d/t} \right) \beta^{-N(d/s + d/t)}. \tag{19.33}$$

The internal energy and the heat capacity of the system are given by

$$E = -\frac{\partial}{\partial\beta} \ln Z = Nd\left(\frac{1}{s} + \frac{1}{t}\right) k_B T, \tag{19.34}$$

$$C = \frac{\partial E}{\partial T} = Nd\left(\frac{1}{s} + \frac{1}{t}\right) k_B. \tag{19.35}$$

This is a generalization of the equipartition theorem, and we note that E and C depends on the powers of momenta and coordinates in the Hamiltonian, but not on the coeffecients A_i, B_i.

For usual harmonic oscillator Hamiltonian in 3 dimensions, we set $s = t = 2$, and we get $E = 3Nk_BT$ and $C = 3Nk_B$.

20 Lecture 20: (19/03/2025)

In canonical ensemble, the energy of the system changes by adding or removing heat from the system. But we want to study systems where the energy of the system changes by chemical work. In Grand Canonical Ensemble, we allow the system to exchange particles and heat with the reservoir, but the system can't do external work.

§20.1 Grand Canonical Ensemble

In Grand Canonical Ensemble, the ensemble is defined by the macrostate $M \equiv (T, \boldsymbol{\mu}, \mathbf{x})$. The microstate of the system is defined by (μ_S, N), where μ_S is the microstate of the system of N particles. Consider a system of N_1 particles in a volume V_1 and a reservoir of N_2 particles in a volume V_2. The total number of particles $N = N_1 + N_2$ is constant and the total volume $V = V_1 + V_2$ is constant. Assuming the system and reservoir are weakly interacting, so we can write the combined Hamiltonian as

$$\mathcal{H}(\mathbf{q}, \mathbf{p}, N) = \mathcal{H}_1(\mathbf{q}_1, \mathbf{p}_1, N_1) + \mathcal{H}_2(\mathbf{q}_2, \mathbf{p}_2, N_2) \tag{20.1}$$

where $\mathcal{H}_1$ and $\mathcal{H}_2$ are the Hamiltonians of the system and reservoir respectively (note that both Hamiltonians are of same form *i.e.* system and reservoir are composed of same type of particles). The system and reservoir are in thermal equilibrium, so we can write the canonical partition function for the combined system as

$$Z(T, V, N) = \sum_{N_1=0}^{N} Z_1(T, V_1, N_1) Z_2(T, V_2, N_2) \tag{20.2}$$

$$= \sum_{N_1=0}^{N} \frac{1}{h^{3N_1}} \frac{1}{N_1!} \int \mathrm{d}\mathbf{q}_1 \, \mathrm{d}\mathbf{p}_1 \, e^{-\beta \mathcal{H}_1(\mathbf{q}_1, \mathbf{p}_1, N_1)} Z_2(T, V_2, N_2) \tag{20.3}$$

rearranging the terms, we get

$$1 = \sum_{N_1=0}^{N} \int \mathrm{d}\mathbf{q}_1 \, \mathrm{d}\mathbf{p}_1 \, \frac{Z_2(T, V_2, N_2)}{Z(T, V, N)} \frac{e^{-\beta \mathcal{H}_1(\mathbf{q}_1, \mathbf{p}_1, N_1)}}{h^{3N_1} N_1!} \tag{20.4}$$

since we are summing over all possible microstates of the system, this equation is equivalent to normalizing condition of the probability distribution. So we can write the probability of finding the system in a microstate $(\mathbf{q}_1, \mathbf{p}_1, N_1)$ as

$$p(\mathbf{q}_1, \mathbf{p}_1, N_1) = \frac{Z_2(T, V_2, N_2)}{Z(T, V, N)} \frac{e^{-\beta \mathcal{H}_1(\mathbf{q}_1, \mathbf{p}_1, N_1)}}{h^{3N_1} N_1!}. \tag{20.5}$$

Since Z is the canonical partition function, we can write it as

$$Z(T, V, N) = e^{-\beta F(T, V, N)} \tag{20.6}$$

where F is the Helmholtz free energy. Since the reservoir and system are composed of same type of particles, the Helmholtz free energy of the system and reservoir are of same form. So we can write the probability distribution as

$$p(\mathbf{q}_1, \mathbf{p}_1, N_1) = \frac{e^{-\beta F(T, V_2, N_2)}}{e^{-\beta F(T, V, N)}} \frac{e^{-\beta \mathcal{H}_1}}{h^{3N_1} N_1!} \tag{20.7}$$

$$= e^{-\beta [F(T, V_2, N_2) - F(T, V, N)]} \frac{e^{-\beta \mathcal{H}_1}}{h^{3N_1} N_1!} \tag{20.8}$$

$$= e^{-\beta [F(T, V - V_1, N - N_1) - F(T, V, N)]} \frac{e^{-\beta \mathcal{H}_1}}{h^{3N_1} N_1!}, \tag{20.9}$$

since the reservoir is very large compared to the system *i.e.* $V_2 \gg V_1$ and $N_2 \gg N_1$, we treat the Helmholtz free energy of the reservoir as the small perturbation to the Helmholtz free energy of combined system. So we can have

$$F(T, V - V_1, N - N_1) - F(T, V, N) \approx -\left(\frac{\partial F}{\partial V}\right)V_1 - \left(\frac{\partial F}{\partial N}\right)N_1, \tag{20.10}$$

define $\boldsymbol{\mu} = \frac{\partial F}{\partial N}$ as the chemical potential and $P = -\frac{\partial F}{\partial V}$ as the pressure, we can write the probability distribution as (dropping the subscript 1)

$$p(\mathbf{q}, \mathbf{p}, N) = \frac{e^{\beta \boldsymbol{\mu} N} e^{-\beta P V}}{h^{3N} N!} e^{-\beta \mathcal{H}(\mathbf{q}, \mathbf{p}, N)}. \tag{20.11}$$

Define fugacity $\mathfrak{z} = e^{\beta \boldsymbol{\mu}}$, we can write the probability distribution as

$$p(\mathbf{q}, \mathbf{p}, N) = \frac{\mathfrak{z}^N e^{-\beta P V}}{h^{3N} N!} e^{-\beta \mathcal{H}(\mathbf{q}, \mathbf{p}, N)}. \tag{20.12}$$

The normalization condition of the probability distribution is

$$1 = \sum_{N=0}^{\infty} \mathfrak{z}^N e^{-\beta P V} \underbrace{\boxed{\frac{1}{h^{3N}} \frac{1}{N!} \int \mathrm{d}\mathbf{q}\, \mathrm{d}\mathbf{p}\, e^{-\beta \mathcal{H}(\mathbf{q}, \mathbf{p}, N)}}}_{\text{Canonical partition function}}. \tag{20.13}$$

Define the '*grand canonical partition function*' as

$$\mathcal{Q}(\mathfrak{z}, T, V) = \sum_{N=0}^{\infty} \mathfrak{z}^N Z(T, V, N) \tag{20.14}$$

where $Z(T, V, N)$ is the canonical partition function. The grand canonical partition function is the sum of all possible canonical partition functions weighted by the fugacity $\mathfrak{z}$. Thus we can write the normalization condition as

$$e^{\beta P V} = \sum_{N=0}^{\infty} \mathfrak{z}^N Z(T, V, N) = \mathcal{Q}(\mathfrak{z}, T, V). \tag{20.15}$$

So we get the following relation between the grand canonical partition function and pressure (similar to the ideal gas law)

$$\frac{PV}{k_B T} = \ln \mathcal{Q}(\mathfrak{z}, T, V). \tag{20.16}$$

Since number of particles in the system is itself a random variable, thus the average number of particles is a relevant quantity to talk.

$$\langle N \rangle = \frac{\sum_{N=0}^{\infty} N \mathfrak{z}^N e^{-\beta P V} Z}{\sum_{N=0}^{\infty} \mathfrak{z}^N e^{-\beta P V} Z} \tag{20.17}$$

$$= \frac{1}{\mathcal{Q}} \sum_{N=0}^{\infty} N \mathfrak{z}^N Z \tag{20.18}$$

$$= \mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Q}(\mathfrak{z}, T, V) = \frac{\mathfrak{z}}{\mathcal{Q}} \frac{\partial \mathcal{Q}}{\partial \mathfrak{z}} \tag{20.19}$$

Let's calculate the fluctuations in the number of particles in the system. Consider

$$\mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \left(\mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Q} \right) = \mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Q} + \mathfrak{z}^2 \frac{\partial^2}{\partial \mathfrak{z}^2} \ln \mathcal{Q} \tag{20.20}$$

$$= \langle N \rangle + \mathfrak{z}^2 \frac{\partial}{\partial \mathfrak{z}} \left(\frac{1}{\mathcal{Q}} \frac{\partial \mathcal{Q}}{\partial \mathfrak{z}} \right) \tag{20.21}$$

$$= \langle N \rangle + \mathfrak{z}^2 \left[-\frac{1}{\mathcal{Q}^2} \left(\frac{\partial \mathcal{Q}}{\partial \mathfrak{z}} \right)^2 + \frac{1}{\mathcal{Q}} \frac{\partial^2 \mathcal{Q}}{\partial \mathfrak{z}^2} \right] \tag{20.22}$$

$$= \langle N \rangle - \langle N \rangle^2 + \frac{1}{\mathcal{Q}} \sum_{N=0}^{\infty} N(N-1) \mathfrak{z}^N Z \tag{20.23}$$

$$= \langle N \rangle - \langle N \rangle^2 + \frac{1}{\mathcal{Q}} \sum_{N=0}^{\infty} N^2 \mathfrak{z}^N Z - \frac{1}{\mathcal{Q}} \sum_{N=0}^{\infty} N \mathfrak{z}^N Z \tag{20.24}$$

$$= \langle N \rangle - \langle N \rangle^2 + \langle N^2 \rangle - \langle N \rangle \tag{20.25}$$

$$= \langle N^2 \rangle - \langle N \rangle^2 \tag{20.26}$$

Thus we can write the fluctuations in the number of particles in the system as

$$\langle N^2 \rangle - \langle N \rangle^2 = \mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \left(\mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Q} \right). \tag{20.27}$$

Recall that the fugacity is defined as $\mathfrak{z} = e^{\beta \boldsymbol{\mu}}$, thus we have $\frac{\partial}{\partial \mathfrak{z}} \equiv \frac{\partial \boldsymbol{\mu}}{\partial \mathfrak{z}} \frac{\partial}{\partial \boldsymbol{\mu}} = \frac{1}{\beta \mathfrak{z}} \frac{\partial}{\partial \boldsymbol{\mu}}$. Thus, we can write the fluctuations in the number of particles in the system as

$$\langle N^2 \rangle - \langle N \rangle^2 = \frac{1}{\beta^2} \frac{\partial^2}{\partial \boldsymbol{\mu}^2} \ln \mathcal{Q}(\mathfrak{z}, T, V) = \frac{1}{\beta^2} \frac{\partial^2}{\partial \boldsymbol{\mu}^2} (\beta P V) = k_B T V \frac{\partial^2 P}{\partial \boldsymbol{\mu}^2}. \tag{20.28}$$

21 Lecture 21: (20/03/2025)

§21.1 Fluctuations in Particle Number (Grand Canonical Ensemble)

To express the fluctuations in particle number (see eqn. (20.28)) in terms conveniently measurable quantities, assume that the Helmbholtz free energy of the system, being an extensive quantity, can be written in the form

$$F(T,V,N) = Nf(T,v), \quad \text{where } v = \frac{V}{N} \tag{21.1}$$

where $f(T,v)$ is the Helmholtz free energy per particle. Then we can express chemical potential and pressure as

$$\boldsymbol{\mu} = \frac{\partial F}{\partial N} = f(T,v) + N\frac{\partial f}{\partial N} = f(T,v) + N\frac{\partial f}{\partial v}\frac{\partial v}{\partial N} = f(T,v) - v\frac{\partial f}{\partial v}, \tag{21.2}$$

$$P = -\frac{\partial F}{\partial V} = -N\frac{\partial f}{\partial V} = -N\frac{\partial f}{\partial v}\frac{\partial v}{\partial V} = -\frac{\partial f}{\partial v}. \tag{21.3}$$

Now, considering both pressure and chemical potential as functions of T and v, we obtain

$$\frac{\partial \boldsymbol{\mu}}{\partial v} = \frac{\partial f}{\partial v} - \frac{\partial f}{\partial v} - v\frac{\partial^2 f}{\partial v^2} = -v\frac{\partial^2 f}{\partial v^2}, \tag{21.4}$$

$$\frac{\partial P}{\partial \boldsymbol{\mu}} = -\frac{\partial}{\partial \boldsymbol{\mu}}\frac{\partial f}{\partial v} = -\frac{\partial^2 f}{\partial v^2}\frac{\partial v}{\partial \boldsymbol{\mu}} = -\frac{\partial^2 f}{\partial v^2}\left(-v\frac{\partial^2 f}{\partial v^2}\right)^{-1} = \frac{1}{v} \equiv \frac{N}{V}. \tag{21.5}$$

Thus, we can compute the derivatives of pressure with respect to chemical potential to express the fluctuations in particle number,

$$\frac{\partial^2 P}{\partial \boldsymbol{\mu}^2} = -\frac{1}{v^2}\frac{\partial v}{\partial \boldsymbol{\mu}} = -\frac{1}{v^2}\left(-v\frac{\partial^2 f}{\partial v^2}\right)^{-1} \tag{21.6}$$

$$= \frac{1}{v^3}\left(\frac{\partial^2 f}{\partial v^2}\right)^{-1} \tag{21.7}$$

$$= -\frac{1}{v^3}\left(\frac{\partial P}{\partial v}\right)^{-1} = -\frac{1}{v^3}\frac{1}{(\partial P/\partial v)}. \tag{21.8}$$

Thus, we can express the fluctuations in particle number as

$$\langle N^2\rangle - \langle N\rangle^2 = k_B T V\frac{\partial^2 P}{\partial \boldsymbol{\mu}^2} = -k_B T V\frac{1}{v^3}\frac{1}{(\partial P/\partial v)} = \langle N\rangle\frac{k_B T}{v}\frac{1}{(-v\,\partial P/\partial v)} = \langle N\rangle\frac{k_B T}{v}\kappa_T \tag{21.9}$$

where κ_T is the isothermal compressibility defined as

$$\kappa_T = -\frac{1}{v\,\partial P/\partial v}. \tag{21.10}$$

§21.2 Spin Chain in a Magnetic Field

Consider a spin chain of N spin-1 particles in a magnetic field B along the z-direction. The Hamiltonian of the system is given by

$$\mathcal{H} = J\sum_{i=1}^{N}\mathbf{S}_i\cdot\mathbf{S}_i - \mu_B B\sum_{i=1}^{N}S_i^z \tag{21.11}$$

where J is self-interaction energy, μ_B is the Bohr magneton, and S_i^z is the z-component of the spin operator. Since each spin are independent, we can write the partition function as

$$Z(N,T,B) = Z_1(T,B)^N \tag{21.12}$$

where $Z_1(T,B)$ is the partition function of a single spin. The partition function of a single spin is given by

$$Z_1(T,B) = \sum_{S^z=-1}^{1} e^{-\beta\left(J(S^z)^2-\mu_B B S^z\right)} = 1 + e^{-\beta(J-\mu_B B)} + e^{-\beta(J+\mu_B B)} = 1 + 2e^{-\beta J}\cosh(\beta\mu_B B). \quad (21.13)$$

Thus, the partition function of the system is given by

$$Z = \Big(1 + 2e^{-\beta J}\cosh(\beta\mu_B B)\Big)^N. \quad (21.14)$$

The Gibbs free energy of the system is given by

$$G(N,T,B) = -k_B T \ln Z = -N k_B T \ln\Big(1 + 2e^{-\beta J}\cosh(\beta\mu_B B)\Big). \quad (21.15)$$

Now, consider the case where the magnetic field is very weak *i.e.* $\mu_B B \ll J$. In this case, we can expand the Gibbs free energy in terms of $\mu_B B$ as

$$G(N,T,B) = -N k_B T \ln\left(1 + 2e^{-\beta J}\left(1 + \frac{(\beta\mu_B B)^2}{2}\right)\right). \quad (21.16)$$

Now, we can compute the average magnetization of the system as

$$\langle M\rangle = -\frac{\partial G}{\partial B} = 2NB\beta\mu_B^2 \frac{e^{-\beta J}}{1 + 2e^{-\beta J}\left(1 + \frac{(\beta\mu_B B)^2}{2}\right)}. \quad (21.17)$$

Thus, we can calculate susceptibility of the system as

$$\chi(T) = \left[\frac{\partial M}{\partial B}\right|_{B=0} = 2N\beta\mu_B^2 \frac{e^{-\beta J}}{1 + 2e^{-\beta J}}. \quad (21.18)$$

22 Lecture 22: (24/03/2025)

Till now, we have only seen the evolution of a statistically large classical system, but in nature, all systems obey Quantum Mechanics. Thus, we will now explore these notions of statistics for Quantum systems.

§22.1 Closer Look at Quantum Mechanics

In quantum mechanics, the state of the system is a vector $|\Psi\rangle$ in Hilbert space $\mathscr{H}$. An observable of a system is associated with a Hermitian operator that operates on a Hilbert space. Say $|q\rangle$ is an eigenvector of the position operator, then $\langle q|\Psi\rangle = \Psi(q)$ is the wave function of the system in the state $|\Psi\rangle$.

At any time t, the state vector $|\Psi; t\rangle$ of a truly isolated system can be expressed as a linear superposition of a completely orthonormal set of stationary states $\{|\Phi_n\rangle\}$:

$$|\Psi; t\rangle = \sum_n c_n(t)\,|\Phi_n\rangle \tag{22.1}$$

where $c_n(t) \in \mathbb{C}$. The index n stands for a set of quantum numbers, which are eigenvalues of certain chosen dynamical operators of the system. The square modulus $|c_n|^2$ is the probability that a measurement performed on the system will find it to have the quantum numbers n.

Suppose $\hat{\mathcal{O}}$ is an operator corresponding to an observable of the system. According to Copenhagen's interpretation, the average result of a large number of measurements of this observable is instantaneously given by the expectation value

$$\mathbb{E}[\hat{\mathcal{O}}] = \frac{\sum_{n,m} c_m^* c_n \langle \Phi_m|\hat{\mathcal{O}}|\Phi_n\rangle}{\sum_n c_n^* c_n}. \tag{22.2}$$

But when we measure an observable in the laboratory, we measure not its instantaneous value but a time average. Thus, the directly measurable quantity is not eqn. (22.2) but the following quantity (this also implies that the quantum mechanics is not ergodic):

$$\langle\hat{\mathcal{O}}\rangle = \frac{\overline{\langle\Psi|\hat{\mathcal{O}}|\Psi\rangle}}{\langle\Psi|\Psi\rangle} = \frac{\sum_{n,m} \overline{c_m^* c_n}\,\langle\Phi_m|\hat{\mathcal{O}}|\Phi_n\rangle}{\sum_n \overline{c_n^* c_n}} \tag{22.3}$$

where $\overline{c_m^* c_n}$ is the time average of $c_m^* c_n$ over a time interval that is short compared to the resolving time of the measuring apparatus but long compared to molecular times (*e.g.* collision times or periods of molecular motion). The denominator is identical as $\sum_n |c_n|^2$ is time independent.

The postulate of quantum statistical mechanics are postulates concering the coefficients $\overline{c_m^* c_n}$ when eqn. (22.3) refers to a macroscopic observable of a macroscopic system in thermodynamic equilibrium.

§22.1.1 Postulates of Quantum Statistical Mechanics

Consider a macroscopic system that weakly interacts with the external world (*i.e.* the energy of the system is approximately constant). Let the number of particles in the system be N confined to a volume V, and its energy lie between E and $E + \Delta E$. Let $\hat{\mathcal{H}}$ be the Hamiltonian operator of the system. For convenience, choose a standard set of complete orthonormal states $\{|\Phi_n\rangle\}$ such that each $|\Phi_n\rangle$ is a state for N particles in the volume V with energy E_n *i.e.*

$$\hat{\mathcal{H}}\,|\Phi_n\rangle = E_n\,|\Phi_n\rangle\,. \tag{22.4}$$

The postulates of quantum statistical mechanics are:

Postulate of Equal a Priori Probability

$$\overline{c_n^* c_n} = \begin{cases} 1 & \text{if } E \le E_n \le E + \Delta E \\ 0 & \text{otherwise} \end{cases} \tag{22.5}$$

Postulate of Random Phases

$$\overline{c_m^* c_n} = 0 \quad \text{for } m \neq n \tag{22.6}$$

Using these postulates, we have an effective state vector of the system:

$$|\Psi\rangle = \sum_n b_n |\Phi_n\rangle \quad \text{where } b_n = \begin{cases} \text{const.} & \text{if } E \leq E_n \leq E + \Delta E \\ 0 & \text{otherwise} \end{cases}. \tag{22.7}$$

The average value of an observable $\hat{\mathcal{O}}$ is given by

$$\langle \hat{\mathcal{O}} \rangle = \frac{\sum_n |b_n|^2 \langle \Phi_n | \hat{\mathcal{O}} | \Phi_n \rangle}{\sum_n |b_n|^2}. \tag{22.8}$$

The postulate of random phases implies that the state of a system in equilibrium may be regarded as an incoherent superposition of eigenstates.

§22.1.2 Density Matrix

Density matrix is a formalism that allows us to describe a state without specifying the random phases. We can define a density matrix $\hat{\rho}$ as

$$\rho_{mn} = \langle \Phi_m | \hat{\rho} | \Phi_n \rangle \equiv \delta_{mn} |b_n|^2 \quad \text{where } |b_n|^2 = \begin{cases} \text{const.} & \text{if } E \leq E_n \leq E + \Delta E \\ 0 & \text{otherwise} \end{cases} \tag{22.9}$$

In terms of the density matrix, the average value of an observable $\hat{\mathcal{O}}$ is given by

$$\langle \hat{\mathcal{O}} \rangle = \frac{\sum_n \langle \Phi_n | \hat{\mathcal{O}} \hat{\rho} | \Phi_n \rangle}{\sum_n \langle \Phi_n | \hat{\rho} | \Phi_n \rangle} = \frac{\mathrm{Tr}\left(\hat{\mathcal{O}} \hat{\rho}\right)}{\mathrm{Tr}(\hat{\rho})}. \tag{22.10}$$

The density operator $\hat{\rho}$ contains all the information about the ensemble. We have the following equation of motion for the density operator

$$\frac{\mathrm{d}\hat{\rho}}{\mathrm{d}t} = \frac{1}{i\hbar}[\hat{\rho}, \hat{\mathcal{H}}] \tag{22.11}$$

which is the quantum mechanical version of Liouville's equation. Formally we can represent the density operator $\hat{\rho}$ as

$$\hat{\rho} = \sum_n |\Phi_n\rangle |b_n|^2 \langle \Phi_n| \tag{22.12}$$

§22.2 Ensembles in Quantum Statistical Mechanics

Recall the definition of density matrix eqn. (22.9), now based on the value of $|b_n|^2$, we can define different ensembles.

§22.2.1 Microcanonical Ensemble

This ensemble is specified by the macrostate $M \equiv (E, \mathbf{x})$, where E is the energy of the system and $\mathbf{x}$ is generalized displacement. Since for a microcanonical ensemble, all the microstates are equally probable, we have

$$\rho_{mn} = \delta_{mn} |b_n|^2 \quad \text{where } |b_n|^2 = \begin{cases} 1 & \text{if } E \leq E_n \leq E + \Delta E \\ 0 & \text{otherwise} \end{cases}. \tag{22.13}$$

Thus, the density operator is given by

$$\hat{\rho} = \sum_n |\Phi_n\rangle |b_n|^2 \langle \Phi_n| = \sum_{E \leq E_n \leq E + \Delta E} |\Phi_n\rangle\langle\Phi_n|. \tag{22.14}$$

The trace of the density operator is given by

$$\mathrm{Tr}(\hat{\rho}) = \sum_n \rho_{nn} = \sum_{E \leq E_n \leq E + \Delta E} 1 = \Omega(E, \mathbf{x}) \tag{22.15}$$

> Remark 22.1. With this new definition of the number of states, no Gibbs paradox arises because the correct counting of states is automatically implied by the new definition of Ω.

In limit $\Delta \ll E$, we can write

$$\Omega(E, \mathbf{x}) = g(E, \mathbf{x})\Delta E \tag{22.16}$$

where $g(E, \mathbf{x})$ is the density of states at energy E. With this, we can identify the entropy of the system as

$$S(E, \mathbf{x}) = k_B \ln \Omega(E, \mathbf{x}) \tag{22.17}$$

where k_B is the Boltzmann constant. From then on, all the relations become exactly similar to those in the classical case.

§22.2.2 Canonical Ensemble

Canonical ensemble is specified by the macrostate $M \equiv (T, \mathbf{x})$, where T is the temperature of the system. In quantum case, we just replace integration over phase space to sum over all the states of the system *i.e.*

$$\int \mathrm{d}\mu \to \sum_n \tag{22.18}$$

The canonical ensemble is defined by the density matrix

$$\rho_{mn} = \delta_{mn} e^{-\beta E_n}, \tag{22.19}$$

so the density operator becomes

$$\hat{\rho} = \sum_n |\Phi_n\rangle\, e^{-\beta E_n}\, \langle\Phi_n| = \sum_n e^{-\beta\hat{\mathcal{H}}}\, |\Phi_n\rangle\langle\Phi_n| = e^{-\beta\hat{\mathcal{H}}} \sum_n |\Phi_n\rangle\langle\Phi_n| = e^{-\beta\hat{\mathcal{H}}} \tag{22.20}$$

where $\hat{\mathcal{H}}$ is the Hamiltonian operator of the system, and last equality follows from the completeness relation $\sum_n |\Phi_n\rangle\langle\Phi_n| = \hat{\mathbb{1}}$.

> Remark 22.2. It must be noted that the sum here is over all the states and not over energy eigenvalues.

The partition function is defined as

$$Z(T, \mathbf{x}) = \mathrm{Tr}\, \hat{\rho} \equiv \mathrm{Tr}\left(e^{-\beta\hat{\mathcal{H}}}\right) = \sum_n e^{-\beta E_n} \tag{22.21}$$

With this, the ensemble average of an observable $\hat{\mathcal{O}}$ is given by

$$\langle\hat{\mathcal{O}}\rangle = \frac{\mathrm{Tr}\left(\hat{\mathcal{O}} e^{-\beta\hat{\mathcal{H}}}\right)}{\mathrm{Tr}\left(e^{-\beta\hat{\mathcal{H}}}\right)} \tag{22.22}$$

§22.2.3 Grand Canonical Ensemble

Grand canonical ensemble is specified by the macrostate $M \equiv (T, \mu, \mathbf{x})$, where T is the temperature of the system and μ is the chemical potential. For the grand canonical ensemble the density operator p operates on a Hilbert space with an indefinite number of particles.

Since it is enough to calculate the 'grand canonical partition function' for further calculations, we define it as

$$\mathcal{Q}(\mathfrak{z}, T, \mathbf{x}) = \sum_{N=0}^{\infty} \mathfrak{z}^N Z(N, T, \mathbf{x}) \tag{22.23}$$

where $\mathfrak{z} = e^{\beta\mu}$ is the fugacity of the system and $Z(N, T, \mathbf{x})$ is the canonical partition function for ensemble with macrostate $(N, T, \mathbf{x})$. The ensemble average of an observable $\hat{\mathcal{O}}$ is given by

$$\langle\hat{\mathcal{O}}\rangle = \frac{1}{\mathcal{Q}} \sum_{N=0}^{\infty} \mathfrak{z}^N \langle\hat{\mathcal{O}}\rangle_N \tag{22.24}$$

where $\langle \hat{\mathcal{O}} \rangle_N$ is the ensemble average of the observable $\hat{\mathcal{O}}$ in the canonical ensemble for N particles.

Let's say $\hat{N}$ is the number operator of the system which commutes with the Hamiltonian and counts the number of particles in the system. The grand canonical partition function is given by

$$\mathcal{Q}(\mathfrak{z}, T, \mathbf{x}) = \mathrm{Tr}\left(e^{-\beta(\hat{\mathcal{H}} - \mu \hat{N})}\right) \tag{22.25}$$

and the ensemble average of an observable $\hat{\mathcal{O}}$ is given by

$$\langle \hat{\mathcal{O}} \rangle = \frac{1}{\mathcal{Q}} \mathrm{Tr}\left(\hat{\mathcal{O}} e^{-\beta(\hat{\mathcal{H}} - \mu \hat{N})}\right). \tag{22.26}$$

§22.3 Maxwell-Boltzmann, Bose-Einstein, and Fermi-Dirac Statistics

In the previous sections, we have seen how to define different ensembles in quantum statistical mechanics. Now, we will see how different properties of the constituents particles of the system lead to different statistics.

Remark 22.3 (Identical vs Indistinguishable Particles). Particles that cannot be distinguished from each other in the given system (*i.e.* for the given Hamiltonian all the particles are same), but we can perform *some experiment* to distinguish them (*i.e.* we can label them) are called *identical and distinguishable* particles.

Particles that cannot be distinguished from each other in the given system, and *no physical experiment* can be performed to distinguish them (*i.e.* we cannot label them) are called *identical and indistinguishable* particles.

Now, with this distinction, we can define the following statistics:

- **Maxwell-Boltzmann Statistics:** Particles that are identical and distinguishable are said to follow Maxwell-Boltzmann statistics.
- **Bose-Einstein Statistics:** Particles that are identical and indistinguishable and there eigenstates are symmetric under exchange are said to follow Bose-Einstein statistics. These particles are called *bosons*.
- **Fermi-Dirac Statistics:** Particles that are identical and indistinguishable and there eigenstates are antisymmetric under exchange are said to follow Fermi-Dirac statistics. These particles are called *fermions*.

Example 22.4 (Some explicit calculations)

Let us consider 2 identical particles which need to be placed in 2 different states "1" and "2".

- Maxwell-Boltzmann Statistics: Since the particles are distinguishable, we can label them as A and B. Thus, the possible states are

"1"	"2"
AB	
	AB
A	B
B	A

  We get a total of 4 states.

- Bose-Einstein Statistics: Since the particles are indistinguishable, both particles are labelled as A. Thus, the possible states are

"1"	"2"
AA	
	AA
A	A

We get a total of 3 states.

- Fermi-Dirac Statistics: Since the particles are indistinguishable, both particles are labelled as A. But due to Pauli's exclusion principle, both particles cannot occupy the same state. Thus, the possible states are

"1"	"2"
A	A

Thus, we get a total of 1 state.

With this, simple example we can see that how the indistinguishability of particles influences the number of states of the system.

Remark 22.5 (Boson vs Fermions). Due to this antisymmetric property of eigenstates of fermions, no two fermions can occupy the same state *i.e.* no two fermions can have the same set of quantum numbers. This is known as *Pauli's exclusion principle*. Thus, the maximum number of fermions that can occupy a state is 1. On the other hand, bosons can occupy the same state, and thus there is no restriction on the number of bosons that can occupy a state.

Appart from this, all the integer spin particles are bosons and all the half-integer spin particles are fermions.

Now since we have understood the importance of indistinguishability of particles, we will now see how to calculate the distribution function for different statistics.

Consider a gas of N identical particles in a volume V at temperature T. We shall use the following notation:

- Label the possible quantum states of a single particle by r or s and the corresponding energy by ϵ_r or ϵ_s.
- Denote the number of particles in the state s by n_s.
- Label the possible quantum states of the system by R or S and the corresponding energy by E_R or E_S.

So the state R of the system is defined by the set of occupation numbers $\{n_s\}$, and the energy of the system is given by

$$E_R = \sum_s n_s \epsilon_s. \tag{22.27}$$

Since the total number of particles in the system is N, we have

$$\sum_s n_s = N. \tag{22.28}$$

The partition function for the system is given by

$$Z = \sum_R e^{-\beta E_R} = \sum_{\{n_s\}} e^{-\beta \sum_s n_s \epsilon_s} \tag{22.29}$$

where the sum is over all the possible sets of occupation numbers $\{n_s\}$ that satisfy the constraint $\sum_s n_s = N$. The probability of finding the system in the state R is given by

$$P_R = \frac{e^{-\beta E_R}}{Z} = \frac{e^{-\beta \sum_s n_s \epsilon_s}}{Z} \tag{22.30}$$

where Z is the partition function of the system. The mean number of particles in the state s is given by

$$\langle n_s \rangle = \sum_R n_s P_R = \frac{1}{Z} \sum_R n_s e^{-\beta E_R} \tag{22.31}$$

We can rewrite this as

$$\langle n_s \rangle = \frac{1}{Z} \sum_R -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} e^{-\beta E_R} = -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} \ln Z. \tag{22.32}$$

§22.3.1 Maxwell-Boltzmann Statistics

The partition function for the Maxwell-Boltzmann statistics is given by

$$Z = \sum_{\{n_s\}} e^{-\beta \sum_s n_s \epsilon_s} \tag{22.33}$$

where the sum is over all the possible sets of occupation numbers $\{n_s\}$ that satisfy the constraint $\sum_s n_s = N$.

Since the particles are distinguishable. Thus, any permutation of two particles in different states must be counted as a distinct state. For a set of occupation numbers $\{n_s\}$, the number of states with same energy is given by

$$\frac{N!}{\prod_s n_s!}. \tag{22.34}$$

Thus, the partition function becomes

$$Z = \sum_{n_1, n_2, \ldots} \frac{N!}{\prod_s n_s!} e^{-\beta \sum_s n_s \epsilon_s} \tag{22.35}$$

$$= \sum_{n_1, n_2, \ldots} \frac{N!}{\prod_s n_s!} \prod_s \left(e^{-\beta \epsilon_s}\right)^{n_s} \tag{22.36}$$

since $\{n_s\}$ is constrained by $\sum_s n_s = N$, the last expression is just a multinomial expansion. Thus, we can write

$$= \left(\sum_s e^{-\beta \epsilon_s}\right)^N = Z_1^N \tag{22.37}$$

where Z_1 is single particle partition function. Thus, we have

$$\ln Z = N \ln \left(\sum_s e^{-\beta \epsilon_s}\right) \tag{22.38}$$

Now we can calculate the mean occupation number of the state s as

$$\langle n_s \rangle = -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} \ln Z = -\frac{1}{\beta} N \frac{-\beta e^{-\beta \epsilon_s}}{\sum_r e^{-\beta \epsilon_r}} = N \frac{e^{-\beta \epsilon_s}}{\sum_r e^{-\beta \epsilon_r}} = N \frac{e^{-\beta \epsilon_s}}{Z_1}. \tag{22.39}$$

This is called the "*Maxwell-Boltzmann distribution.*"

§22.3.2 Planck and Bose-Einstein Statistics

Here the particles are indistinguishable, and each state can be occupied by any number of particles. Thus, the partition function is given by

$$Z = \sum_{\{n_s\}} e^{-\beta \sum_s n_s \epsilon_s}, \quad \text{where } n_s = 0, 1, 2, \ldots \tag{22.40}$$

This sum can be rewritten as

$$Z = \sum_{n_1, n_2, \ldots} \prod_s e^{-\beta n_s \epsilon_s} \tag{22.41}$$

$$= \prod_s \left[\sum_{n_s=0}^{\infty} e^{-\beta n_s \epsilon_s}\right] \tag{22.42}$$

$$= \prod_s \frac{1}{1 - e^{-\beta \epsilon_s}} \tag{22.43}$$

Thus, we have

$$\ln Z = -\sum_s \ln\left(1 - e^{-\beta \epsilon_s}\right). \tag{22.44}$$

Now we can calculate the mean occupation number of the state s as

$$\langle n_s \rangle = -\frac{1}{\beta}\frac{\partial}{\partial \epsilon_s} \ln Z = \frac{e^{-\beta\epsilon_s}}{1-e^{-\beta\epsilon_s}} = \frac{1}{e^{\beta\epsilon_s}-1}. \tag{22.45}$$

This is called the "*Plank distribution.*"

Photons and phonons are examples of bosons, but their number is not conserved because they can be absorbed or emitted. Other bosons, such as helium-4 atoms, do have their number conserved. If the number of bosons is conserved, then $\langle n_s \rangle$ must satisfy the constraint

$$\sum_s \langle n_s \rangle = N \tag{22.46}$$

where N is the number of bosons in the system. In order to satisfy this constraint, we need to modify the distribution function as

$$\langle n_s \rangle = \frac{1}{e^{\beta(\epsilon_s-\mu)}-1} \tag{22.47}$$

where μ is the chemical potential of the system. This is called the "*Bose-Einstein distribution.*" The chemical potential μ is determined by the constraint $\sum_s \langle n_s \rangle = N$. The Bose-Einstein distribution is valid for all bosons, but it is only necessary to include the chemical potential for bosons with conserved number.

With this the correct partition function for the Bose-Einstein statistics is given by

$$\ln \mathcal{Z} = -\sum_s \ln\left(1-e^{-\beta(\epsilon_s-\mu)}\right) \tag{22.48}$$

where $\mathcal{Z}$ is the grand canonical partition function. With this we can calculate the average number of particles in the system as

$$\begin{aligned}
\langle N \rangle &= \frac{1}{\mathcal{Z}}\sum_R N e^{-\beta E_R} e^{\beta\mu N} && (22.49)\\
&= \frac{1}{\beta}\frac{1}{\mathcal{Z}}\frac{\partial}{\partial\mu}\sum_R e^{-\beta E_R} e^{\beta\mu N} && (22.50)\\
&= \frac{1}{\beta}\frac{\partial}{\partial\mu}\ln\mathcal{Z} && (22.51)\\
&= \frac{1}{\beta}\frac{\partial}{\partial\mu}\left[-\sum_s \ln\left(1-e^{-\beta(\epsilon_s-\mu)}\right)\right] && (22.52)\\
&= \frac{1}{\beta}\sum_s \frac{\beta e^{-\beta(\epsilon_s-\mu)}}{1-e^{-\beta(\epsilon_s-\mu)}} && (22.53)\\
&= \sum_s \frac{1}{e^{\beta(\epsilon_s-\mu)}-1} && (22.54)
\end{aligned}$$

By setting $\langle N \rangle = N$, we can fix the chemical potential μ. Finally, the average number of particles in the state s is given by

$$\langle n_s \rangle = -\frac{1}{\beta}\frac{\partial}{\partial\epsilon_s}\ln\mathcal{Z} = \frac{1}{e^{\beta(\epsilon_s-\mu)}-1} \tag{22.55}$$

Note that $\langle N \rangle = \sum_s \langle n_s \rangle$ is satisfied.

For bosons with non-conserved number, we can set $\mu = 0$ in the above equation.

§22.3.3 Fermi-Dirac Statistics

Here the particles are indistinguishable, and each state can be occupied by at most one particle *i.e.* $n_r = 0, 1$ for each r. And since the particle number is also not conserved in general, we can use the grand canonical partition function

$$\begin{aligned}
\mathcal{Z} &= \sum_R e^{-\beta(E_R-\mu N)} && (22.56)\\
&= \sum_{n_1,n_2,\ldots} e^{-\beta\sum_s n_s\epsilon_s} e^{\beta\mu\sum_s n_s} && (22.57)
\end{aligned}$$

$$= \sum_{n_1, n_2, \ldots} \prod_s e^{-\beta(\epsilon_s - \mu) n_s} \tag{22.58}$$

$$= \prod_s \left[\sum_{n_s=0}^{1} e^{-\beta(\epsilon_s - \mu) n_s} \right] \tag{22.59}$$

$$= \prod_s \left[1 + e^{-\beta(\epsilon_s - \mu)} \right] \tag{22.60}$$

Thus, we have

$$\ln \mathcal{Z} = \sum_s \ln\Big(1 + e^{-\beta(\epsilon_s - \mu)}\Big). \tag{22.61}$$

So the average number of particles in the state s is given by

$$\langle n_s \rangle = -\frac{1}{\beta} \frac{\partial}{\partial \epsilon_s} \ln \mathcal{Z} = \frac{1}{e^{\beta(\epsilon_s - \mu)} + 1} \tag{22.62}$$

This is called the "*Fermi-Dirac distribution*". And the average number of particles in the system is given by

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln \mathcal{Z} = \sum_s \frac{1}{e^{\beta(\epsilon_s - \mu)} + 1} \tag{22.63}$$

$$= \sum_s \langle n_s \rangle \tag{22.64}$$

By setting $\langle N \rangle = N$, we can fix the chemical potential μ.

23 Lecture 23: (26/03/2025)

§23.1 Summary

In the grand-canonical ensemble, we have the partition function

$$\mathcal{Z} = \begin{cases} \sum_{N=0}^{\infty} \mathfrak{z}^N Z(N,V,T) & \text{for distinguishable particles} \\ \prod_s \left(\frac{1}{1-\mathfrak{z}e^{-\beta\epsilon_s}} \right) & \text{for bosons} \\ \prod_s \left(1+\mathfrak{z}e^{-\beta\epsilon_s} \right) & \text{for fermions} \end{cases} \tag{23.1}$$

where $\mathfrak{z} = e^{\beta\mu}$ is the fugacity and $Z(N,V,T)$ is the canonical partition function for N particles. And the distribution function is given by

$$\langle n_s \rangle = \begin{cases} \mathfrak{z}e^{-\beta\epsilon_s} & \text{for distinguishable particles} \\ \frac{1}{\mathfrak{z}^{-1}e^{\beta\epsilon_s}-1} & \text{for bosons} \\ \frac{1}{\mathfrak{z}^{-1}e^{\beta\epsilon_s}+1} & \text{for fermions} \end{cases} \tag{23.2}$$

where $\langle n_s \rangle$ is the mean occupation number of the state s with energy ϵ_s.

§23.2 Classical Limit

Let us consider 2 limiting cases:

1. Low density limit *i.e.* $\langle n_s \rangle \ll 1$ so that $\exp[\beta(\epsilon_s - \mu)] \gg 1$.
2. High temperature limit *i.e.* $\epsilon_s \gg \mu$ so that $\exp[\beta(\epsilon_s - \mu)] \gg 1$.

In both cases, we can reduce the mean occupation number $\langle n_s \rangle$ to

$$\langle n_s \rangle = e^{-\beta(\epsilon_s - \mu)} \tag{23.3}$$

Plugging this into $\sum_s \langle n_s \rangle = N$ gives us

$$\sum_s e^{-\beta(\epsilon_s-\mu)} = N \quad \Rightarrow \quad e^{\beta\mu} = \frac{N}{\sum_s e^{-\beta\epsilon_s}}. \tag{23.4}$$

Using this, we can rewrite the mean occupation number as

$$\langle n_s \rangle = N \frac{e^{-\beta\epsilon_s}}{\sum_s e^{-\beta\epsilon_s}} \tag{23.5}$$

which is the Boltzmann distribution. Hence, in the classical limit of low density or high temperature, both the Fermi-Dirac and Bose-Einstein distributions reduce to the Boltzmann distribution.

§23.3 Relationship between Z and $\mathcal{Z}$

In deriving the Bose-Einstein and Fermi-Dirac distributions, we have used the grand-canonical partition function $\mathcal{Z}$. We can relate this to the canonical partition function Z using the following relation:

$$\mathcal{Z} = \sum_R e^{-\beta(E_R - \mu N_R)} \tag{23.6}$$

$$= \sum_{N=0}^{\infty} e^{\beta\mu N} \sum_{R(N)} e^{-\beta E_R} \tag{23.7}$$

where $R(N)$ is the set of all states with N particles

$$\tag{23.8}$$

$$= \sum_{N=0}^{\infty} e^{\beta\mu N} Z(N) \tag{23.9}$$

where $Z(N)$ is the canonical partition function for N particles. Since $Z(N)$ is a rapidly increasing function of N and $e^{\beta\mu N}$ is a rapidly decreasing function of N (for $\mu < 0$), the grand partition function is sharply peaked at $N = \langle N \rangle$. So we can write

$$\mathcal{Z} \approx e^{\beta\mu\langle N\rangle} Z(\langle N\rangle)\, \delta[N] \tag{23.10}$$

where $\delta[N]$ is the width of the peak. Thus,

$$\ln \mathcal{Z} = \beta\mu \langle N\rangle + \ln Z(\langle N\rangle) + \ln \delta[N] \tag{23.11}$$

$$\approx \beta\mu \langle N\rangle + \ln Z(\langle N\rangle) \tag{23.12}$$

since $\ln \delta[N]$ is negligible. Or

$$\ln Z = \ln \mathcal{Z} - \beta\mu N \tag{23.13}$$

This is the relation between the partition function Z and the grand partition function $\mathcal{Z}$.

§23.4 Gibbs Paradox: Revisited

Plugging the formula for the grand partition function into eqn. (23.13) gives us (for both bosons and fermions)

$$\ln Z = -\beta\mu N \pm \sum_s \ln\left(1 \pm e^{-\beta(\epsilon_s - \mu)}\right) \tag{23.14}$$

where $\pm$ is $+$ for fermions and $-$ for bosons. At high temperature (*i.e.* $\epsilon_s \gg \mu$), we can use the Taylor expansion of $\ln(1 \pm x)$ to get

$$\ln Z = -\beta\mu N \pm \sum_s \left(\pm e^{-\beta(\epsilon_s - \mu)}\right) = -\beta\mu N + N \tag{23.15}$$

where in the last step we used the Maxwell-Boltzmann distribution $\langle n_s\rangle = e^{-\beta(\epsilon_s-\mu)}$ to get $N = \sum_s \langle n_s\rangle = \sum_s e^{-\beta(\epsilon_s-\mu)}$. We have

$$N = e^{\beta\mu} \sum_s e^{-\beta\epsilon_s} \tag{23.16}$$

take the logarithm of both sides, and we get

$$\ln N = \beta\mu + \ln\left(\sum_s e^{-\beta\epsilon_s}\right) \quad \Rightarrow \quad -\beta\mu = \ln\left(\sum_s e^{-\beta\epsilon_s}\right) - \ln N \tag{23.17}$$

Plugging this into eqn. (23.15) gives us

$$\ln Z = N \ln\left(\sum_s e^{-\beta\epsilon_s}\right) - N\ln N + N \tag{23.18}$$

first term is the canonical partition function for N distinguishable particles, the second term is $\ln N!$ after Stirling's approximation. So we have

$$\ln Z = \ln Z_{\text{MB}} - \ln N! \quad \Rightarrow \quad Z = \frac{Z_{\text{MB}}}{N!} \tag{23.19}$$

This was the resolution to the Gibbs paradox. Without quantum mechanics we had to put in the factor $N!$ by hand. But now we see that the indistinguishability of the particles comes out naturally.

§23.5 Ideal Gas: Maxwell-Boltzmann Statistics

Let's consider an ideal gas of identical but distinguishable particles in a volume V at temperature T. The grand canonical partition function is given by

$$\mathcal{Z} = \sum_{N=0}^{\infty} \mathfrak{z}^N Z(N,V,T) \tag{23.20}$$

where $\mathfrak{z} = e^{\beta\mu}$ is the fugacity and $Z(N,V,T)$ is the canonical partition function for N particles. The canonical partition function is given by

$$Z(N,V,T) = \frac{1}{N!} Z_1(V,T)^N \tag{23.21}$$

where $Z_1(V,T)$ is the canonical partition function for a single particle. For an ideal gas, we have

$$\begin{aligned}
Z_1(V,T) &= \int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{h^3} e^{-\beta\mathcal{H}(q,p)} && (23.22)\\
&= \frac{V}{h^3}\int \mathrm{d}^3\mathbf{p}\, e^{-\beta\frac{|\mathbf{p}|^2}{2m}} && (23.23)\\
&= \frac{V}{h^3}\int_0^\infty \mathrm{d}p\,(4\pi p^2) e^{-\beta\frac{p^2}{2m}} && (23.24)\\
&= \frac{V}{(2\pi\hbar)^3}(4\pi)\int_0^\infty \mathrm{d}p\, p^2 e^{-\frac{p^2}{2mk_BT}} && (23.25)
\end{aligned}$$

where we have used $\hbar = h/(2\pi)$ and $\beta = 1/(k_BT)$. Now, we can use the substitution $x^2 = \frac{p^2}{2mk_BT}$ to get

$$= \frac{V}{2\pi^2\hbar^3}(2mk_BT)^{3/2}\int_0^\infty \mathrm{d}x\, x^2 e^{-x^2} \tag{23.26}$$

again using the substitution $x^2 = t$ gives us

$$\begin{aligned}
&= \frac{V}{2\pi^2\hbar^3}(2mk_BT)^{3/2}\frac{1}{2}\int_0^\infty \mathrm{d}t\,\sqrt{t}e^{-t} && (23.27)\\
&= \frac{V}{2\pi^2\hbar^3}(2mk_BT)^{3/2}\frac{1}{2}\Gamma(3/2) && (23.28)\\
&= \frac{V}{2\pi^2\hbar^3}(2mk_BT)^{3/2}\frac{\sqrt{\pi}}{4} && (23.29)\\
&= \frac{V}{\lambda^3(T)} && (23.30)
\end{aligned}$$

where $\lambda(T) = \sqrt{\frac{2\pi\hbar^2}{mk_BT}}$ is the thermal de Broglie wavelength. So we have

$$Z_1(V,T) = \frac{V}{\lambda^3(T)} \quad\Rightarrow\quad Z(N,V,T) = \frac{V^N}{\lambda^{3N}(T)} \tag{23.31}$$

Say the average number of particles in the system is $\langle N\rangle$. Then, we have the following constraint

$$\sum_s \langle n_s\rangle = \langle N\rangle \quad\Rightarrow\quad \mathfrak{z}\sum_s e^{-\beta\epsilon_s} = \langle N\rangle \tag{23.32}$$

where $\langle n_s\rangle = \mathfrak{z}e^{-\beta\epsilon_s}$ is the mean occupation number of the state s with energy ϵ_s. So we have

$$\mathfrak{z} = \langle N\rangle\,\frac{\lambda^3(T)}{V}. \tag{23.33}$$

Plugging this into the grand canonical partition function gives us

$$\begin{aligned}
\mathcal{Z} &= \sum_{N=0}^{\infty}\frac{\mathfrak{z}^N}{N!}Z(N,V,T) && (23.34)\\
&= \sum_{N=0}^{\infty}\frac{1}{N!}\left(\langle N\rangle\,\frac{\lambda^3(T)}{V}\right)^N\frac{V^N}{\lambda^{3N}(T)} && (23.35)
\end{aligned}$$

$$= \sum_{N=0}^{\infty} \frac{1}{N!} (\langle N \rangle)^N \tag{23.36}$$

$$= e^{\langle N \rangle}. \tag{23.37}$$

Thus, the equation of state for an ideal gas is given by

$$PV = k_B T \ln \mathcal{Z} \quad \Rightarrow \quad PV = \langle N \rangle k_B T. \tag{23.38}$$

Let's compute the average energy of the system. The average energy is given by

$$\begin{aligned}
\langle E \rangle = U &= \sum_s \epsilon_s \langle n_s \rangle \\
&= \mathfrak{z} \sum_s \epsilon_s e^{-\beta \epsilon_s} \\
&= \mathfrak{z} \int \frac{\mathrm{d}^3\mathbf{q}\, \mathrm{d}^3\mathbf{p}}{h^3} \frac{|\mathbf{p}|^2}{2m} e^{-\beta \frac{|\mathbf{p}|^2}{2m}} \\
&= \mathfrak{z} \frac{V}{h^3} (4\pi) \int_0^\infty \mathrm{d}p\, \frac{p^4}{2m} e^{-\beta \frac{p^2}{2m}}
\end{aligned}$$

substituting $x^2 = \beta p^2/2m$ gives

$$\begin{aligned}
&= \mathfrak{z} \frac{V}{2\pi^2 \hbar^3} (2mk_BT)^{3/2} (k_BT) \int_0^\infty \mathrm{d}x\, x^4 e^{-x^2} \\
&= \mathfrak{z} \frac{V}{2\pi^2 \hbar^3} (2mk_BT)^{3/2} (k_BT) \frac{3\sqrt{\pi}}{8}
\end{aligned}$$

where we used $\int_0^\infty \mathrm{d}x\, x^4 e^{-x^2} = \frac{3\sqrt{\pi}}{8}$ and the value of fugacity $\mathfrak{z}$ to get

$$U = \frac{3}{2} \langle N \rangle k_B T \tag{23.39}$$

Recall the grand canonical partition function is given by

$$\mathcal{Z} = \sum_{N=0}^{\infty} \frac{\mathfrak{z}^N}{N!} \left(\frac{V}{\lambda^3(T)} \right)^N = \exp\left[\mathfrak{z} \frac{V}{\lambda^3(T)} \right]. \tag{23.40}$$

Thus, the grand potential is given by

$$\Omega = -k_B T \ln \mathcal{Z} \tag{23.41}$$

$$= -k_B T \frac{V}{\lambda^3(T)} \mathfrak{z} \tag{23.42}$$

The entropy is given by

$$S = -\left(\frac{\partial \Omega}{\partial T} \right)_{V,\mu} \tag{23.43}$$

$$= k_B \frac{V}{\lambda^3(T)} \mathfrak{z} + k_B T \mathfrak{z} V \frac{\partial}{\partial T} \left(\frac{1}{\lambda^3(T)} \right) + k_B T \frac{V}{\lambda^3(T)} \frac{\partial \mathfrak{z}}{\partial T} \tag{23.44}$$

$$= \langle N \rangle k_B + k_B T \mathfrak{z} V \frac{3}{2} \frac{1}{T \lambda^3(T)} + k_B T \frac{V}{\lambda^3(T)} \frac{\partial}{\partial T} \left(e^{\mu/(k_B T)} \right) \tag{23.45}$$

$$= \langle N \rangle k_B + \frac{3}{2} \langle N \rangle k_B + k_B T \frac{V}{\lambda^3(T)} \left(-\mathfrak{z} \frac{\mu}{k_B T^2} \right) \tag{23.46}$$

$$= \frac{5}{2} \langle N \rangle k_B - \langle N \rangle \frac{\mu}{T} \tag{23.47}$$

using the relation $\langle N \rangle = \frac{V}{\lambda^3(T)} \mathfrak{z}$, we can rewrite the last term as

$$= \frac{5}{2} \langle N \rangle k_B + \langle N \rangle k_B \ln \left(\frac{V}{\langle N \rangle} \frac{1}{\lambda^3(T)} \right) \tag{23.48}$$

So the entropy is given by

$$S = \langle N \rangle k_B \left[\frac{5}{2} + \ln \left(\frac{V}{\langle N \rangle} \frac{1}{\lambda^3(T)} \right) \right]. \tag{23.49}$$

This is the Sackur-Tetrode equation for an ideal gas. It is to be noted that this entropy does not satisfy the third law of thermodynamics. This should cause no concern, because a Boltzmann gas is not a physical system. It is only a model toward which the Bose and Fermi gas converge at high temperatures.

24 Lecture 24: (27/03/2025)

§24.1 Ideal Fermi Gas

Let's start with the equation of state for an ideal Fermi gas. The equation of state is given by:

$$\frac{PV}{k_B T} = \ln \mathcal{Z} \tag{24.1}$$

Let's calculate the RHS of the equation of state:

$$\ln \mathcal{Z}(\mathfrak{z}, V, T) = \sum_s \ln\left(1 + \mathfrak{z} e^{-\beta \epsilon_s}\right) = \int \frac{\mathrm{d}^3\mathbf{q}\, \mathrm{d}^3\mathbf{p}}{h^3} \ln\left(1 + \mathfrak{z} e^{-\beta |\mathbf{p}|^2/2m}\right) \tag{24.2}$$

$$= \frac{V}{h^3}(4\pi) \int_0^\infty \mathrm{d}p\, p^2 \ln\left(1 + \mathfrak{z} e^{-\beta p^2/2m}\right) \tag{24.3}$$

substituting $x^2 = \beta p^2/2m$

$$= \frac{V}{\lambda^3(T)} \frac{4}{\sqrt{\pi}} \int_0^\infty \mathrm{d}x\, x^2 \ln\left(1 + \mathfrak{z} e^{-x^2}\right) \tag{24.4}$$

We can't integrate this function analytically, but this integral will be a function of $\mathfrak{z}$. Define,

$$f_{5/2}(\mathfrak{z}) = \frac{4}{\sqrt{\pi}} \int_0^\infty \mathrm{d}x\, x^2 \ln\left(1 + \mathfrak{z} e^{-x^2}\right). \tag{24.5}$$

We can taylor expand the logarithm, and we have a series expansion for $f_{5/2}(\mathfrak{z})$:

$$\begin{aligned}
f_{5/2}(\mathfrak{z}) &= \frac{4}{\sqrt{\pi}} \int_0^\infty \mathrm{d}x\, x^2 \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n} e^{-nx^2} \\
&= \frac{4}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n} \int_0^\infty \mathrm{d}x\, x^2 e^{-nx^2} \\
&= \frac{4}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n} \left[-\frac{\mathrm{d}}{\mathrm{d}n}\left(\int_0^\infty \mathrm{d}x\, e^{-nx^2}\right)\right] \\
&= \frac{4}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n} \left[-\frac{\mathrm{d}}{\mathrm{d}n}\left(\frac{1}{2}\sqrt{\frac{\pi}{n}}\right)\right] \\
&= \frac{4}{\sqrt{\pi}} \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n} \left[\frac{\sqrt{\pi}}{4} \frac{1}{n^{3/2}}\right] \\
&= \sum_{n=1}^\infty (-1)^{n+1} \frac{\mathfrak{z}^n}{n^{5/2}}
\end{aligned} \tag{24.6}$$

So the equation of state for an ideal Fermi gas is:

$$\ln \mathcal{Z}(\mathfrak{z}, V, T) = \frac{V}{\lambda^3(T)} f_{5/2}(\mathfrak{z}) \quad \Rightarrow \quad \beta P = \frac{1}{\lambda^3(T)} f_{5/2}(\mathfrak{z}). \tag{24.7}$$

Now, let's calculate the average number of particles in the system, which is given by:

$$\langle N \rangle = \sum_s \frac{1}{\mathfrak{z}^{-1} e^{\beta \epsilon_s} + 1} = \frac{1}{\beta} \frac{\partial}{\partial \mu} \ln \mathcal{Z} \tag{24.8}$$

Now rewritting the last derivative in terms of the fugacity:

$$\langle N \rangle = \frac{1}{\beta} \frac{\partial \mathfrak{z}}{\partial \mu} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Z} = \mathfrak{z} \frac{\partial}{\partial \mathfrak{z}} \ln \mathcal{Z}. \tag{24.9}$$

Also, we have

$$\langle N \rangle = \sum_s \frac{1}{\mathfrak{z}^{-1} e^{\beta \epsilon_s} + 1} \tag{24.10}$$

$$= \int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{h^3} \frac{1}{\mathfrak{z}^{-1}e^{\beta|\mathbf{p}|^2/2m}+1} \tag{24.11}$$

$$= \frac{V}{h^3}(4\pi)\int_0^\infty \mathrm{d}p\, p^2 \frac{1}{\mathfrak{z}^{-1}e^{\beta p^2/2m}+1} \tag{24.12}$$

substituting $x^2 = \beta p^2/2m$

$$= \frac{V}{\lambda^3(T)}\frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, \frac{x^2}{\mathfrak{z}^{-1}e^{-x^2}+1} \tag{24.13}$$

This integral also doesn't have a closed form, so call this integral $f_{3/2}(\mathfrak{z})$:

$$f_{3/2}(\mathfrak{z}) = \frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, \frac{x^2}{\mathfrak{z}^{-1}e^{-x^2}+1}. \tag{24.14}$$

So we have:

$$\langle N\rangle = \frac{V}{\lambda^3(T)} f_{3/2}(\mathfrak{z}) \quad \Rightarrow \quad \frac{\langle N\rangle}{V} = \frac{1}{v} = \frac{1}{\lambda^3(T)} f_{3/2}(\mathfrak{z}), \tag{24.15}$$

where v is the volume per particle. Comparing eqns. (24.8) and (24.15), we have the following relation:

$$f_{3/2}(\mathfrak{z}) = \mathfrak{z}\frac{\partial}{\partial \mathfrak{z}} f_{5/2}(\mathfrak{z}). \tag{24.16}$$

§24.2 Ideal Bose Gas

Recall the mean occupation number in state s:

$$\langle n_s\rangle = \frac{1}{e^{\beta(\epsilon_s-\mu)}-1} \tag{24.17}$$

for $\epsilon_s - \mu < 0$, the mean occupation number is negative, which is unphysical. So we have to restrict the chemical potential such that $\mu < \epsilon_s$ for all states. Since the ground state has zero energy, we have $\mu < 0$, hence $\beta\mu < 0$.

Let's take a closer look at the mean occupation number for ground state $s = 0$ *i.e.* $\epsilon_0 = 0$ in the limit $-\beta\mu \gg 1$

$$\langle n_0\rangle = \frac{1}{e^{-\beta\mu}-1} = \frac{1}{(1-\beta\mu+\cdots)-1} \approx -\frac{1}{\beta\mu}, \tag{24.18}$$

which means that $\langle n_0\rangle$ diverges *i.e.* the ground state may hence be *macroscopically* occupied. This is the case when

$$\frac{1}{\beta|\mu|} \sim N, \qquad \text{or} \qquad |\mu| \sim \frac{k_BT}{N} \qquad \Rightarrow \qquad 1-\mathfrak{z} \sim \frac{1}{N}. \tag{24.19}$$

Thus, we need to consider the ground state separately. The partition function is given by:

$$\ln\mathcal{Z} = -\sum_r \ln\Big(1-\mathfrak{z}e^{-\beta\epsilon_r}\Big) = -\ln(1-\mathfrak{z}) - \sum_{r\neq 0}\ln\Big(1-\mathfrak{z}e^{-\beta\epsilon_r}\Big). \tag{24.20}$$

Let's explicitly calculate the partition function,

$$\ln\mathcal{Z} = -\ln(1-\mathfrak{z}) - \int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{h^3}\ln\Big(1-\mathfrak{z}e^{-\beta|\mathbf{p}|^2/2m}\Big) \tag{24.21}$$

$$= -\ln(1-\mathfrak{z}) - \frac{V}{h^3}(4\pi)\int_0^\infty \mathrm{d}p\, p^2 \ln\Big(1-\mathfrak{z}e^{-\beta p^2/2m}\Big) \tag{24.22}$$

substituting $x^2 = \beta p^2/2m$

$$= -\ln(1-\mathfrak{z}) - \frac{V}{\lambda^3(T)}\frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, x^2 \ln\Big(1-\mathfrak{z}e^{-x^2}\Big). \tag{24.23}$$

This integral also doesn't have a closed form, so call this integral $g_{5/2}(\mathfrak{z})$:

$$g_{5/2}(\mathfrak{z}) = -\frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, x^2 \ln\Big(1-\mathfrak{z}e^{-x^2}\Big). \tag{24.24}$$

Taylor expanding the logarithm in the integral, we have:

$$\begin{aligned}
g_{5/2}(\mathfrak{z}) &= -\frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, x^2\left[-\sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n}e^{-nx^2}\right] \\
&= \frac{4}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n}\int_0^\infty \mathrm{d}x\, x^2 e^{-nx^2} \\
&= \frac{4}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n}\left[-\frac{\mathrm{d}}{\mathrm{d}n}\left(\int_0^\infty \mathrm{d}x\, e^{-nx^2}\right)\right] \\
&= \frac{4}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n}\left[-\frac{\mathrm{d}}{\mathrm{d}n}\left(\frac{1}{2}\sqrt{\frac{\pi}{n}}\right)\right] \\
&= \frac{4}{\sqrt{\pi}}\sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n}\left[\frac{\sqrt{\pi}}{4}\frac{1}{n^{3/2}}\right] \\
&= \sum_{n=1}^{\infty}\frac{\mathfrak{z}^n}{n^{5/2}}
\end{aligned} \tag{24.25}$$

So, the equation of state for an ideal Bose gas is:

$$\ln\mathcal{Z} = \frac{V}{\lambda^3(T)}g_{5/2}(\mathfrak{z}) - \ln(1-\mathfrak{z}) \quad \Rightarrow \quad \beta P = \frac{1}{\lambda^3(T)}g_{5/2}(\mathfrak{z}) - \frac{1}{V}\ln(1-\mathfrak{z}). \tag{24.26}$$

Now, let's calculate the average number of particles in the system, which is given by:

$$\langle N\rangle = \frac{4\pi V}{h^3}\int_0^\infty \mathrm{d}p\, p^2 \frac{1}{\mathfrak{z}^{-1}e^{\beta p^2/2m}-1} + \frac{1}{\mathfrak{z}^{-1}-1} \tag{24.27}$$

Now substituting $x^2 = \beta p^2/2m$, we have

$$\langle N\rangle = \frac{V}{\lambda^3(T)}\frac{4}{\sqrt{\pi}}\int_0^\infty \mathrm{d}x\, \frac{x^2}{\mathfrak{z}^{-1}e^{x^2}-1} + \frac{\mathfrak{z}}{1-\mathfrak{z}} \tag{24.28}$$

this integral also doesn't have a closed form, so call this integral $g_{3/2}(\mathfrak{z})$

$$\langle N\rangle = \frac{V}{\lambda^3(T)}g_{3/2}(\mathfrak{z}) + \frac{\mathfrak{z}}{1-\mathfrak{z}}. \tag{24.29}$$

We also know that $\langle N\rangle = \mathfrak{z}\frac{\partial}{\partial\mathfrak{z}}\ln\mathcal{Z}$, so we have

$$\langle N\rangle = \mathfrak{z}\frac{\partial}{\partial\mathfrak{z}}\ln\mathcal{Z} = \frac{V}{\lambda^3(T)}\left[\mathfrak{z}\frac{\partial}{\partial\mathfrak{z}}\Big(g_{5/2}(\mathfrak{z})\Big)\right] + \frac{\mathfrak{z}}{1-\mathfrak{z}} \tag{24.30}$$

Comparing eqns. (24.29) and (24.30), we have the following relation:

$$g_{3/2}(\mathfrak{z}) = \mathfrak{z}\frac{\partial}{\partial\mathfrak{z}}g_{5/2}(\mathfrak{z}) \tag{24.31}$$

From the above equations, we can see that the average number of particles in the ground state is given by

$$\langle n_0\rangle = \frac{\mathfrak{z}}{1-\mathfrak{z}}. \tag{24.32}$$

25 Lecture 25: (02/04/2025)

§25.1 Bose-Einstein Condensation

Recall that for convergence of the grand partition function we require $\mu < 0$ so that $\mathfrak{z} \in [0, 1]$ and the specific volume of ideal Bose gas

$$\frac{1}{v} = \frac{\langle N \rangle}{V} = \frac{1}{\lambda^3(T)} g_{3/2}(\mathfrak{z}) + \frac{1}{V} \frac{\mathfrak{z}}{1 - \mathfrak{z}} \tag{25.1}$$

where $\lambda(T) = \sqrt{2\pi\hbar^2/mk_BT}$ is the thermal de Broglie wavelength, and $g_{3/2}(\mathfrak{z})$ is the Bose-Einstein integral defined as

$$g_n(\mathfrak{z}) = \sum_{l=1}^{\infty} \frac{\mathfrak{z}^l}{l^n} = \frac{1}{\Gamma(n)} \int_0^{\infty} \frac{x^{n-1}}{\mathfrak{z}^{-1}e^x - 1} \, \mathrm{d}x \,. \tag{25.2}$$

Moreover, at $\mathfrak{z} = 1$, it is equal to the Riemann zeta function

$$g_n(1) = \zeta(n) = \sum_{l=1}^{\infty} \frac{1}{l^n}. \tag{25.3}$$

For our particular problem it will be useful to know that $g_{3/2}(1) = \zeta(3/2) \approx 2.612$.

Let's look at two extreme temperature limits *i.e.* $T \to 0$ and $T \to \infty$.

- **High temperature limit** $T \to \infty$: When $T \to \infty$, $\beta \to 0$ so naively it looks as if $\mathfrak{z} \to 1$. But this is too naive. In this limit, the average occupation number for the ground state will be negligible, and the average number of particles in the system is given by

$$\langle N \rangle = \frac{V}{\lambda^3(T)} g_{3/2}(\mathfrak{z}) \quad \Rightarrow \quad \frac{\langle N \rangle}{V} = \frac{1}{\lambda^3(T)} \left(\mathfrak{z} + \frac{\mathfrak{z}^2}{2^{3/2}} + \frac{\mathfrak{z}^3}{3^{3/2}} + \cdots \right) \tag{25.4}$$

  If we keep the average particle number $\langle N \rangle$ fixed in eqn. (24.27) then μ must vary as we change the temperature, and it turns out that $\mu \to -\infty$ faster than $\beta \to 0$. To see this, notice that to leading order we need $\mathfrak{z}/\lambda^3(T)$ to be constant, so $\mathfrak{z} \sim T^{-3/2}$

  Thus, as $T \to \infty$ we have $\mathfrak{z} \to 0$.

- **Low temperature limit** $T \to 0$: Since high temperature limit is $\mathfrak{z} \to 0$, we'll anticipate that low temperature limit is $\mathfrak{z} \to 1$.

  Recall that in the low temperature limit, we have seen that the ground state is macroscopically occupied, *i.e.*

$$\langle n_0 \rangle = \frac{1}{\mathfrak{z}^{-1} - 1} \sim N \quad \Rightarrow \quad \mathfrak{z} \sim 1 - \frac{1}{N}. \tag{25.5}$$

  So zero temperature limit is $\mathfrak{z} \to 1$ in thermodynamic limit.

Rewrite the expression in eqn. (25.1), we have

$$\lambda^3(T) \frac{\langle n_0 \rangle}{V} = \frac{\lambda^3(T)}{v} - g_{3/2}(\mathfrak{z}) \tag{25.6}$$

since LHS is non-negative, we have

$$\frac{\lambda^3(T)}{v} \geq g_{3/2}(\mathfrak{z}) \tag{25.7}$$

From the power series expansion of $g_{3/2}(\mathfrak{z})$, we know that $g_{3/2}$ is monotonically increasing function of $\mathfrak{z}$. So, we have

$$\frac{\lambda^3(T)}{v} \geq g_{3/2}(1). \tag{25.8}$$

We define a transition between a gas phase and a condensed phase when $\lambda^3(T)/v = g_{3/2}(1)$. This is the condition for Bose-Einstein condensation. This condition gives us a temperature scale T_c *i.e.* at T_c, we have

$$\frac{\lambda^3(T_c)}{v} = g_{3/2}(1) = \zeta(3/2) \tag{25.9}$$

using the definition of $\lambda(T)$, we have

$$T_c = \left(\frac{2\pi\hbar^2}{mk_B}\right)\left(\frac{1}{\zeta(3/2)v}\right)^{2/3} = \left(\frac{2\pi\hbar^2}{mk_B}\right)\left(\frac{1}{\zeta(3/2)}\frac{\langle N\rangle}{V}\right)^{2/3} \tag{25.10}$$

Now, let's calculate the fraction of particles in the ground state when $T < T_c$. For this set $\mathfrak{z} = 1$ in eqn. (25.6), we have

$$\frac{\langle n_0\rangle}{\langle N\rangle} = 1 - \frac{1}{\lambda^3(T)}\frac{V}{\langle N\rangle}\zeta(3/2) = 1 - \left(\frac{\lambda(T_c)}{\lambda(T)}\right)^3 = 1 - \left(\frac{T}{T_c}\right)^{3/2} \tag{25.11}$$

At temperatures $T < T_c$, a macroscopic number of atoms discard their individual identities and merge into a communist collective, a single quantum state so large that it can be seen with the naked eye. This is known as the Bose-Einstein condensate. It provides an exquisitely precise playground in which many quantum phenomena can be tested.

§25.2 Equipartition Theorem: Revisit

The equipartition theorem states that each degree of freedom contributes $\frac{1}{2}k_BT$ to the average energy of the system. For a system of N particles, the total energy is given by

$$\langle E\rangle = \frac{3}{2}Nk_BT. \tag{25.12}$$

The ideal Boltzmann gas follows the equipartition theorem, see eqn. (23.39).

Let's revisit the equipartition theorem for ideal Fermi and Bose gases. Recall the expression for the average energy for grand canonical ensemble:

$$\langle E\rangle = U = \left(-\frac{\partial}{\partial\beta}\ln\mathcal{Z}\right)_{\mathfrak{z},V} \tag{25.13}$$

where $\mathcal{Z}(\mathfrak{z}, V, T)$ is the grand partition function. We have

$$\ln\mathcal{Z} = \begin{cases} \dfrac{V}{\lambda^3(T)}f_{5/2}(\mathfrak{z}) & \text{for Fermi gas} \\ \dfrac{V}{\lambda^3(T)}g_{5/2}(\mathfrak{z}) - \ln(1-\mathfrak{z}) & \text{for Bose gas} \end{cases}. \tag{25.14}$$

The temperature dependence in the grand partition function is through $\lambda^3(T)$, so we have

$$\lambda(T) = \sqrt{\frac{2\pi\hbar^2}{mk_BT}} = \sqrt{\frac{2\pi\hbar^2\beta}{m}} \quad\Rightarrow\quad \frac{1}{\lambda^3(T)} \sim \beta^{-3/2} \tag{25.15}$$

Thus, we have

$$-\frac{\partial}{\partial\beta}\left(\frac{1}{\lambda^3(T)}\right) \sim \frac{3}{2}\beta^{-5/2} \tag{25.16}$$

Now, let's calculate the average energy for Quantum gases

$$U = \left(-\frac{\partial}{\partial\beta}\ln\mathcal{Z}\right)_{\mathfrak{z},V} = \left\{\begin{matrix} \dfrac{3}{2}\dfrac{k_BT}{\lambda^3(T)}Vf_{5/2}(\mathfrak{z}) & \text{for Fermi gas} \\ \dfrac{3}{2}\dfrac{k_BT}{\lambda^3(T)}Vg_{5/2}(\mathfrak{z}) & \text{for Bose gas} \end{matrix}\right\} \approx \frac{3}{2}PV \tag{25.17}$$

The factor $\frac{3}{2}$ gives a similarity to the classical ideal gas, but the energy per degree of freedom is not $\frac{1}{2}k_BT$ but depends on the statistical distribution of the particles.

26 Lecture 26: (03/04/2025)

§26.1 Density of States

Say we have a particle in a box of volume V with periodic boundary conditions. The energy eigenstates are plane waves of the form

$$\psi(\mathbf{x}) = \frac{1}{\sqrt{V}} e^{i\mathbf{k}\cdot\mathbf{x}}, \tag{26.1}$$

where $\mathbf{k}$ is the wavevector. Due to the periodic boundary conditions, the wavevector $\mathbf{k}$ is quantized as

$$\mathbf{k} = \frac{2\pi}{L}\mathbf{n} = \frac{2\pi}{L}(n_1, n_2, n_3), \qquad n_i \in \mathbb{Z}, \tag{26.2}$$

where L is the length of the box in each dimension. And the energy of the state is given by

$$E_{\mathbf{n}} = \frac{\hbar^2 k^2}{2m} = \frac{4\pi^2\hbar^2}{2mL^2}(n_1^2 + n_2^2 + n_3^2). \tag{26.3}$$

The single particle canonical partition function is given by

$$Z_1 = \sum_{\mathbf{n}} e^{-\beta E_{\mathbf{n}}}. \tag{26.4}$$

The simplest way to compute this sum is to convert it into an integral over the wavevector $\mathbf{k}$,

$$\sum_{\mathbf{n}} \approx \frac{V}{(2\pi)^3}\int \mathrm{d}^3\mathbf{k} = \frac{(4\pi)V}{(2\pi)^3}\int_0^\infty k^2\,\mathrm{d}k\,. \tag{26.5}$$

For future applications, it will prove more useful to change integration variables at this stage. We work instead with the energy,

$$E_{\mathbf{k}} = \frac{\hbar^2 k^2}{2m} \quad \Rightarrow \quad \mathrm{d}E = \frac{\hbar^2}{m}k\,\mathrm{d}k\,. \tag{26.6}$$

Now the integral becomes

$$\frac{(4\pi)V}{(2\pi)^3}\int_0^\infty k^2\,\mathrm{d}k = \frac{V}{2\pi^2}\int \mathrm{d}E\sqrt{\frac{2mE}{\hbar^2}}\frac{m}{\hbar^2} \equiv \int g(E)\,\mathrm{d}E\,, \tag{26.7}$$

where we have defined the density of states $g(E)$ as

$$g(E) = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}. \tag{26.8}$$

We can arrive at the same result by considering a classical gas particle in a box of volume V,

$$\int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{h^3} = \frac{V}{h^3}\int \mathrm{d}^3\mathbf{p} \tag{26.9}$$

now after integrating over the angular part of the momentum space, we have

$$\frac{(4\pi)V}{(2\pi\hbar)^3}\int_0^\infty p^2\,\mathrm{d}p\,, \tag{26.10}$$

and we can change variables to energy as before,

$$\frac{V}{4\pi^2}\int \mathrm{d}E\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E} \equiv \int g(E)\,\mathrm{d}E\,, \tag{26.11}$$

where we have defined the density of states $g(E)$ as

$$g(E) = \frac{V}{4\pi^2}\left(\frac{2m}{\hbar^2}\right)^{3/2}\sqrt{E}. \tag{26.12}$$

§26.1.1 Relativistic Particles

A relativistic particle moving in $3+1$ spacetime dimensions has the energy-momentum relation

$$E^2 = p^2c^2 + m^2c^4 \quad \Rightarrow \quad p^2 = \frac{E^2 - m^2c^4}{c^2}. \tag{26.13}$$

Starting with the same setup as before, we have

$$\int \frac{\mathrm{d}^3\mathbf{q}\,\mathrm{d}^3\mathbf{p}}{h^3} = \frac{V}{h^3}\int \mathrm{d}^3\mathbf{p} = \frac{(4\pi)V}{(2\pi\hbar)^3}\int_0^\infty p^2\,\mathrm{d}p\,, \tag{26.14}$$

now using the energy-momentum relation, we can change variables to energy,

$$\frac{V}{2\pi^2\hbar^3}\int \mathrm{d}E\left(\frac{E^2 - m^2c^4}{c^2}\right)\frac{E}{c}\frac{\mathrm{d}E}{\sqrt{E^2 - m^2c^4}} \equiv \int g(E)\,\mathrm{d}E\,, \tag{26.15}$$

where we have defined the density of states $g(E)$ as

$$g(E) = \frac{V}{2\pi^2\hbar^3c^3}E\sqrt{E^2 - m^2c^4}. \tag{26.16}$$

In particular, for massless particles, we have

$$g(E) = \frac{V}{2\pi^2\hbar^3c^3}E^2. \tag{26.17}$$

> Remark 26.1 (Density of States as a Measure). $g(E)\,\mathrm{d}E$ counts the number of states with energy between E and $E + \mathrm{d}E$. Notice that we haven't actually done the integral over E in eqns. (26.7), (26.11) and (26.15); instead this is to be viewed as a measure which we can integrate over any function $f(E)$ of our choosing.

§26.2 Photon Gas: Blackbody Radiation

To start with, the energy of a photon is determined by its wavelength λ or, equivalently, its frequency $\nu = \frac{c}{\lambda}$,

$$E = h\nu = \frac{hc}{\lambda}. \tag{26.18}$$

In terms of the angular frequency $\omega = 2\pi\nu$, we have

$$E = \hbar\omega = \hbar\frac{2\pi c}{\lambda}. \tag{26.19}$$

Since photons have two polarizations states (one for each dimension transverse to the direction of propagation). To account for this, the density of states should be multiplied by a factor of two, thus the number of states with energy between E and $E + \mathrm{d}E$ is given by

$$g(E)\,\mathrm{d}E = \frac{V}{\pi^2\hbar^3c^3}E^2\,\mathrm{d}E\,. \tag{26.20}$$

Equivalently, we can express this in terms of the frequency,

$$g(E)\,\mathrm{d}E = g(\omega)\,\mathrm{d}\omega = \frac{V}{\pi^2c^3}\omega^2\,\mathrm{d}\omega\,. \tag{26.21}$$

Because photon number is not conserved, we're unable to define a chemical potential for photons. We should instead use the grand canonical ensemble with chemical potential $\mu = 0$ or equivalently, the fugacity $\mathfrak{z} = e^{\beta\mu} = 1$. This means that we should consider states with any number of photons. Thus, the grand canonical partition function is given by

$$\mathcal{Z} = \prod_\omega (Z_\omega)^{g(\omega)} \tag{26.22}$$

where Z_ω is the partition function for photons at fixed frequency ω,

$$Z_\omega = \sum_{N=0}^{\infty} e^{-N\beta\hbar\omega} = \frac{1}{1 - e^{-\beta\hbar\omega}}. \tag{26.23}$$

The log of the grand canonical partition function is given by

$$\ln\mathcal{Z} = \int \mathrm{d}\omega\, g(\omega)\ln Z_\omega = -\frac{V}{\pi^2c^3}\int_0^\infty \mathrm{d}\omega\,\omega^2\ln\left(1 - e^{-\beta\hbar\omega}\right). \tag{26.24}$$

§26.2.1 Planck Distribution

From the partition function, we can compute the average energy of the photon gas,

$$E = -\frac{\partial \ln \mathcal{Z}}{\partial \beta} = \frac{V\hbar}{\pi^2 c^3} \int_0^\infty \mathrm{d}\omega \, \frac{\omega^3}{e^{\beta\hbar\omega} - 1}. \tag{26.25}$$

Since the integral is over all frequencies, we can extranct one more information from the integrand, *i.e.* the average energy carried by photons with frequency between ω and $\omega + \mathrm{d}\omega$,

$$E(\omega)\,\mathrm{d}\omega = \frac{V\hbar}{\pi^2 c^3} \frac{\omega^3}{e^{\beta\hbar\omega} - 1}\,\mathrm{d}\omega\,. \tag{26.26}$$

This is the Planck distribution, which gives the energy density of a photon gas at temperature T.

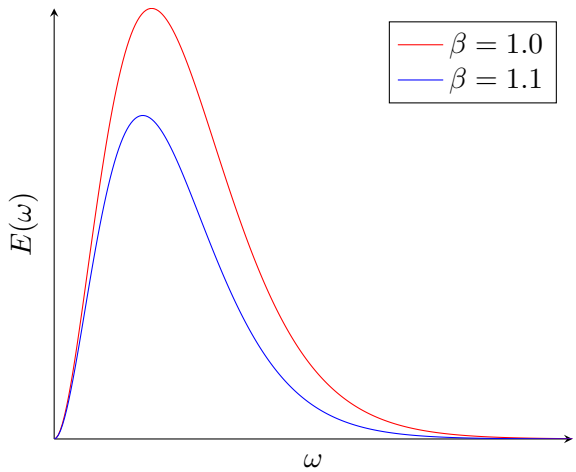


Figure 26.1: The Planck distribution for different temperatures.

We can see that the Planck distribution peaks at a frequency that increases with temperature. Let's determine the peak frequency of the Planck distribution. To do this, we can differentiate the Planck distribution with respect to ω and set it equal to zero,

$$\frac{\mathrm{d}E(\omega)}{\mathrm{d}\omega} = 0 \quad \Rightarrow \quad \frac{V\hbar}{\pi^2 c^3}\left[\frac{3\omega^2}{e^{\beta\hbar\omega} - 1} - \frac{\omega^3(\beta\hbar e^{\beta\hbar\omega})}{(e^{\beta\hbar\omega} - 1)^2}\right] = 0. \tag{26.27}$$

Say we define $x = \beta\hbar\omega$, then after some algebra, we can rewrite the equation as

$$3 - x = 3e^{-x} \tag{26.28}$$

This is a transcendental equation, and we can solve it numerically. The solution is approximately $x \approx 2.822$. Thus, we have

$$\omega_{\max} = 2.822\frac{k_B T}{\hbar} \quad \Rightarrow \quad \lambda_{\max} = \frac{1}{2.822}\frac{hc}{k_B T}. \tag{26.29}$$

This is known as *Wien's Displacement Law*, and it tells us that the peak wavelength of the Planck distribution is inversely proportional to the temperature. In other words, as the temperature increases, the peak wavelength decreases.

To compute the total energy in the photon gas, we can integrate the Planck distribution over all frequencies,

$$E = \frac{V\hbar}{\pi^2 c^3} \int_0^\infty \mathrm{d}\omega \, \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \xrightarrow{\text{substituting } x=\beta\hbar\omega} \frac{V}{\pi^2 c^3}\frac{(k_B T)^4}{\hbar^3} \int_0^\infty \mathrm{d}x \, \frac{x^3}{e^x - 1}. \tag{26.30}$$

Recall the generalized Bose-Einstein integral (see eqn. (25.2)), we retrieve the above integral for $n = 4$ and $\mathfrak{z} = 1$,

$$g_n(z) = \sum_{k=1}^\infty \frac{z^k}{k^n} = \frac{1}{\Gamma(n)} \int_0^\infty \frac{x^{n-1}}{z^{-1}e^x - 1}\,\mathrm{d}x \xrightarrow{n=4,\ z=1} \frac{1}{\Gamma(4)} \int_0^\infty \frac{x^3}{e^x - 1}\,\mathrm{d}x = g_4(1) = \zeta(4). \tag{26.31}$$

Since $\zeta(4) = \frac{\pi^4}{90}$, we have the integral to be $\Gamma(4)\zeta(4) = \frac{\pi^4}{15}$. Thus, we have

$$E = \frac{V}{\pi^2 c^3}\frac{(k_B T)^4}{\hbar^3}\frac{\pi^4}{15} = \frac{V\pi^2}{15\hbar^3 c^3}(k_B T)^4. \tag{26.32}$$

Thus, the energy density $\mathcal{E} = E/V$ in the photon gas is given by

$$\mathcal{E} = \frac{\pi^2 k_B^4}{15\hbar^3 c^3}T^4. \tag{26.33}$$

§26.2.2 Stefan-Boltzmann Law

With this we can compute the energy flux from a black body at temperature T, which is defined as

$$\text{Energy Flux} = \frac{\mathcal{E}c}{4} \equiv \sigma T^4, \tag{26.34}$$

where σ is the Stefan-Boltzmann constant given by

$$\sigma = \frac{\pi^2 k_B^4}{60\hbar^3 c^2} \approx 5.67 \times 10^{-8}\ \mathrm{Js^{-1}m^{-2}K^{-4}}. \tag{26.35}$$

This is known as the *Stefan-Boltzmann Law*, and it tells us that the energy flux from a black body is proportional to the fourth power of its temperature.

The factor of 1/4 comes because we're not considering the flux emitted by a point source, but rather by an actual object whose size is bigger than the wavelength of individual photons. Moreover, we only care about the velocity perpendicular to the object, which is $(c\cos\theta)$ where θ is the angle the photon makes with the normal. This means that rather than filling out a sphere of area 4π surrounding the object, the actual flux of photons from any point on the object's surface is given by

$$\frac{1}{4\pi}\int_0^{2\pi} \mathrm{d}\phi \int_0^{\pi/2} \mathrm{d}\theta \sin\theta (c\cos\theta) = \frac{c}{4}. \tag{26.36}$$

§26.2.3 Radiation Pressure

Let's first compute the exact expression for the partition function of the photon gas. Recall the grand canonical partition function for the photon gas (eqn. (26.24)),

$$\ln \mathcal{Z} = -\frac{V}{\pi^2 c^3}\int_0^\infty \mathrm{d}\omega\, \omega^2 \ln\left(1 - e^{-\beta\hbar\omega}\right). \tag{26.37}$$

We can rewrite this as

$$\ln \mathcal{Z} = -\frac{V}{\pi^2 c^3}\int \mathrm{d}\left(\frac{\omega^3}{3}\right)\ln\left(1 - e^{-\beta\hbar\omega}\right) \xrightarrow{\text{integration by parts}}$$
$$-\frac{V}{\pi^2 c^3}\left\{\left[\frac{\omega^3}{3}\ln\left(1 - e^{-\beta\hbar\omega}\right)\right]_0^\infty - \int_0^\infty \mathrm{d}\omega\, \frac{\omega^3}{3}\frac{e^{-\beta\hbar\omega}(\beta\hbar)}{1 - e^{-\beta\hbar\omega}}\,\mathrm{d}\omega\right\} \tag{26.38}$$

The first term vanishes at both limits, and we can rewrite the second term as

$$\ln \mathcal{Z} = \frac{V\beta\hbar}{3\pi^2 c^3}\int_0^\infty \mathrm{d}\omega\, \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \xrightarrow{\text{substituting } x = \beta\hbar\omega} \frac{V}{3\pi^2 c^3}\frac{(k_B T)^3}{\hbar^3}\int_0^\infty \mathrm{d}x\, \frac{x^3}{e^x - 1}. \tag{26.39}$$

Recall that $\int_0^\infty \mathrm{d}x\, \frac{x^3}{e^x - 1} = \Gamma(4)\zeta(4) = \frac{\pi^4}{15}$, thus we have

$$\ln \mathcal{Z} = \frac{V}{3\pi^2 c^3}\frac{(k_B T)^3}{\hbar^3}\frac{\pi^4}{15} = \frac{V\pi^2}{45\hbar^3 c^3}(k_B T)^3. \tag{26.40}$$

Thus, we have

$$\frac{PV}{k_B T} = \ln \mathcal{Z} \quad \Rightarrow \quad P = \frac{\pi^2}{45\hbar^3 c^3}(k_B T)^4 \quad \Rightarrow \quad \boxed{E = 3PV}. \tag{26.41}$$

27 Lecture 27: (07/04/2025)

We know intuitively that solids are much different from gases, but it is quite surprising that we can employ our study of gases to accurately understand certain properties of solids. The atoms in a solid are not free to move around, but they can vibrate about their equilibrium positions (which is sound waves).

§27.1 Phonons

Quantum mechanics turns electromagnetic waves into discrete packets of energy called photons. In exactly the same way, sounds waves in solids also come in discrete packets. They are called *phonons*. We will take the energy of a phonon to be

$$E = \hbar\omega = pc_s, \tag{27.1}$$

where c_s is the speed of sound in the solid.

The density of states for phonons is same as that of massless relativistic particles (see eqn. (26.17)) with two exceptions:

- we must replace the speed of light c with the speed of sound c_s;
- phonons have three polarizations (two transverse and one longitudinal).

Thus, the density of states for phonons is given by

$$g(\omega)\,\mathrm{d}\omega = \frac{3V}{2\pi^2 c_s^3}\omega^2\,\mathrm{d}\omega\,. \tag{27.2}$$

There is another major difference between phonons and photons. Which is that photons can have arbitrarly large frequencies, but phonons have a maximum frequency. This is because as the frequency increases, the wavelength decreases. But there is a minimum wavelength in the based on the spacing between atoms; it is not possible for sound waves to propagate through a solid with wavelength smaller than the atomic spacing because there's nothing in the middle there to shake. Let's denote this maximum frequency as ω_D (the Debye frequency) and the corresponding wavelength as λ_D.

We expect the minimum wavelength to be on the order of the atomic spacing *a i.e.* $\lambda_D \sim (V/N)^{1/3}$, thus the maximum frequency is of the order of $\omega_D \sim (N/V)^{1/3}c_s$. To find the correct prefactor, we will use Debye's "*clever arguement*". According to Debye, the number of phonon states is equal to the number of degree of freedom in the solid *i.e.* $3N$. Thus, we have

$$\int_0^{\omega_D} g(\omega)\,\mathrm{d}\omega = 3N \quad\Rightarrow\quad \frac{V\omega_D^3}{2\pi^2 c_s^3} = 3N \quad\Rightarrow\quad \omega_D = \left(\frac{6\pi^2 N}{V}\right)^{1/3} c_s. \tag{27.3}$$

From the maximum frequency ω_D, we construct an associated energy scale $\hbar\omega_D$, and temperature scale,

$$T_D = \frac{\hbar\omega_D}{k_B}. \tag{27.4}$$

This is called the *Debye temperature*. It provides a way of characterising all solids: it is the temperature at which the highest frequency phonon starts to become excited.

§27.2 Heat Capacity of Solids

Like photons, the number of phonons in a solid is not conserved. Thus, we will use the grand canonical ensemble with $\mathfrak{z} = 1$. Let's start by calculating the partition function for phonons of same frequency ω:

$$Z_\omega = \sum_{N=0}^{\infty} e^{-\beta N\hbar\omega} = \frac{1}{1 - e^{-\beta\hbar\omega}}. \tag{27.5}$$

The grand canonical partition function is given by

$$\mathcal{Z}_{\text{phonon}} = \prod_{\omega} (Z_\omega)^{g(\omega)} \quad \Rightarrow \quad \ln \mathcal{Z}_{\text{phonon}} = \int_0^{\omega_D} g(\omega) \ln Z_\omega \, \mathrm{d}\omega \tag{27.6}$$

The total energy is sound waves is given by

$$U = -\frac{\partial \ln \mathcal{Z}_{\text{phonon}}}{\partial \beta} = \int_0^{\omega_D} g(\omega) \frac{\hbar\omega}{e^{\beta\hbar\omega} - 1} \, \mathrm{d}\omega \,. \tag{27.7}$$

So, we have

$$U = \frac{3V\hbar}{2\pi^2 c_s^3} \int_0^{\omega_D} \frac{\omega^3}{e^{\beta\hbar\omega} - 1} \, \mathrm{d}\omega \xrightarrow{\text{substitution } x=\beta\hbar\omega} = \frac{3Vk_B^4}{2\pi^2 c_s^3 \hbar^3} T^4 \int_0^{T_D/T} \frac{x^3}{e^x - 1} \, \mathrm{d}x \,. \tag{27.8}$$

This integral is a function of T_D/T and has no analytic expression. Let's study the two limits:

- In the limit $T \ll T_D$, we can replace the upper limit of the integral by infinity (since $T_D/T \gg 1$). Thus, we have

$$U = \frac{3Vk_B^4}{2\pi^2 c_s^3 \hbar^3} T^4 \int_0^{\infty} \frac{x^3}{e^x - 1} \, \mathrm{d}x = \frac{3Vk_B^4}{2\pi^2 c_s^3 \hbar^3} T^4 \cdot \frac{\pi^4}{15} = \frac{\pi^2 V k_B^4}{10\hbar^3 c_s^3} T^4. \tag{27.9}$$

  The heat capacity is given by

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{2\pi^2 V k_B^4}{5\hbar^3 c_s^3} T^3 \xrightarrow{\text{in terms of } T_D} C_V = N k_B \frac{12\pi^4}{5} \left(\frac{T}{T_D}\right)^3 \tag{27.10}$$

- In the limit $T \gg T_D$, we only integrate over small values of x, allowing us to use the approximation $e^x \approx 1 + x$. Thus, we have

$$\int_0^{T_D/T} \frac{x^3}{e^x - 1} \, \mathrm{d}x \approx \int_0^{T_D/T} \frac{x^3}{(1+x) - 1} \, \mathrm{d}x = \int_0^{T_D/T} x^2 \, \mathrm{d}x = \frac{1}{3}\left(\frac{T_D}{T}\right)^3. \tag{27.11}$$

  Thus, we have

$$U = \frac{3Vk_B^4}{2\pi^2 c_s^3 \hbar^3} T^4 \cdot \frac{1}{3}\left(\frac{T_D}{T}\right)^3 = \frac{V k_B^4 T_D^3}{2\pi^2 c_s^3 \hbar^3} T \xrightarrow{\text{in terms of } N} U = 3Nk_B T \tag{27.12}$$

  The heat capacity is given by

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V = \frac{V k_B^4 T_D^3}{2\pi^2 c_s^3 \hbar^3} = 3Nk_B. \tag{27.13}$$

  This high-temperature behaviour has been known experimentally since the early 1800s. It is called the *Dulong-Petit law*.

§27.2.1 What's wrong with Debye's model?: A Closer Look at dispersion relation

As we've seen that the Debye model is remarkably accurate in capturing the heat capacities of solids. But there is one major flaw in the model. The Debye model assumes that the dispersion relation is linear, but it turns out that the dispersion relation for phonons is a little more complicated. It is a bit complicated to calculate the dispersion relation for phonons, but we can see why this model fails with the help of a simple example.

Let's consider a one-dimensional periodic lattice of N atoms with spacing a. The equilibrium of lth atom is given by $x_l = la$ and to impose periodicity, we have $x_{N+1} = x_1$. Let u_l be the displacement of the lth atom from its equilibrium position. If we approximate the bonds between the atoms as harmonic oscillators, we can write the Hamiltonian as

$$\mathcal{H} = \frac{1}{2m} \sum_{l=1}^{N} \dot{u}_l^2 + \frac{\alpha}{2} \sum_{l=1}^{N} (u_{l+1} - u_l)^2, \tag{27.14}$$

where α is a constant that determines the strength of the bond between the atoms. The equation of motion is

$$\ddot{u}_l = \frac{\alpha}{m}(2u_l - u_{l+1} - u_{l-1}). \tag{27.15}$$

This is easily solved by the discrete Fourier transform. We make the ansatz

$$u_l = \frac{1}{\sqrt{N}} \sum_k \tilde{u}_k e^{i(kla - \omega_k t)}, \tag{27.16}$$

Plugging this into the equation of motion gives the dispersion relation

$$\omega_k = 2\sqrt{\frac{\alpha}{m}} \left| \sin\left(\frac{ka}{2}\right) \right|. \tag{27.17}$$

We can see that the dispersion relation is linear for small k (long wavelengths), but it is not linear for large k (short wavelengths).

§27.3 Low Temperature Limit of Fermi Distribution

The Fermi distribution is given by

$$f(\epsilon) = \frac{1}{e^{(\epsilon-\mu)/k_B T} + 1}. \tag{27.18}$$

Define the quantity $\delta = (\epsilon - \mu)/k_B$. The Fermi distribution can be rewritten as

$$f(\delta) = \frac{1}{e^{\delta/T} + 1}. \tag{27.19}$$

Consider the two case $\delta > 0$ and $\delta < 0$:

- For $\delta > 0$, as $T \to 0$, we have $\delta/T \to \infty$ and hence $e^{\delta/T} \to \infty$. Thus, we have

$$f(\delta) = \frac{1}{e^{\delta/T} + 1} \to 0. \tag{27.20}$$

- For $\delta < 0$, as $T \to 0$, we have $\delta/T \to -\infty$ and hence $e^{\delta/T} \to 0$. Thus, we have

$$f(\delta) = \frac{1}{e^{\delta/T} + 1} \to 1. \tag{27.21}$$

And for $\delta = 0$, at finite temperature, we have

$$f(0) = \frac{1}{2}. \tag{27.22}$$

Thus, in low temperature limit, we have

$$f(\epsilon) = \begin{cases} 1 & \text{if } \epsilon < \mu \\ \frac{1}{2} & \text{if } \epsilon = \mu \\ 0 & \text{if } \epsilon > \mu \end{cases} \tag{27.23}$$

We define the *Fermi energy* ϵ_F as the energy of the highest occupied state at $T = 0$. Thus, we have

$$\epsilon_F = \mu(T = 0). \tag{27.24}$$

In this way, the fermions fill out a ball in momentum space. The momentum of the final fermion is called the *Fermi momentum* and is related to the Fermi energy in the usual way: $\hbar k_F = \sqrt{2m\epsilon_F}$. All states with wavevector $\left|\vec{\mathbf{k}}\right| \leq k_F$ are filled and are said to form the *Fermi sea* or *Fermi sphere*. Those states with $\left|\vec{\mathbf{k}}\right| = k_F$ lie on the edge of the Fermi sea. They are said to form the *Fermi surface*.

Alongside the Fermi energy, we can also define the *Fermi temperature* T_F as

$$T_F = \frac{\epsilon_F}{k_B} \tag{27.25}$$

This gives us a virtual temperature scale with which we can the quantum behaviour of the system. For example, if $T \ll T_F$, we can ignore the thermal excitations and treat the system as a degenerate Fermi gas. If $T \sim T_F$, only particles near the Fermi surface are excited, and we can treat the system as a *quasi-degenerate* Fermi gas. If $T \gg T_F$, we can ignore the quantum effects and treat the system as a classical gas.

§27.4 Degenerate Fermi Gas

Till now, we have added the degeneracy factor g_s to the density of states by hand *i.e.* in case of photons, we have $g_s = 2$ and in case of phonons, we have $g_s = 3$. But for fermions, there is a more systematic way of including the degeneracy factor. The density of states for fermions is given by

$$g(E) = g_s \frac{V}{4\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \sqrt{E}, \tag{27.26}$$

where g_s is the degeneracy factor due to spin, and it is equal to $2s+1$ where s is the spin of the fermion. In low temperature limit, the Fermi distribution is given by

$$f(E) = \begin{cases} 1 & \text{if } E < \epsilon_F \\ 0 & \text{if } E > \epsilon_F \end{cases} \tag{27.27}$$

Now we can use the density of states to calculate the number of particles in the system. The number of particles in the system is given by

$$N = \int_0^\infty g(E) f(E)\, \mathrm{d}E = \int_0^{\epsilon_F} g(E)\, \mathrm{d}E = g_s \frac{V}{6\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \epsilon_F^{3/2}. \tag{27.28}$$

This gives us the Fermi energy in terms of the number of particles in the system:

$$\epsilon_F = \frac{\hbar^2}{2m} \left(\frac{6\pi^2}{g_s} \frac{N}{V}\right)^{2/3}. \tag{27.29}$$

We can also calculate the total energy of the system. The total energy of the system is given by

$$U = \int_0^\infty g(E) f(E) E\, \mathrm{d}E = \int_0^{\epsilon_F} g(E) E\, \mathrm{d}E = g_s \frac{V}{10\pi^2} \left(\frac{2m}{\hbar^2}\right)^{3/2} \epsilon_F^{5/2}. \tag{27.30}$$

In terms of the number of particles in the system, we have

$$U = \frac{3}{5} N \epsilon_F. \tag{27.31}$$

Recall the relationship between total energy and pressure (see eqn. (25.17))

$$U = \frac{3}{2} PV = \frac{3}{5} N \epsilon_F \quad \Rightarrow \quad P = \frac{2}{5} \frac{N}{V} \epsilon_F. \tag{27.32}$$

Even at zero temperature, the gas has non-zero pressure, known as *degeneracy pressure*. It is a consequence of the Pauli exclusion principle and is important in the astrophysics of white dwarf stars and neutron stars as it prevents them from collapsing under their own gravity.

A Tutorials

§A.1 Tutorial 01: (16/01/2025)

Problem A.1.1 (Gaussian Integral).
Show that following holds for $a > 0$:

$$\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2} = \sqrt{\frac{\pi}{a}}. \tag{A.1}$$

Solution.
We can write the integral I as:

$$\begin{aligned} I^2 &= \left(\int_{-\infty}^{\infty} \mathrm{d}x\, e^{-ax^2}\right)\left(\int_{-\infty}^{\infty} \mathrm{d}y\, e^{-ay^2}\right) \\ &= \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y\, e^{-a(x^2+y^2)}. \end{aligned}$$

Now, we can convert the integral to polar coordinates:

$$\begin{aligned} \int_{-\infty}^{\infty} \mathrm{d}x \int_{-\infty}^{\infty} \mathrm{d}y\, e^{-a(x^2+y^2)} &= \int_0^{2\pi} \mathrm{d}\theta \int_0^{\infty} \mathrm{d}r\, re^{-ar^2} \\ &= 2\pi \int_0^{\infty} \mathrm{d}r\, re^{-ar^2}. \end{aligned}$$

Now, we can substitute $u = ar^2$ to get:

$$\begin{aligned} I^2 = 2\pi \int_0^{\infty} \mathrm{d}r\, re^{-ar^2} &= \frac{\pi}{a} \int_0^{\infty} \mathrm{d}u\, e^{-u} \\ &= \frac{\pi}{a}[-e^{-u}|_0^{\infty} \\ &= \frac{\pi}{a}. \end{aligned}$$

Thus, we have $I = \sqrt{\pi/a}$. ◀

Problem A.1.2 (Continuous Memoryless Distribution).
Prove that the Exponential Distribution (see eqn. (3.13)) follows the Markov property (memoryless property). That is, show that

$$P(T > t + s \mid T > s) = P(T > t). \tag{A.2}$$

Solution.
We can write the LHS as:

$$\begin{aligned} P(T > t + s \mid T > s) &= \frac{P([T > t + s] \cap [T > s])}{P(T > s)} \\ &= \frac{P(T > t + s)}{P(T > s)} \end{aligned}$$

Let $P(t)$ be the CDF of the Exponential Distribution (see eqn. (3.14)) *i.e.* $P(t) = \text{prob}(T \leq t)$. So $P(T > t) = 1 - P(t)$.

$$\begin{aligned} &= \frac{e^{-\lambda(t+s)}}{e^{-\lambda s}} = e^{-\lambda t} \\ &= P(T > t). \end{aligned}$$

Thus, we have $P(T > t + s \mid T > s) = P(T > t)$. Hence, the Exponential Distribution follows the Markov property. ◀

Problem A.1.3 (Change of Variables).

Let X be a random variable with the probability density function

$$p_X(x) = \begin{cases} 3(1-x)^2, & 0 \le x \le 1, \\ 0, & \text{otherwise.} \end{cases} \tag{A.3}$$

Let $Y = (1-X)^3$. Find the probability density function of Y.

Solution.

We can write the CDF of Y as:

$$\begin{aligned} P(Y \le y) &= P((1-X)^3 \le y) = P(1-X \le y^{1/3}) = P(X \ge 1 - y^{1/3}) \\ &= 1 - P(X < 1 - y^{1/3}) = 1 - \int_0^{1-y^{1/3}} 3(1-x)^2 \, \mathrm{d}x \,. \end{aligned}$$

Thus, computing the integral, we get:

$$\begin{aligned} P(Y \le y) &= 1 - \int_0^{1-y^{1/3}} 3(1-x)^2 \, \mathrm{d}x \\ &= 1 - \left[-(1-x)^3\right|_0^{1-y^{1/3}} \\ &= 1 - [(1) - (y)] \\ &= y. \end{aligned}$$

Thus, the CDF of Y is $P(Y \le y) = y$. Differentiating with respect to y, we get the PDF of Y:

$$p_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} y = 1. \tag{A.4}$$

◀

§A.2 Tutorial 02: (30/01/2025)

Problem A.2.1.

Let a system of N particles with no interactions be in a potential $V(x) = V_\circ \exp(-x^2/w^2)$, where $V_\circ$ and w are constants. The single-particle distribution function (for equilibrium) satisfies following boundary conditions:

$$f_1^{\mathrm{eq}}(|x| \to \infty, p) = \exp\left(-\beta \frac{p^2}{2m}\right) \tag{A.5}$$

where $\beta = 1/(k_B T)$, k_B is the Boltzmann constant, T is the temperature, and m is the mass of the particle. Find the single-particle distribution function $f_1^{\mathrm{eq}}(x, p)$.

Problem A.2.2 (Liouville's Theorem).

Show that the volume of a phase space element is conserved under Hamiltonian dynamics.

Problem A.2.3.

Cauchy distribution is given by

$$f(x) = \frac{1}{\pi} \frac{\gamma}{(x - x_0)^2 + \gamma^2} \tag{A.6}$$

where x_0 is the location parameter and γ is the scale parameter. Find the expectation value of x.

Solution.

Naive approach:

$$\langle x \rangle = \int_{-\infty}^{\infty} x f(x) \, \mathrm{d}x \tag{A.7}$$

Since the PDF is even, so the integrand is odd.

$$= 0 \tag{A.8}$$

But this is not true. The correct approach is to use the definition of expectation value:

$$\int |xf(x)|\,\mathrm{d}x < \infty \quad \Rightarrow \quad \langle x \rangle = \int xf(x)\,\mathrm{d}x\,. \tag{A.9}$$

In this case, the integral does not converge absolutely, so the expectation value does not exist. ◀

B Exams

§B.1 Class Test 01: (13/02/2025)

Problem B.1.1 (8 Marks).

Consider a chemical reaction

$$X \underset{k_1}{\overset{k_2}{\rightleftharpoons}} A, \tag{B.1}$$

and assume that the number of molecules of $A = a$ is fixed and the number of molecules of X is random variable n. Let the rate of the forward reaction be $k_2 n$ and the rate of the backward reaction be $k_1 a$.

a) Write the master equation for the probability $P(n, t)$ of finding n molecules of X at time t.

b) Find the steady state solution of the master equation.

c) Find the characteristic function of the steady state solution.

d) Calculate the mean and variance of the steady state distribution.

Problem B.1.2 (7 Marks).

Consider a system of N indistinguishable non-interacting particles. Each particle can either be in the ground state with energy $-\epsilon$ or in the excited state with energy ϵ. Let the total energy of the system be E and define the macrostate of the system by (E, N).

Let $x = \frac{E}{N\epsilon}$ and T be the temperature of the system. Find the relation between x and T *i.e.*

$$\frac{2\epsilon}{k_B T} = f(x). \tag{B.2}$$

§B.2 Mid-Semester Exam: (28/02/2025)

Problem B.2.1 (5 Marks).

Find the leading term of the asymptotics of the following integral for $\lambda \to \infty$:

$$I(\lambda) = \int_{-5}^{4} e^{-\lambda x^2} \ln\left(1 + x^2\right) \mathrm{d}x\,. \tag{B.3}$$

Problem B.2.2 (5 Marks).

Suppose there are two independent discrete random variables K_1 and K_2, each drawn from a Poisson distribution with parameters λ_1 and λ_2 respectively. Define the random variable $K := K_1 + K_2$, determine the characteristic function of the random variable K. Hence, find the mean and the variance of K.

Problem B.2.3 (4 Marks).

Let the internal energy of a particular system be given by

$$U = A\frac{N^3}{V^2} \exp\left(\frac{S}{Nk_B}\right) \tag{B.4}$$

where A is an unitful constant. Initially the system is at temperature $T = 320$ K and at pressure $P = 2 \times 10^5$ Pa. The system expands reversibly, such that the final pressure is $P = 1 \times 10^5$ Pa, by an isoentropic process. Find the final temperature of the system.

Problem B.2.4 (6 Marks).

Consider a collection of N hard spheres confined in a box of volume V and each sphere occupies a tiny volume ε and apart from hardcore exclusion, there are no other interactions between the spheres. Calculate the heat capacity ratio $\gamma = \frac{C_V}{C_P}$ of the system.

Useful Formulae

$$\Gamma(z) = \int_0^\infty t^{z-1} e^{-t}\,\mathrm{d}t \qquad C_P = T\left(\frac{\partial S}{\partial T}\right)_{P,N} \qquad C_V = T\left(\frac{\partial S}{\partial T}\right)_{V,N}$$

§B.3 Class Test 02: (09/04/2025)

Problem B.3.1 (Bose Magnetism: 7 Marks).

A 3D gas of non-interacting bosons with spin $S^{(z)} \equiv S = \{-1, 0, +1\}$ has the following Hamiltonian:

$$\mathcal{H} = \sum_i \frac{|\vec{\mathbf{p}}_i|^2}{2m} - \mu_0 B S_i^{(z)}. \tag{B.5}$$

Here, B is a constant magnetic field in the z-direction.

1. Calculate the number of bosons $N_+,\ N_0,\ N_-$ in the three different spin states for some temperature $T \gg 0$.
2. Find the average magnetization.

Problem B.3.2 (4 Marks).

Given a Hamiltonian $\mathcal{H}(\vec{\mathbf{q}}, \vec{\mathbf{p}})$, show that

$$\left\langle q_i \frac{\partial \mathcal{H}}{\partial q_i} \right\rangle = \frac{1}{\beta} \tag{B.6}$$

Problem B.3.3 (4 Marks).

Let a classical system given by the Hamiltonian $\mathcal{H}_\circ$ has Helmholtz free energy $F_\circ = F(\mathcal{H}_\circ)$. Now consider a Hamiltonian $\mathcal{H} = \mathcal{H}_\circ + \lambda \mathcal{H}_1$. Obtain a series expansion for the Helmholtz free energy $F(\mathcal{H})$ written in terms of the cumulants of $\mathcal{H}_1$ upto the second order in λ.

Useful Formulae

$$g_{5/2}(z) = -\frac{4}{\sqrt{\pi}} \int_0^\infty \mathrm{d}x\, x^2 \ln\left(1 - z e^{-x^2}\right) \qquad g_{3/2}(z) = z \frac{\partial}{\partial z} g_{5/2}(z)$$

$$\lambda = \sqrt{\frac{2\pi\hbar^2}{m k_B T}} \qquad \int_0^\infty \mathrm{d}x\, \frac{x^3}{e^x - 1} = \Gamma(4)\zeta(4) = \frac{\pi^4}{15}$$

§B.4 End-Semester Exam: (01/05/2025)

Problem B.4.1 (8 Marks).

Consider a probability distribution

$$f(\epsilon) = \frac{2}{\sqrt{\pi}} \sqrt{\epsilon} e^{-\epsilon}, \quad \epsilon \in [0, \infty). \tag{B.7}$$

Find and sketch the probability distribution function of the transformed random variable $\nu(\epsilon) = \sqrt{\epsilon}$.

Problem B.4.2 (10 Marks).

A bosonic gas has density of states $g(E) = AE^{\sigma}$ per volume, where A is a constant. Empirical observations indicate that $T_c \sim N^{3/7}$. Assuming the scenario of ideal Bose-Einstein condensation, estimate σ.

Problem B.4.3 (12 Marks).

For a gas of fermions with spin states $s \equiv \pm 1$, in volume V at temperature T, and in background magnetic field B,

(a) Write the occupation number of fermions N_+ and N_- in the spin states $s = +1$ and $s = -1$ respectively.

(b) Find the average magnetization in the high temperature limit, where $f_{3/2}(x) \approx x$.

(c) Show that magnetic susceptibility in the limit of vanishing B satisfies the Curie's law.

Problem B.4.4 (10 Marks).

Write the density of states for acoustic phonons in 2D lattice given that number of possible polarizations is 2. Hence estimate the maximum allowed phonon frequency ω_D following the argument of Debye.

Problem B.4.5 (10 Marks).

Consider an ideal gas in the isobaric ensemble described by the macrostate $M \equiv (N, T, P)$ and the microstate $\boldsymbol{\mu} \equiv \{\mathbf{q}_i, \mathbf{p}_i, V\}$ with $\mathbf{q}_i, \mathbf{p}_i \in \mathbb{R}^3$ and $V \in \mathbb{R}_{>0}$ for $i = 1, 2, \ldots, N$. Write the partition function and derive the enthalpy.

References

[1] Luca Donati. *From Chapman-Kolmogorov equation to master equation and Fokker-Planck equation*. 2023. URL: https://www.zib.de/userpage/donati/stochastics2023/03/lecture_notes/L03_dCKeq.pdf.

[2] Kerson Huang. *Statistical Mechanics*. 2nd ed. John Wiley & Sons, 1987. ISBN: 978-0-471-81518-1.

[3] Mehran Kardar. *Statistical Physics of Particles*. Cambridge University Press, 2007. ISBN: 978-0-521-87342-0. DOI: 10.1017/CBO9780511815898.

[4] J Marcinkiewicz. "Sur une propriété de la loi de Gauß". fr. In: *Mathematische Zeitschrift* 44.1 (Dec. 1939), pp. 612–618. DOI: 10.1007/BF01210677.

[5] Jonathan Noval. *Pólya's Random Walk Theorem*. 2013. URL: https://math.mit.edu/classes/18.095/lect2/notes.pdf.